In the table of 100 simulated sets, in what proportion were there 7 or more heads? Give a decimal.

Heads in 10 flips	0	1	2	3	4	5	6	7	8	9	10
Number of simulated trials	0	1	4	12	20	25	20	12	4	2	0

Proportion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F2-Q4

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 23 • FORWARD 2 • Q4/5

801

QUESTION ID
S23-F2-Q4

TYPE
Short answer

CORRECT ANSWER
0.18

GRADING NOTE (12 + 4 + 2)/100.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F2-Q4

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 23 • FORWARD 2 • Q4/5

True or false?

If a model is consistent with the data, the simulation proves that the model is true.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F2-Q5

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 23 • FORWARD 2 • Q5/5

802

QUESTION ID

S23-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It only fails to rule the model out.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F2-Q5

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 23 • FORWARD 2 • Q5/5

Decide whether a model is consistent with data

MAIN	Probability & Statistics main question	**HSS-IC.A.2**
BACKWARD 1	Read the results of a simulation	**HSS-IC.A.2**
BACKWARD 2	Compare observed frequencies with a probability model	**7.SP.C.7.b**
FORWARD 1	Decide using results extreme in either direction	**HSS-IC.A.2**
FORWARD 2	Judge a treatment difference by re-randomization	**HSS-IC.B.5**

SET 24 • HSS-IC.A.2

MAIN

Probability & Statistics main question • High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2

The table shows 100 simulated sets of 10 fair-coin flips. In what proportion of the sets were there 8 or more heads? Give a decimal.

Heads in 10 flips	0	1	2	3	4	5	6	7	8	9	10
Number of simulated trials	0	1	4	12	20	25	20	12	4	2	0

Proportion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-M-Q1

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 24 • MAIN • Q1/5

805

QUESTION ID
S24-M-Q1

TYPE
Short answer

CORRECT ANSWER
0.06

GRADING NOTE (4 + 2 + 0)/100.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S24-M-Q1**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 24 • MAIN • Q1/5

A coin claimed to be fair gave 8 heads in 10 flips. Use the table of 100 simulated sets. Decision rule: the model is a fair coin. Count the simulated results that are 8 or more heads (a one-sided check: only an excess of heads counts against the model). If they make up fewer than 5% of the simulated results, the model is not consistent with the data; otherwise it is consistent. Which decision is correct?

Heads in 10 flips	0	1	2	3	4	5	6	7	8	9	10
Number of simulated trials	0	1	4	12	20	25	20	12	4	2	0

A. Not consistent: a fair coin should give 5 heads every time.

B. Consistent: 8 heads happened in exactly 4% of the sets.

C. Not consistent: 8 or more heads happened in only 6% of the sets.

D. Consistent: 8 or more heads happened in 6% of the simulated sets.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-M-Q2

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 24 • MAIN • Q2/5

806

QUESTION ID

S24-M-Q2

TYPE

Multiple choice

CORRECT ANSWER

D. Consistent: 8 or more heads happened in 6% of the simulated sets.

GRADING NOTE 6% is not fewer than 5%. The others misapply the rule, expect exactly 5 heads, or use "exactly 8" (4%) instead of "8 or more".

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-M-Q2

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 24 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

(Hypothetical simulation results.) A die is claimed to be fair. In 60 rolls it showed 18 sixes. In 200 simulated sets of 60 fair rolls, 18 or more sixes happened 4 times. What proportion of the simulated sets is that? Give a decimal.

Proportion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-M-Q3

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 24 • MAIN • Q3/5

807

QUESTION ID
S24-M-Q3

TYPE
Short answer

CORRECT ANSWER
0.02

GRADING NOTE 4/200.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S24-M-Q3**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 24 • MAIN • Q3/5

(Hypothetical simulation results.) A die claimed to be fair showed 18 sixes in 60 rolls. In 200 simulated sets of 60 fair rolls, 18 or more sixes happened 4 times. Decision rule: the model is a fair die. Count the simulated results that are 18 or more sixes (a one-sided check: only an excess of sixes counts against the model). If they make up fewer than 5% of the simulated results, the model is not consistent with the data; otherwise it is consistent. Which decision is correct?

A. Consistent: 18 sixes is possible with a fair die.

B. Not consistent: so many sixes happened in only 2% of simulated sets.

C. Not consistent: a fair die must give exactly 10 sixes.

D. Consistent: 4 simulated sets had 18 or more sixes.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-M-Q4

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 24 • MAIN • Q4/5

808

QUESTION ID

S24-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. Not consistent: so many sixes happened in only 2% of simulated sets.

GRADING NOTE 2% < 5%. Possible is not the test; the proportion is.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S24-M-Q4**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 24 • MAIN • Q4/5

(Hypothetical simulation results.) A spinner is claimed to land on red 25% of the time. In 40 spins it landed on red 6 times. In 1,000 simulated sets of 40 spins, 6 or fewer reds happened 120 times. Decision rule: the model is a spinner that lands on red 25% of the time. Count the simulated results that are 6 or fewer reds (a one-sided check: only too few reds counts against the model). If they make up fewer than 5% of the simulated results, the model is not consistent with the data; otherwise it is consistent. Which decision is correct?

A. Consistent: the spinner landed on red exactly 25% of the time.

B. Consistent: 6 or fewer reds happened in 12% of the sets.

C. Not consistent: only 120 of the sets had 6 or fewer.

D. Not consistent: 6 is fewer than the expected 10 reds.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-M-Q5

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 24 • MAIN • Q5/5

809

QUESTION ID

S24-M-Q5

TYPE

Multiple choice

CORRECT ANSWER

B. Consistent: 6 or fewer reds happened in 12% of the sets.

GRADING NOTE 120/1000 = 12% ≥ 5%. 6/40 = 15%, not 25%.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S24-M-Q5**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 24 • MAIN • Q5/5

SET 24 • HSS-IC.A.2

BACKWARD 1

Read the results of a simulation • High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2

ANSWER KEY Reduced copy of the question page, with its answer

In the table of 50 simulated families of 4 children, how many families had exactly 2 girls?

Girls in 4 births	0	1	2	3	4
Number of simulated trials	3	12	19	13	3

Families: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B1-Q1

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 24 • BACKWARD 1 • Q1/5

811

QUESTION ID

S24-B1-Q1

TYPE

Short answer

CORRECT ANSWER

19

GRADING NOTE Read the table.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S24-B1-Q1**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 24 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

In the table of 50 simulated families, what proportion had 1 or fewer girls? Give a decimal.

Girls in 4 births	0	1	2	3	4
Number of simulated trials	3	12	19	13	3

Proportion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B1-Q2

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 24 • BACKWARD 1 • Q2/5

812

QUESTION ID

S24-B1-Q2

TYPE

Short answer

CORRECT ANSWER

0.3

GRADING NOTE (3 + 12)/50.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S24-B1-Q2**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 24 • BACKWARD 1 • Q2/5

True or false?

The table of 100 simulated sets of 10 fair-coin flips shows 25 sets with exactly 5 heads.

Heads in 10 flips	0	1	2	3	4	5	6	7	8	9	10
Number of simulated trials	0	1	4	12	20	25	20	12	4	2	0

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B1-Q3

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 24 • BACKWARD 1 • Q3/5

813

QUESTION ID

S24-B1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Read it.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B1-Q3

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 24 • BACKWARD 1 • Q3/5

In the table of 100 simulated sets of 10 fair-coin flips, how many sets had 3 or fewer heads?

Heads in 10 flips	0	1	2	3	4	5	6	7	8	9	10
Number of simulated trials	0	1	4	12	20	25	20	12	4	2	0

A. 12

B. 17

C. 5

D. 29

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B1-Q4

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 24 • BACKWARD 1 • Q4/5

814

QUESTION ID

S24-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 17

GRADING NOTE 0 + 1 + 4 + 12.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B1-Q4

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 24 • BACKWARD 1 • Q4/5

True or false?

In the table of 100 simulated sets of 10 fair-coin flips, more sets had 6 heads than 4 heads.

Heads in 10 flips	0	1	2	3	4	5	6	7	8	9	10
Number of simulated trials	0	1	4	12	20	25	20	12	4	2	0

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B1-Q5

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 24 • BACKWARD 1 • Q5/5

815

QUESTION ID

S24-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 20 each.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B1-Q5

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 24 • BACKWARD 1 • Q5/5

SET 24 • HSS-IC.A.2

BACKWARD 2

Compare observed frequencies with a probability model • Grade 7 • Statistics & Probability • 7.SP.C.7.b

True or false?

A coin that lands heads 520 times in 1,000 flips is reasonably consistent with a fair coin.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B2-Q1

Grade 7 • Statistics & Probability • 7.SP.C.7.b • Set 24 • BACKWARD 2 • Q1/5

817

QUESTION ID

S24-B2-Q1

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE 52% is close to 50%.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B2-Q1

Grade 7 • Statistics & Probability • 7.SP.C.7.b • Set 24 • BACKWARD 2 • Q1/5

A fair die is rolled 600 times, independently. About how many sixes does the fair model predict?

A. 160

B. 60

C. 100

D. 600

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B2-Q2

Grade 7 • Statistics & Probability • 7.SP.C.7.b • Set 24 • BACKWARD 2 • Q2/5

818

QUESTION ID

S24-B2-Q2

TYPE

Multiple choice

CORRECT ANSWER

C. 100

GRADING NOTE 600/6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B2-Q2

Grade 7 • Statistics & Probability • 7.SP.C.7.b • Set 24 • BACKWARD 2 • Q2/5

How many heads does a fair-coin model predict on average in 10 flips?

Heads: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B2-Q3

Grade 7 • Statistics & Probability • 7.SP.C.7.b • Set 24 • BACKWARD 2 • Q3/5

819

QUESTION ID

S24-B2-Q3

TYPE

Short answer

CORRECT ANSWER

5

GRADING NOTE $10 \cdot 0.5$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B2-Q3

Grade 7 • Statistics & Probability • 7.SP.C.7.b • Set 24 • BACKWARD 2 • Q3/5

True or false?

A fair die rolled 60 times, independently, must show each face exactly 10 times.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B2-Q4

Grade 7 • Statistics & Probability • 7.SP.C.7.b • Set 24 • BACKWARD 2 • Q4/5

820

QUESTION ID

S24-B2-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Results vary by chance.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B2-Q4

Grade 7 • Statistics & Probability • 7.SP.C.7.b • Set 24 • BACKWARD 2 • Q4/5

A spinner is claimed to land on red 50% of the time. Which result in 100 spins most suggests the claim is wrong?

A. 82 red

B. 50 red

C. 47 red

D. 53 red

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B2-Q5

Grade 7 • Statistics & Probability • 7.SP.C.7.b • Set 24 • BACKWARD 2 • Q5/5

821

QUESTION ID

S24-B2-Q5

TYPE

Multiple choice

CORRECT ANSWER

A. 82 red

GRADING NOTE Farthest from 50.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B2-Q5

Grade 7 • Statistics & Probability • 7.SP.C.7.b • Set 24 • BACKWARD 2 • Q5/5

SET 24 • HSS-IC.A.2

FORWARD 1

Decide using results extreme in either direction • High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2

In the table of 100 simulated sets of 10 fair flips, what proportion had 9 or more heads or 1 or fewer heads? Give a decimal.

Heads in 10 flips	0	1	2	3	4	5	6	7	8	9	10
Number of simulated trials	0	1	4	12	20	25	20	12	4	2	0

Proportion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F1-Q1

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 24 • FORWARD 1 • Q1/5

823

QUESTION ID

S24-F1-Q1

TYPE

Short answer

CORRECT ANSWER

0.03

GRADING NOTE (2 + 0) + (0 + 1) = 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S24-F1-Q1**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 24 • FORWARD 1 • Q1/5

A coin claimed to be fair gave 1 head in 10 flips. Use the table of 100 simulated sets of 10 fair-coin flips; results at least as far from 5 heads as 1 head are in either tail. Decision rule: the model is a fair coin. Count the simulated results that are 9 or more heads or 1 or fewer heads (a two-sided check: too many or too few heads both count). If they make up fewer than 5% of the simulated results, the model is not consistent with the data; otherwise it is consistent. Which decision is correct?

Heads in 10 flips	0	1	2	3	4	5	6	7	8	9	10
Number of simulated trials	0	1	4	12	20	25	20	12	4	2	0

A. Not consistent: a fair coin cannot give 1 head.

B. Not consistent: such results happened in 3% of the sets.

C. Consistent: 1 head happened in exactly 1 set.

D. Consistent: such results happened in 3% of the sets.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F1-Q2

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 24 • FORWARD 1 • Q2/5

QUESTION ID

S24-F1-Q2

TYPE

Multiple choice

CORRECT ANSWER

B. Not consistent: such results happened in 3% of the sets.

GRADING NOTE 3% < 5%.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S24-F1-Q2**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 24 • FORWARD 1 • Q2/5

True or false?

Under the 5% rule, the smaller the proportion of simulated results as extreme as the data, the stronger the evidence against the model.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F1-Q3

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 24 • FORWARD 1 • Q3/5

825

QUESTION ID

S24-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Rarer under the model means stronger evidence.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F1-Q3

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 24 • FORWARD 1 • Q3/5

A coin claimed to be fair gave 7 heads in 10 flips. In the table, what proportion of sets are at least as far from 5 heads in either direction (7 or more, or 3 or fewer)? Give a decimal.

Heads in 10 flips	0	1	2	3	4	5	6	7	8	9	10
Number of simulated trials	0	1	4	12	20	25	20	12	4	2	0

Proportion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F1-Q4

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 24 • FORWARD 1 • Q4/5

826

QUESTION ID

S24-F1-Q4

TYPE

Short answer

CORRECT ANSWER

0.35

GRADING NOTE $(12 + 4 + 2) + (12 + 4 + 1) = 35$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S24-F1-Q4**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 24 • FORWARD 1 • Q4/5

True or false?

If a model is judged not consistent under the 5% rule, there is no chance that the model is true.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F1-Q5

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 24 • FORWARD 1 • Q5/5

827

QUESTION ID

S24-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Rare results can still happen.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F1-Q5

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 24 • FORWARD 1 • Q5/5

SET 24 • HSS-IC.A.2

FORWARD 2

Judge a treatment difference by re-randomization • High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5

ANSWER KEY Reduced copy of the question page, with its answer

(Hypothetical simulation results.) In 100 re-randomizations of an experiment's results, a difference in means of 6 or more appeared 3 times. What proportion is that? Give a decimal.

Proportion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F2-Q1

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 24 • FORWARD 2 • Q1/5

829

QUESTION ID

S24-F2-Q1

TYPE

Short answer

CORRECT ANSWER

0.03

GRADING NOTE 3/100.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S24-F2-Q1**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 24 • FORWARD 2 • Q1/5

(Hypothetical simulation results.) In a randomized experiment, the treatment group's mean minus the control group's mean was 6. In 100 re-randomizations of the results, differences (treatment minus control) of 6 or more appeared 3 times. Decision rule: the model is the "no effect" model (the treatment does not change the results). Count the simulated results that are differences of 6 or more (a one-sided check). If they make up fewer than 5% of the simulated results, the "no effect" model is not consistent with the data and the difference is significant; otherwise it is consistent, and the difference is not significant. Which decision is correct?

A. The difference is significant: every difference is significant.

B. The difference is not significant: it appeared 3 times.

C. The difference is significant: chance alone gave it 3% of the time.

D. The difference is not significant: re-randomizing changes the data.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F2-Q2

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 24 • FORWARD 2 • Q2/5

830

QUESTION ID

S24-F2-Q2

TYPE

Multiple choice

CORRECT ANSWER

C. The difference is significant: chance alone gave it 3% of the time.

GRADING NOTE 3% < 5%.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S24-F2-Q2**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 24 • FORWARD 2 • Q2/5

True or false?

A re-randomization simulation assumes that the treatment has no effect.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F2-Q3

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 24 • FORWARD 2 • Q3/5

831

QUESTION ID

S24-F2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE It reshuffles the responses as if labels did not matter.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F2-Q3

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 24 • FORWARD 2 • Q3/5

In an experiment, the treatment group's mean was 24 and the control group's mean was 19. What is the difference (treatment minus control)?

Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F2-Q4

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 24 • FORWARD 2 • Q4/5

832

QUESTION ID

S24-F2-Q4

TYPE

Short answer

CORRECT ANSWER

5

GRADING NOTE 24 − 19.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S24-F2-Q4**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 24 • FORWARD 2 • Q4/5

True or false?

If the observed difference appeared in 40% of re-randomizations, the treatment effect is significant under the 5% rule.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F2-Q5

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 24 • FORWARD 2 • Q5/5

833

QUESTION ID
S24-F2-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE 40% ≥ 5%: not significant.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F2-Q5

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 24 • FORWARD 2 • Q5/5

SET 25 • HSS-IC.B.3 • MAKING INFERENCES AND JUSTIFYING CONCLUSIONS

Sample surveys, experiments and observational studies

MAIN	Probability & Statistics main question	**HSS-IC.B.3**
BACKWARD 1	Identify the population, sample, parameter and statistic	**HSS-IC.A.1**
BACKWARD 2	Separate association from causation	**HSS-ID.C.9**
FORWARD 1	Identify the parts of an experiment	**HSS-IC.B.3**
FORWARD 2	Choose a study design for a question	**HSS-IC.B.3**

SET 25 • HSS-IC.B.3

MAIN

Probability & Statistics main question • High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3

ANSWER KEY Reduced copy of the question page, with its answer

A researcher randomly assigns 60 volunteers to a new sleep routine or their usual routine and compares their test scores. What kind of study is this?

A. census

B. sample survey

C. experiment

D. observational study

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-M-Q1

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 25 • MAIN • Q1/5

836

QUESTION ID

S25-M-Q1

TYPE

Multiple choice

CORRECT ANSWER

C. experiment

GRADING NOTE Treatments are imposed and assigned.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-M-Q1

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 25 • MAIN • Q1/5

A polling company asks a random sample of 1,000 adults which candidate they prefer. What kind of study is this?

A. experiment

B. simulation

C. sample survey

D. observational study comparing groups

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-M-Q2

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 25 • MAIN • Q2/5

837

QUESTION ID

S25-M-Q2

TYPE

Multiple choice

CORRECT ANSWER

C. sample survey

GRADING NOTE It estimates a population characteristic by asking a sample.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S25-M-Q2**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 25 • MAIN • Q2/5

Researchers compare lung health of people who already smoke with people who do not. What kind of study is this?

A. experiment

B. census

C. observational study

D. sample survey of opinions

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-M-Q3

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 25 • MAIN • Q3/5

838

QUESTION ID

S25-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. observational study

GRADING NOTE Groups form on their own; nothing is imposed.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-M-Q3

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 25 • MAIN • Q3/5

Which kind of study is designed mainly to estimate a characteristic of a population?

A. an observational study of two groups

B. a sample survey

C. a simulation of a model

D. an experiment

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-M-Q4

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 25 • MAIN • Q4/5

839

QUESTION ID

S25-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. a sample survey

GRADING NOTE Surveys estimate; experiments test treatments.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-M-Q4

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 25 • MAIN • Q4/5

Why can an experiment support a cause-and-effect conclusion when an observational study cannot?

A. The researcher imposes treatments, so other variables can be controlled.

B. Observational studies never collect any numerical data.

C. Experiments do not need any comparison or control group.

D. Experiments always use more subjects than observational studies.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-M-Q5

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 25 • MAIN • Q5/5

840

QUESTION ID

S25-M-Q5

TYPE

Multiple choice

CORRECT ANSWER

A. The researcher imposes treatments, so other variables can be controlled.

GRADING NOTE Imposed, randomly assigned treatments control lurking variables. The others are false claims about data, sizes or control groups.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S25-M-Q5**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 25 • MAIN • Q5/5

SET 25 • HSS-IC.B.3

BACKWARD 1

Identify the population, sample, parameter and statistic • High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1

300 randomly chosen city residents are asked how often they ride the bus. What is the population?

A. the 300 residents

B. all city residents

C. the city's buses

D. bus riders only

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B1-Q1

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 25 • BACKWARD 1 • Q1/5

842

QUESTION ID

S25-B1-Q1

TYPE

Multiple choice

CORRECT ANSWER

B. all city residents

GRADING NOTE The whole group.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B1-Q1

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 25 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

300 randomly chosen city residents are asked how often they ride the bus. What is the sample size?

Size: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B1-Q2

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 25 • BACKWARD 1 • Q2/5

843

QUESTION ID
S25-B1-Q2

TYPE
Short answer

CORRECT ANSWER
300

GRADING NOTE Count of people asked.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S25-B1-Q2**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 25 • BACKWARD 1 • Q2/5

True or false?

A statistic describes a sample.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B1-Q3

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 25 • BACKWARD 1 • Q3/5

844

QUESTION ID

S25-B1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B1-Q3

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 25 • BACKWARD 1 • Q3/5

True or false?

A census collects data from only part of the population.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B1-Q4

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 25 • BACKWARD 1 • Q4/5

845

QUESTION ID

S25-B1-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE A census uses everyone.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B1-Q4

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 25 • BACKWARD 1 • Q4/5

Which is a parameter?

A. the mean height of 50 sampled students

B. the number 50

C. the proportion in the sample

D. the mean height of all students in a state

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B1-Q5

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 25 • BACKWARD 1 • Q5/5

846

QUESTION ID

S25-B1-Q5

TYPE

Multiple choice

CORRECT ANSWER

D. the mean height of all students in a state

GRADING NOTE Describes the population.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B1-Q5

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 25 • BACKWARD 1 • Q5/5

SET 25 • HSS-IC.B.3

BACKWARD 2

Separate association from causation • High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9

True or false?

An association seen without assigning treatments may be caused by a lurking variable.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B2-Q1

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9 • Set 25 • BACKWARD 2 • Q1/5

848

QUESTION ID

S25-B2-Q1

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Observational data.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B2-Q1

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9 • Set 25 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Towns with more ice-cream shops have more drownings. What is a likely lurking variable?

A. hot weather

B. the number of shops

C. the price of ice cream

D. the town's name

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B2-Q2

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9 • Set 25 • BACKWARD 2 • Q2/5

849

QUESTION ID

S25-B2-Q2

TYPE

Multiple choice

CORRECT ANSWER

A. hot weather

GRADING NOTE Affects both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B2-Q2

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9 • Set 25 • BACKWARD 2 • Q2/5

True or false?

A correlation always implies causation when the study is large.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B2-Q3

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9 • Set 25 • BACKWARD 2 • Q3/5

850

QUESTION ID

S25-B2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Size does not change it.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B2-Q3

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9 • Set 25 • BACKWARD 2 • Q3/5

Which conclusion requires an experiment?

A. The mean recovery time was 6 days.

B. Recovery time and vitamin use are associated.

C. Half of the patients took the vitamin.

D. A vitamin causes faster recovery.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B2-Q4

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9 • Set 25 • BACKWARD 2 • Q4/5

851

QUESTION ID

S25-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. A vitamin causes faster recovery.

GRADING NOTE A causal claim.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B2-Q4

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9 • Set 25 • BACKWARD 2 • Q4/5

True or false?

Random assignment helps rule out lurking variables.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B2-Q5

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9 • Set 25 • BACKWARD 2 • Q5/5

852

QUESTION ID

S25-B2-Q5

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE It balances them.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B2-Q5

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9 • Set 25 • BACKWARD 2 • Q5/5

SET 25 • HSS-IC.B.3

FORWARD 1

Identify the parts of an experiment • High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3

ANSWER KEY Reduced copy of the question page, with its answer

An experiment compares two fertilizers on 40 plots and measures crop yield. What is the response variable?

A. the 40 plots

B. the two fertilizers

C. the farmer

D. crop yield

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F1-Q1

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 25 • FORWARD 1 • Q1/5

854

QUESTION ID

S25-F1-Q1

TYPE

Multiple choice

CORRECT ANSWER

D. crop yield

GRADING NOTE What is measured.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F1-Q1

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 25 • FORWARD 1 • Q1/5

An experiment tests 3 doses of a medicine and a placebo. How many treatments are there?

Treatments: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F1-Q2

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 25 • FORWARD 1 • Q2/5

855

QUESTION ID

S25-F1-Q2

TYPE

Short answer

CORRECT ANSWER

4

GRADING NOTE 3 + 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S25-F1-Q2**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 25 • FORWARD 1 • Q2/5

True or false?

A control group receives the standard treatment or a placebo so the new treatment can be compared.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F1-Q3

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 25 • FORWARD 1 • Q3/5

856

QUESTION ID

S25-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Purpose of control.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F1-Q3

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 25 • FORWARD 1 • Q3/5

Why do experiments on medicines often give some subjects a placebo?

A. to increase the number of different treatments

B. to make the experiment cheaper to run

C. to account for the effect of believing one is being treated

D. to make sure the results come out positive

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F1-Q4

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 25 • FORWARD 1 • Q4/5

857

QUESTION ID

S25-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. to account for the effect of believing one is being treated

GRADING NOTE Placebo effect. The others are false: a placebo does not lower cost, add treatments, or fix the outcome.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F1-Q4

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 25 • FORWARD 1 • Q4/5

True or false?

In a double-blind experiment, only the subjects do not know which treatment they receive.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F1-Q5

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 25 • FORWARD 1 • Q5/5

858

QUESTION ID

S25-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The people who evaluate them do not know either.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F1-Q5

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 25 • FORWARD 1 • Q5/5

SET 25 • HSS-IC.B.3

FORWARD 2

Choose a study design for a question • High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3

Question: does a new math app raise test scores? Which design fits best?

A. a randomized experiment

B. a census of all students

C. a sample survey

D. an observational study of app users

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F2-Q1

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 25 • FORWARD 2 • Q1/5

860

QUESTION ID

S25-F2-Q1

TYPE

Multiple choice

CORRECT ANSWER

A. a randomized experiment

GRADING NOTE Cause needs an experiment.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F2-Q1

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 25 • FORWARD 2 • Q1/5

Question: what percent of a town's residents own a bicycle? Which design fits best?

A. a randomized experiment

B. a sample survey

C. a simulation

D. an observational study of cyclists

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F2-Q2

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 25 • FORWARD 2 • Q2/5

861

QUESTION ID

S25-F2-Q2

TYPE

Multiple choice

CORRECT ANSWER

B. a sample survey

GRADING NOTE Estimate a population percent.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S25-F2-Q2**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 25 • FORWARD 2 • Q2/5

Question: do people who live near highways have more asthma? Assigning where people live is not possible. Which design fits best?

A. an observational study

B. a sample survey of opinions

C. a simulation

D. a randomized experiment

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F2-Q3

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 25 • FORWARD 2 • Q3/5

862

QUESTION ID

S25-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. an observational study

GRADING NOTE Treatment cannot be assigned.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S25-F2-Q3**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 25 • FORWARD 2 • Q3/5

True or false?

It is ethical to randomly assign people to smoke in an experiment on lung disease.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F2-Q4

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 25 • FORWARD 2 • Q4/5

863

QUESTION ID

S25-F2-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It would cause harm.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F2-Q4

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 25 • FORWARD 2 • Q4/5

True or false?

An observational study can be the best choice when assigning a treatment would be unethical.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F2-Q5

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 25 • FORWARD 2 • Q5/5

QUESTION ID

S25-F2-Q5

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Then observation is what is possible.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F2-Q5

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 25 • FORWARD 2 • Q5/5

How randomization relates to each kind of study

MAIN	Probability & Statistics main question	**HSS-IC.B.3**
BACKWARD 1	Classify a study	**HSS-IC.B.3**
BACKWARD 2	Select a simple random sample	**7.SP.A.1**
FORWARD 1	Decide the scope of conclusions	**HSS-IC.B.3**
FORWARD 2	Evaluate a report's conclusion from its design	**HSS-IC.B.6**

SET 26 • HSS-IC.B.3

MAIN

Probability & Statistics main question • High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3

ANSWER KEY Reduced copy of the question page, with its answer

A class has 40 students numbered 01 to 40. Read line 1 of the table as two-digit numbers, skipping 00, numbers above 40 and repeats. Which three students are chosen first?

Line	Digits	Line	Digits
1	15881 39986 98797 30911	5	17784 86488 76634 40029
2	41707 53645 56815 18844	6	78753 47787 68045 71784
3	72940 55761 69871 68976	7	72671 47513 62701 07981
4	84697 88093 20819 67273	8	93319 08452 05997 28379

First: _____

Second: _____

Third: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-M-Q1

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 26 • MAIN • Q1/5

867

QUESTION ID

S26-M-Q1

TYPE

Short answer

CORRECT ANSWER

15; 13; 9

GRADING NOTE Form required: all three, in order. Line 1: 15, 88, 13, 99, 86, 98, 79, 73, 09 → 15, 13, 9.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-M-Q1

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 26 • MAIN • Q1/5

What is the main purpose of selecting a sample at random in a survey?

A. so the sample is as small as possible

B. so that every answer comes out the same

C. so a treatment can be shown to cause a change

D. so the results can be generalized to the population

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-M-Q2

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 26 • MAIN • Q2/5

868

QUESTION ID

S26-M-Q2

TYPE

Multiple choice

CORRECT ANSWER

D. so the results can be generalized to the population

GRADING NOTE Random sampling supports generalization. The others describe random assignment (cause), or claim it makes answers identical or the sample small.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S26-M-Q2**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 26 • MAIN • Q2/5

What is the main purpose of assigning subjects to treatments at random in an experiment?

A. to guarantee that the treatment will work

B. to help balance other variables between the groups

C. to avoid needing any control or comparison group

D. to let the results describe everyone in the country

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-M-Q3

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 26 • MAIN • Q3/5

869

QUESTION ID

S26-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. to help balance other variables between the groups

GRADING NOTE Random assignment tends to balance other variables, so a difference larger than chance variation can be credited to the treatment; it does not guarantee identical groups. The others describe random sampling (generalization) or claim it guarantees success or replaces a control group.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S26-M-Q3**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 26 • MAIN • Q3/5

A study randomly assigned 80 volunteers to two diets and found a significant difference in weight loss. The volunteers were not a random sample. What can be concluded?

A. The diet causes the same difference for everyone in the country.

B. The diet caused the difference for volunteers like these.

C. Nothing, because the subjects were volunteers.

D. Only that the two diets are associated with weight.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-M-Q4

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 26 • MAIN • Q4/5

870

QUESTION ID

S26-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. The diet caused the difference for volunteers like these.

GRADING NOTE Random assignment gives cause; no random sample, so no broad generalization. The others generalize to the country, ignore the random assignment, or claim only association.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-M-Q4

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 26 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Plots in a field are numbered 01 to 60. Read line 3 of the table as two-digit numbers, skipping 00, numbers above 60 and repeats. Which three plots are assigned fertilizer A first?

Line	Digits	Line	Digits
1	15881 39986 98797 30911	5	17784 86488 76634 40029
2	41707 53645 56815 18844	6	78753 47787 68045 71784
3	72940 55761 69871 68976	7	72671 47513 62701 07981
4	84697 88093 20819 67273	8	93319 08452 05997 28379

First: _____

Second: _____

Third: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-M-Q5

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 26 • MAIN • Q5/5

871

QUESTION ID

S26-M-Q5

TYPE

Short answer

CORRECT ANSWER

5; 57; 16

GRADING NOTE Form required: all three, in order. Line 3: 72, 94, 05, 57, 61, 69, 87, 16 → 5, 57, 16.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-M-Q5

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 26 • MAIN • Q5/5

SET 26 • HSS-IC.B.3

BACKWARD 1

Classify a study • High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3

ANSWER KEY Reduced copy of the question page, with its answer

A teacher asks a random sample of 50 students how many hours they sleep. What kind of study is this?

A. observational study comparing treatments

B. sample survey

C. census

D. experiment

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B1-Q1

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 26 • BACKWARD 1 • Q1/5

873

QUESTION ID
S26-B1-Q1

TYPE
Multiple choice

CORRECT ANSWER
B. sample survey

GRADING NOTE Asking a sample.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B1-Q1

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 26 • BACKWARD 1 • Q1/5

True or false?

In an experiment, the researcher decides which subjects get which treatment.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B1-Q2

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 26 • BACKWARD 1 • Q2/5

874

QUESTION ID

S26-B1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Imposed.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B1-Q2

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 26 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Doctors compare the heart health of people who already jog with people who do not. What kind of study is this?

A. sample survey

B. simulation

C. observational study

D. experiment

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B1-Q3

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 26 • BACKWARD 1 • Q3/5

875

QUESTION ID

S26-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. observational study

GRADING NOTE Nothing imposed.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B1-Q3

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 26 • BACKWARD 1 • Q3/5

True or false?

A sample survey assigns treatments to people.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B1-Q4

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 26 • BACKWARD 1 • Q4/5

876

QUESTION ID

S26-B1-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It asks or measures only.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B1-Q4

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 26 • BACKWARD 1 • Q4/5

Which study imposes a treatment?

A. giving half of the volunteers a new drug, chosen by lottery

B. asking a random sample of voters for their opinion

C. counting the cars that pass along a busy road

D. recording the diets that people already choose to eat

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B1-Q5

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 26 • BACKWARD 1 • Q5/5

877

QUESTION ID

S26-B1-Q5

TYPE

Multiple choice

CORRECT ANSWER

A. giving half of the volunteers a new drug, chosen by lottery

GRADING NOTE Assignment by chance.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S26-B1-Q5**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 26 • BACKWARD 1 • Q5/5

SET 26 • HSS-IC.B.3

BACKWARD 2

Select a simple random sample • Grade 7 • Statistics & Probability • 7.SP.A.1

True or false?

In a simple random sample, every group of the same size has the same chance of being chosen.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B2-Q1

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 26 • BACKWARD 2 • Q1/5

879

QUESTION ID

S26-B2-Q1

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B2-Q1

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 26 • BACKWARD 2 • Q1/5

Which method gives a simple random sample of 5 students from a class of 30?

A. draw 5 names from a hat holding all 30 names

B. ask for 5 volunteers to raise their hands

C. choose the 5 tallest students in the class

D. choose the first 5 students to arrive

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B2-Q2

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 26 • BACKWARD 2 • Q2/5

880

QUESTION ID

S26-B2-Q2

TYPE

Multiple choice

CORRECT ANSWER

A. draw 5 names from a hat holding all 30 names

GRADING NOTE Chance selection.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B2-Q2

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 26 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Students are numbered 01 to 30. Reading the two-digit numbers 45 07 29 07 33 18, skipping numbers above 30 and repeats, which three students are chosen?

First: _____

Second: _____

Third: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B2-Q3

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 26 • BACKWARD 2 • Q3/5

881

QUESTION ID

S26-B2-Q3

TYPE

Short answer

CORRECT ANSWER

7; 29; 18

GRADING NOTE Form required: all three, in order.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B2-Q3

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 26 • BACKWARD 2 • Q3/5

True or false?

Choosing every student whose name starts with A is a random sample.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B2-Q4

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 26 • BACKWARD 2 • Q4/5

882

QUESTION ID

S26-B2-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Not chance-based.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B2-Q4

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 26 • BACKWARD 2 • Q4/5

Why are repeated numbers skipped when selecting a sample from a random-digit table?

A. skipping them saves time in the selection

B. repeated digits are always printing errors

C. repeated numbers are too large to use

D. a person cannot be chosen twice

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B2-Q5

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 26 • BACKWARD 2 • Q5/5

883

QUESTION ID

S26-B2-Q5

TYPE

Multiple choice

CORRECT ANSWER

D. a person cannot be chosen twice

GRADING NOTE Sampling without replacement. The others are false reasons: repeats are valid digits of the right size.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B2-Q5

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 26 • BACKWARD 2 • Q5/5

SET 26 • HSS-IC.B.3

FORWARD 1

Decide the scope of conclusions • High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3

ANSWER KEY Reduced copy of the question page, with its answer

A study used a random sample from a population but no random assignment. What can be concluded?

A. Nothing at all can be concluded from such a study.

B. Associations generalize to the population, but not cause.

C. Cause can be concluded, but only for the sample.

D. Cause can be concluded for the whole population.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F1-Q1

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 26 • FORWARD 1 • Q1/5

885

QUESTION ID
S26-F1-Q1

TYPE
Multiple choice

CORRECT ANSWER
B. Associations generalize to the population, but not cause.

GRADING NOTE Random sampling → generalize; no random assignment → no cause.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S26-F1-Q1**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 26 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A study used both a random sample from a population and random assignment to treatments. What can be concluded?

A. Cause, but only for the sample.

B. Association only.

C. Cause, generalized to the population.

D. Nothing.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F1-Q2

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 26 • FORWARD 1 • Q2/5

886

QUESTION ID
S26-F1-Q2

TYPE
Multiple choice

CORRECT ANSWER
C. Cause, generalized to the population.

GRADING NOTE Both kinds of randomization.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S26-F1-Q2**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 26 • FORWARD 1 • Q2/5

True or false?

Random assignment allows cause-and-effect conclusions.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F1-Q3

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 26 • FORWARD 1 • Q3/5

887

QUESTION ID
S26-F1-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Its purpose.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S26-F1-Q3**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 26 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

Random sampling allows cause-and-effect conclusions even without random assignment.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F1-Q4

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 26 • FORWARD 1 • Q4/5

888

QUESTION ID

S26-F1-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It allows generalization only.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F1-Q4

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 26 • FORWARD 1 • Q4/5

A study used neither random sampling nor random assignment. What can be concluded?

A. Cause for the group studied.

B. Cause for the population.

C. Associations for the population.

D. Associations in the group studied only.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F1-Q5

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 26 • FORWARD 1 • Q5/5

889

QUESTION ID

S26-F1-Q5

TYPE

Multiple choice

CORRECT ANSWER

D. Associations in the group studied only.

GRADING NOTE No randomization of either kind.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F1-Q5

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 26 • FORWARD 1 • Q5/5

SET 26 • HSS-IC.B.3

FORWARD 2

Evaluate a report's conclusion from its design • High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6

A news report says, "People who drink green tea live longer, so drink tea to live longer." The data came from an observational study. What is the main flaw?

A. The sample was too large to be trusted.

B. Green tea intake cannot be measured at all.

C. Observational studies cannot find associations.

D. It claims cause from an observational study.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F2-Q1

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 26 • FORWARD 2 • Q1/5

891

QUESTION ID

S26-F2-Q1

TYPE

Multiple choice

CORRECT ANSWER

D. It claims cause from an observational study.

GRADING NOTE Cause needs random assignment.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F2-Q1

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 26 • FORWARD 2 • Q1/5

True or false?

A report based on a voluntary online poll can be generalized to all adults.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F2-Q2

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 26 • FORWARD 2 • Q2/5

892

QUESTION ID

S26-F2-Q2

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Volunteers are not a random sample.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F2-Q2

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 26 • FORWARD 2 • Q2/5

A randomized experiment on 200 college athletes found a new drink improved sprint times. To whom does the result apply most directly?

A. all adults

B. college athletes like those studied

C. all children

D. nobody

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F2-Q3

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 26 • FORWARD 2 • Q3/5

893

QUESTION ID
S26-F2-Q3

TYPE
Multiple choice

CORRECT ANSWER
B. college athletes like those studied

GRADING NOTE The subjects define the scope.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F2-Q3

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 26 • FORWARD 2 • Q3/5

True or false?

A report should state how its sample was chosen and whether treatments were randomly assigned.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F2-Q4

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 26 • FORWARD 2 • Q4/5

894

QUESTION ID

S26-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Needed to judge the conclusion.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F2-Q4

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 26 • FORWARD 2 • Q4/5

A survey of 50 gym members found 90% exercise daily, reported as "90% of Americans exercise daily." What is wrong?

A. Surveys can never be used to estimate percents.

B. Gym members are not representative of all Americans.

C. The sample of 50 people is far too large.

D. The percent of 90 was computed incorrectly.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F2-Q5

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 26 • FORWARD 2 • Q5/5

895

QUESTION ID

S26-F2-Q5

TYPE

Multiple choice

CORRECT ANSWER

B. Gym members are not representative of all Americans.

GRADING NOTE Biased sample.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S26-F2-Q5**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 26 • FORWARD 2 • Q5/5

Estimate a population proportion with a margin of error

MAIN	Probability & Statistics main question	**HSS-IC.B.4**
BACKWARD 1	Compute a sample proportion and name it	**HSS-IC.A.1**
BACKWARD 2	Read a standard deviation as typical variation	**HSS-ID.A.2**
FORWARD 1	See how sample size changes the margin of error	**HSS-IC.B.4**
FORWARD 2	Read margins of error in poll reports	**HSS-IC.B.6**

SET 27 • HSS-IC.B.4

MAIN

Probability & Statistics main question • High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4

ANSWER KEY Reduced copy of the question page, with its answer

In a random sample of 400 adults, 248 favor a new park. What is the sample proportion? Give a decimal.

Proportion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-M-Q1

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 27 • MAIN • Q1/5

898

QUESTION ID
S27-M-Q1

TYPE
Short answer

CORRECT ANSWER
0.62

GRADING NOTE 248/400.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S27-M-Q1**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 27 • MAIN • Q1/5

(Hypothetical simulation results.) 500 random samples of size 400 were simulated from a population with proportion 0.62. The simulated sample proportions had standard deviation 0.024. Use margin of error = 2 × (standard deviation of the simulated values). What is the margin of error?

Margin of error: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-M-Q2

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 27 • MAIN • Q2/5

899

QUESTION ID
S27-M-Q2

TYPE
Short answer

CORRECT ANSWER
0.048

GRADING NOTE 2 · 0.024.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S27-M-Q2**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 27 • MAIN • Q2/5

A sample proportion is 0.62 and the margin of error is 0.048. Give the interval of plausible values for the population proportion.

From: _____

To: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-M-Q3

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 27 • MAIN • Q3/5

900

QUESTION ID

S27-M-Q3

TYPE

Short answer

CORRECT ANSWER

0.572; 0.668

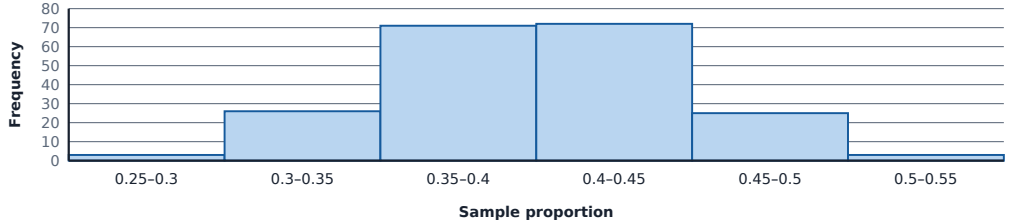
GRADING NOTE Form required: both. 0.62 ± 0.048.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S27-M-Q3**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 27 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

The histogram shows 200 simulated sample proportions. How many of them are between 0.35 and 0.45 (bins 0.35-0.40 and 0.40-0.45)?



Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-M-Q4

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 27 • MAIN • Q4/5

901

QUESTION ID
S27-M-Q4

TYPE
Short answer

CORRECT ANSWER
143

GRADING NOTE 71 + 72.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-M-Q4

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 27 • MAIN • Q4/5

A poll of a random sample reports that 54% of voters support a measure, with a margin of error of 3 percentage points. Which conclusion is best supported?

A. Exactly 54% of all voters support the measure.

B. Support could plausibly be anywhere from 0% to 100%.

C. The poll shows that support is below 50%.

D. A majority plausibly supports it: 51% to 57% are all above 50%.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-M-Q5

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 27 • MAIN • Q5/5

902

QUESTION ID
S27-M-Q5

TYPE
Multiple choice

CORRECT ANSWER
D. A majority plausibly supports it: 51% to 57% are all above 50%.

GRADING NOTE 54 ± 3 gives 51% to 57%.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-M-Q5

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 27 • MAIN • Q5/5

SET 27 • HSS-IC.B.4

BACKWARD 1

Compute a sample proportion and name it • High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1

In a random sample of 250 students, 90 walk to school. What is the sample proportion? Give a decimal.

Proportion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B1-Q1

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 27 • BACKWARD 1 • Q1/5

904

QUESTION ID

S27-B1-Q1

TYPE

Short answer

CORRECT ANSWER

0.36

GRADING NOTE 90/250.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S27-B1-Q1**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 27 • BACKWARD 1 • Q1/5

True or false?

A sample proportion is a statistic.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B1-Q2

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 27 • BACKWARD 1 • Q2/5

905

QUESTION ID

S27-B1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE It describes the sample.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S27-B1-Q2**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 27 • BACKWARD 1 • Q2/5

The proportion of all voters in a state who favor a law is a

A. parameter

B. statistic

C. margin of error

D. sample

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B1-Q3

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 27 • BACKWARD 1 • Q3/5

906

QUESTION ID

S27-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. parameter

GRADING NOTE Describes the population.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S27-B1-Q3**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 27 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

In a random sample of 80 phones, 6 were defective. What is the sample proportion? Give a decimal.

Proportion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B1-Q4

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 27 • BACKWARD 1 • Q4/5

907

QUESTION ID
S27-B1-Q4

TYPE
Short answer

CORRECT ANSWER
0.075

GRADING NOTE 6/80.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S27-B1-Q4**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 27 • BACKWARD 1 • Q4/5

True or false?

Different random samples always give the same sample proportion.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B1-Q5

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 27 • BACKWARD 1 • Q5/5

908

QUESTION ID

S27-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE They vary.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B1-Q5

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 27 • BACKWARD 1 • Q5/5

SET 27 • HSS-IC.B.4

BACKWARD 2

Read a standard deviation as typical variation • High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2

True or false?

A standard deviation measures how far values typically are from their mean.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B2-Q1

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 27 • BACKWARD 2 • Q1/5

910

QUESTION ID

S27-B2-Q1

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Meaning.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B2-Q1

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 27 • BACKWARD 2 • Q1/5

(Hypothetical simulation results.) Simulated proportions from samples of size 100 have SD 0.05; from samples of size 400, SD 0.025. Which vary less?

A. cannot tell

B. the proportions from samples of size 100

C. the proportions from samples of size 400

D. they vary the same

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B2-Q2

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 27 • BACKWARD 2 • Q2/5

911

QUESTION ID

S27-B2-Q2

TYPE

Multiple choice

CORRECT ANSWER

C. the proportions from samples of size 400

GRADING NOTE Smaller SD.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S27-B2-Q2**

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 27 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Compute 2 × 0.031.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B2-Q3

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 27 • BACKWARD 2 • Q3/5

912

QUESTION ID

S27-B2-Q3

TYPE

Short answer

CORRECT ANSWER

0.062

GRADING NOTE Doubling.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B2-Q3

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 27 • BACKWARD 2 • Q3/5

True or false?

A larger standard deviation means the values are more tightly clustered.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B2-Q4

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 27 • BACKWARD 2 • Q4/5

913

QUESTION ID

S27-B2-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE More spread.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B2-Q4

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 27 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

About what percent of values in a roughly normal distribution lie within 2 SDs of the mean?

A. 68%

B. 95%

C. 50%

D. 99.7%

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B2-Q5

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 27 • BACKWARD 2 • Q5/5

914

QUESTION ID

S27-B2-Q5

TYPE

Multiple choice

CORRECT ANSWER

B. 95%

GRADING NOTE The rule behind the convention.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S27-B2-Q5**

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 27 • BACKWARD 2 • Q5/5

SET 27 • HSS-IC.B.4

FORWARD 1

See how sample size changes the margin of error • High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4

For a sample proportion near 0.5, the margin of error is about $2 \cdot \sqrt{(0.5 \cdot 0.5/n)}$. Find it for n = 400 and n = 100.

n = 400: _____

n = 100: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F1-Q1

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 27 • FORWARD 1 • Q1/5

916

QUESTION ID

S27-F1-Q1

TYPE

Short answer

CORRECT ANSWER

0.05; 0.1

GRADING NOTE Form required: both. 2·0.025 and 2·0.05.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S27-F1-Q1**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 27 • FORWARD 1 • Q1/5

True or false?

Making the sample 4 times as large cuts the margin of error about in half.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F1-Q2

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 27 • FORWARD 1 • Q2/5

917

QUESTION ID

S27-F1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $\sqrt{4} = 2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F1-Q2

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 27 • FORWARD 1 • Q2/5

A pollster wants a smaller margin of error. What should change?

A. take a larger random sample

B. ask only volunteers

C. round the results

D. take a smaller random sample

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F1-Q3

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 27 • FORWARD 1 • Q3/5

QUESTION ID

S27-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. take a larger random sample

GRADING NOTE Larger n.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S27-F1-Q3**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 27 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

(Hypothetical simulation results.) Simulated proportions from samples of size 900 have SD 0.016. Use margin of error = 2 × (standard deviation of the simulated values). What is the margin of error?

Margin of error: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F1-Q4

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 27 • FORWARD 1 • Q4/5

919

QUESTION ID
S27-F1-Q4

TYPE
Short answer

CORRECT ANSWER
0.032

GRADING NOTE 2 · 0.016.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S27-F1-Q4**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 27 • FORWARD 1 • Q4/5

True or false?

A larger sample removes the bias of a poorly worded question.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F1-Q5

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 27 • FORWARD 1 • Q5/5

920

QUESTION ID
S27-F1-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE Size does not fix bias.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F1-Q5

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 27 • FORWARD 1 • Q5/5

SET 27 • HSS-IC.B.4

FORWARD 2

Read margins of error in poll reports • High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6

Candidate A has 48% and candidate B has 45%, each with a margin of error of 4 percentage points. What is the best conclusion?

A. Both candidates have under 50%, so both lose.

B. No clear leader: the two intervals overlap a lot.

C. B is certainly behind A by at least 3 points.

D. A is certainly ahead of B by exactly 3 points.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F2-Q1

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 27 • FORWARD 2 • Q1/5

922

QUESTION ID

S27-F2-Q1

TYPE

Multiple choice

CORRECT ANSWER

B. No clear leader: the two intervals overlap a lot.

GRADING NOTE 44%−52% and 41%−49% overlap a lot, so a cautious reading is that the poll shows no clear leader. The others claim certainty the poll cannot give, or confuse sample percents with winning.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F2-Q1

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 27 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A report gives 62% with a margin of error of 5 percentage points. Give the interval in percents.

From: _____

To: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F2-Q2

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 27 • FORWARD 2 • Q2/5

923

QUESTION ID

S27-F2-Q2

TYPE

Short answer

CORRECT ANSWER

57%; 67%

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S27-F2-Q2**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 27 • FORWARD 2 • Q2/5

True or false?

A margin of error describes only random sampling variation, not bias from how the sample was chosen.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F2-Q3

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 27 • FORWARD 2 • Q3/5

924

QUESTION ID

S27-F2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE It assumes a random sample.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S27-F2-Q3**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 27 • FORWARD 2 • Q3/5

A website poll of 10,000 visitors reports 70% ± 1%. Why is the margin of error misleading?

A. The sample is too small for a margin of error.

B. A margin of error must be at least 5%.

C. Percents cannot have a margin of error.

D. The visitors chose themselves, so the sample is not random.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F2-Q4

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 27 • FORWARD 2 • Q4/5

925

QUESTION ID

S27-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. The visitors chose themselves, so the sample is not random.

GRADING NOTE Self-selected sample.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S27-F2-Q4**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 27 • FORWARD 2 • Q4/5

True or false?

If two candidates' intervals overlap, the poll proves that they are tied.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F2-Q5

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 27 • FORWARD 2 • Q5/5

926

QUESTION ID

S27-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It shows the difference is not clear, not that they are tied.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F2-Q5

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 27 • FORWARD 2 • Q5/5

Estimate a population mean with a margin of error

MAIN	Probability & Statistics main question	**HSS-IC.B.4**
BACKWARD 1	Use a margin of error	**HSS-IC.B.4**
BACKWARD 2	Compute a mean	**6.SP.B.5.c**
FORWARD 1	Compare variability of sample means	**HSS-IC.B.4**
FORWARD 2	Compare two treatments' means	**HSS-IC.B.5**

SET 28 • HSS-IC.B.4

MAIN

Probability & Statistics main question • High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4

A random sample of 8 students reported these hours of sleep: 7, 6.5, 8, 7.5, 6, 9, 7, 8. What is the sample mean?

Mean: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-M-Q1

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 28 • MAIN • Q1/5

929

QUESTION ID

S28-M-Q1

TYPE

Short answer

CORRECT ANSWER

7.375

GRADING NOTE 59/8.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S28-M-Q1**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 28 • MAIN • Q1/5

(Hypothetical simulation results.) Simulated random samples of 8 students gave sample means with standard deviation 0.15 hour. Use margin of error = 2 × (standard deviation of the simulated values). What is the margin of error?

Margin of error (hours): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-M-Q2

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 28 • MAIN • Q2/5

930

QUESTION ID
S28-M-Q2

TYPE
Short answer

CORRECT ANSWER
0.3

GRADING NOTE 2 · 0.15.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S28-M-Q2**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 28 • MAIN • Q2/5

A sample mean is 7.4 hours with a margin of error of 0.3 hour. Give the interval of plausible values for the population mean.

From: _____

To: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-M-Q3

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 28 • MAIN • Q3/5

931

QUESTION ID

S28-M-Q3

TYPE

Short answer

CORRECT ANSWER

7.1; 7.7

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S28-M-Q3**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 28 • MAIN • Q3/5

A study reports a mean of 7.4 hours of sleep with a margin of error of 0.3 hour, from a random sample. Which interpretation is correct?

A. Population means from 7.1 to 7.7 hours are plausible.

B. The population mean is exactly 7.4 hours.

C. Half of all students sleep 7.4 hours.

D. Every student sleeps between 7.1 and 7.7 hours.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-M-Q4

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 28 • MAIN • Q4/5

932

QUESTION ID

S28-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. Population means from 7.1 to 7.7 hours are plausible.

GRADING NOTE The interval is for the mean, not for individuals.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S28-M-Q4**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 28 • MAIN • Q4/5

(Hypothetical simulation results.) In 1,000 simulated samples, the middle 95% of sample means ran from 41.2 to 44.8. Give the center of this interval and the margin of error (half its width).

Center: _____

Margin of error: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-M-Q5

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 28 • MAIN • Q5/5

933

QUESTION ID

S28-M-Q5

TYPE

Short answer

CORRECT ANSWER

43; 1.8

GRADING NOTE Form required: both. (41.2 + 44.8)/2 and (44.8 − 41.2)/2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S28-M-Q5**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 28 • MAIN • Q5/5

SET 28 • HSS-IC.B.4

BACKWARD 1

Use a margin of error • High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4

ANSWER KEY Reduced copy of the question page, with its answer

An estimate is 0.40 with a margin of error of 0.06. Give the interval.

From: _____

To: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B1-Q1

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 28 • BACKWARD 1 • Q1/5

935

QUESTION ID

S28-B1-Q1

TYPE

Short answer

CORRECT ANSWER

0.34; 0.46

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S28-B1-Q1**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 28 • BACKWARD 1 • Q1/5

True or false?

Margin of error = 2 × 0.05 = 0.1 when the simulated values have SD 0.05.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B1-Q2

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 28 • BACKWARD 1 • Q2/5

936

QUESTION ID

S28-B1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Convention.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B1-Q2

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 28 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Which estimate has the widest interval?

A. 20 ± 1

B. 20 ± 4

C. 20 ± 2

D. 20 ± 0.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B1-Q3

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 28 • BACKWARD 1 • Q3/5

937

QUESTION ID

S28-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 20 ± 4

GRADING NOTE Largest margin.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S28-B1-Q3**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 28 • BACKWARD 1 • Q3/5

An interval runs from 12 to 18. What is the margin of error?

Margin of error: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B1-Q4

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 28 • BACKWARD 1 • Q4/5

938

QUESTION ID

S28-B1-Q4

TYPE

Short answer

CORRECT ANSWER

3

GRADING NOTE Half of 6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S28-B1-Q4**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 28 • BACKWARD 1 • Q4/5

True or false?

An interval from 0.30 to 0.40 has center 0.30.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B1-Q5

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 28 • BACKWARD 1 • Q5/5

939

QUESTION ID

S28-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Center 0.35.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B1-Q5

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 28 • BACKWARD 1 • Q5/5

SET 28 • HSS-IC.B.4

BACKWARD 2

Compute a mean • Grade 6 • Statistics & Probability • 6.SP.B.5.c

ANSWER KEY Reduced copy of the question page, with its answer

Find the mean of 4, 6, 8, 10.

Mean: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B2-Q1

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 28 • BACKWARD 2 • Q1/5

941

QUESTION ID

S28-B2-Q1

TYPE

Short answer

CORRECT ANSWER

7

GRADING NOTE 28/4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B2-Q1

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 28 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the mean of 6.5, 7, 8.5, rounded to three decimal places.

Mean: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B2-Q2

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 28 • BACKWARD 2 • Q2/5

942

QUESTION ID

S28-B2-Q2

TYPE

Short answer

CORRECT ANSWER

7.333

GRADING NOTE 22/3 = 7.333..., to three decimal places.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B2-Q2

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 28 • BACKWARD 2 • Q2/5

True or false?

The mean of 2, 4, 9 is 5.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B2-Q3

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 28 • BACKWARD 2 • Q3/5

943

QUESTION ID

S28-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE 15/3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B2-Q3

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 28 • BACKWARD 2 • Q3/5

What is the mean of 10, 12, 14, 20?

A. 14

B. 56

C. 13

D. 12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B2-Q4

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 28 • BACKWARD 2 • Q4/5

944

QUESTION ID

S28-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 14

GRADING NOTE 56/4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B2-Q4

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 28 • BACKWARD 2 • Q4/5

True or false?

The mean of a data set must be one of its values.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B2-Q5

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 28 • BACKWARD 2 • Q5/5

945

QUESTION ID

S28-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Not necessarily.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B2-Q5

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 28 • BACKWARD 2 • Q5/5

SET 28 • HSS-IC.B.4

FORWARD 1

Compare variability of sample means • High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4

Sample means from samples of size 9 have SD 2; from samples of size 36, SD 1. Which statement is correct?

A. Means of larger samples vary more, so the margin is larger.

B. Both sample sizes give the same margin of error.

C. Means of larger samples vary less, so the margin of error is smaller.

D. Sample size has no effect at all on sample means.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F1-Q1

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 28 • FORWARD 1 • Q1/5

947

QUESTION ID

S28-F1-Q1

TYPE

Multiple choice

CORRECT ANSWER

C. Means of larger samples vary less, so the margin of error is smaller.

GRADING NOTE Quadrupling n halves the SD.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F1-Q1

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 28 • FORWARD 1 • Q1/5

(Hypothetical simulation results.) Simulated sample means from samples of size 36 have SD 1. Use margin of error = 2 × (standard deviation of the simulated values). What is the margin of error?

Margin of error: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F1-Q2

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 28 • FORWARD 1 • Q2/5

948

QUESTION ID

S28-F1-Q2

TYPE

Short answer

CORRECT ANSWER

2

GRADING NOTE 2 · 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S28-F1-Q2**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 28 • FORWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

Individual values vary more than means of samples of those values.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F1-Q3

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 28 • FORWARD 1 • Q3/5

949

QUESTION ID
S28-F1-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Averaging reduces spread.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F1-Q3

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 28 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

An estimate of 52 has a margin of error of 2. Is 55 a plausible population mean? Enter yes or no.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F1-Q4

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 28 • FORWARD 1 • Q4/5

950

QUESTION ID

S28-F1-Q4

TYPE

Short answer

CORRECT ANSWER

no

GRADING NOTE 50 to 54.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S28-F1-Q4**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 28 • FORWARD 1 • Q4/5

True or false?

A margin of error corrects the bias of a convenience sample.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F1-Q5

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 28 • FORWARD 1 • Q5/5

951

QUESTION ID

S28-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE A margin of error describes only random sampling variation; it does not fix bias from how the sample was chosen.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S28-F1-Q5**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 28 • FORWARD 1 • Q5/5

SET 28 • HSS-IC.B.4

FORWARD 2

Compare two treatments' means • High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5

Plant food A: mean growth 13 cm; plant food B: mean growth 10 cm. What is the difference (A minus B)?

Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F2-Q1

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 28 • FORWARD 2 • Q1/5

953

QUESTION ID

S28-F2-Q1

TYPE

Short answer

CORRECT ANSWER

3

GRADING NOTE 13 − 10.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S28-F2-Q1**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 28 • FORWARD 2 • Q1/5

An experiment's difference in means could be due to the treatments or to chance. What tool checks chance?

A. a re-randomization simulation

B. a census

C. a larger margin of error

D. a scatter plot

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F2-Q2

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 28 • FORWARD 2 • Q2/5

954

QUESTION ID

S28-F2-Q2

TYPE

Multiple choice

CORRECT ANSWER

A. a re-randomization simulation

GRADING NOTE Re-randomization models "no effect".

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S28-F2-Q2**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 28 • FORWARD 2 • Q2/5

True or false?

A difference in means from a randomized experiment is always caused by the treatment.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F2-Q3

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 28 • FORWARD 2 • Q3/5

955

QUESTION ID

S28-F2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It may be chance.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F2-Q3

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 28 • FORWARD 2 • Q3/5

(Hypothetical simulation results.) In 200 re-randomizations, a difference of 3 or more occurred 4 times. What proportion? Give a decimal.

Proportion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F2-Q4

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 28 • FORWARD 2 • Q4/5

956

QUESTION ID

S28-F2-Q4

TYPE

Short answer

CORRECT ANSWER

0.02

GRADING NOTE 4/200.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S28-F2-Q4**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 28 • FORWARD 2 • Q4/5

True or false?

A difference that chance alone produces in only 2% of re-randomizations is significant under a 5% rule.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F2-Q5

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 28 • FORWARD 2 • Q5/5

957

QUESTION ID
S28-F2-Q5

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE 2% < 5%.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F2-Q5

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 28 • FORWARD 2 • Q5/5

Compare two treatments with a randomization simulation

MAIN	Probability & Statistics main question	**HSS-IC.B.5**
BACKWARD 1	Decide whether a model is consistent with data	**HSS-IC.A.2**
BACKWARD 2	Explain random assignment	**HSS-IC.B.3**
BACKWARD 3	Compute means of small groups	**6.SP.B.5.c**
FORWARD 1	Use differences in either direction	**HSS-IC.B.5**
FORWARD 2	Evaluate a report of an experiment	**HSS-IC.B.6**

SET 29 • HSS-IC.B.5

MAIN

Probability & Statistics main question • High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5

ANSWER KEY Reduced copy of the question page, with its answer

Ten plants were randomly assigned to plant food A or B. Find the mean growth for each group and the difference (A minus B).

Plant food	Growth of the 5 plants (cm)
A	14, 11, 13, 12, 15
B	10, 12, 9, 11, 8

Mean A: _____

Mean B: _____

Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-M-Q1

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 29 • MAIN • Q1/5

960

QUESTION ID

S29-M-Q1

TYPE

Short answer

CORRECT ANSWER

13; 10; 3

GRADING NOTE Form required: all three. 65/5, 50/5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S29-M-Q1**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 29 • MAIN • Q1/5

(Hypothetical simulation results.) In an experiment, 10 plants were randomly assigned to two plant foods (5 each), and the difference in mean growth was 3 cm. The 10 results were re-randomized into two groups of 5 a total of 200 times. A difference of 3 or more occurred 4 times. What proportion of the re-randomizations is that? Give a decimal.

Proportion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-M-Q2

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 29 • MAIN • Q2/5

961

QUESTION ID
S29-M-Q2

TYPE
Short answer

CORRECT ANSWER
0.02

GRADING NOTE 4/200.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-M-Q2

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 29 • MAIN • Q2/5

(Hypothetical simulation results.) Ten plants were randomly assigned to plant food A or B (5 each); mean growth A minus B was 3 cm. In 200 re-randomizations, a difference (A minus B) of 3 or more occurred 4 times. Decision rule: the model is the "no difference" model (plant food does not change growth). Count the simulated results that are differences of 3 or more (A minus B; a one-sided check). If they make up fewer than 5% of the simulated results, the model is not consistent with the data and the difference is significant; otherwise it is consistent and the difference is not significant. Which decision is correct?

A. The difference is significant: any difference is significant.

B. The difference is not significant: it happened 4 times by chance.

C. The difference is significant: chance gave it only 2% of the time.

D. The difference is not significant: the groups had only 5 plants.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-M-Q3

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 29 • MAIN • Q3/5

962

QUESTION ID

S29-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. The difference is significant: chance gave it only 2% of the time.

GRADING NOTE 2% < 5%.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S29-M-Q3**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 29 • MAIN • Q3/5

In an experiment, 10 plants were randomly assigned to plant food A or B (5 each). What does one re-randomization of the 10 growth results do?

A. Repeats the whole experiment with many more plants.

B. Swaps the means of the two original groups of plants.

C. Reshuffles the 10 results into groups of 5, ignoring food.

D. Grows 10 new plants, again using the two plant foods.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-M-Q4

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 29 • MAIN • Q4/5

963

QUESTION ID

S29-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. Reshuffles the 10 results into groups of 5, ignoring food.

GRADING NOTE It models the "no difference" assumption.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-M-Q4

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 29 • MAIN • Q4/5

(Hypothetical simulation results.) In another plant-food experiment with random assignment, mean growth A minus B was 1.2 cm. In 200 re-randomizations, differences (A minus B) of 1.2 or more occurred 62 times. Decision rule: the model is the "no difference" model (plant food does not change growth). Count the simulated results that are differences of 1.2 or more (A minus B; a one-sided check). If they make up fewer than 5% of the simulated results, the model is not consistent with the data and the difference is significant; otherwise it is consistent and the difference is not significant. Which decision is correct?

A. Not significant: re-randomizations can never be significant.

B. Significant: 62 is a large number of re-randomizations.

C. Not significant: chance gave such a difference 31% of the time.

D. Significant: the difference 1.2 is greater than 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-M-Q5

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 29 • MAIN • Q5/5

964

QUESTION ID

S29-M-Q5

TYPE

Multiple choice

CORRECT ANSWER

C. Not significant: chance gave such a difference 31% of the time.

GRADING NOTE 62/200 = 31% ≥ 5%.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S29-M-Q5**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 29 • MAIN • Q5/5

SET 29 • HSS-IC.B.5

BACKWARD 1

Decide whether a model is consistent with data • High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2

True or false?

A result that would happen rarely under a model is evidence against the model.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B1-Q1

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 29 • BACKWARD 1 • Q1/5

966

QUESTION ID

S29-B1-Q1

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Logic of the test.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B1-Q1

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 29 • BACKWARD 1 • Q1/5

(Hypothetical simulation results.) A result at least as extreme happened in 3 of 150 simulated trials. What proportion? Give a decimal.

Proportion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B1-Q2

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 29 • BACKWARD 1 • Q2/5

967

QUESTION ID

S29-B1-Q2

TYPE

Short answer

CORRECT ANSWER

0.02

GRADING NOTE 3/150.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S29-B1-Q2**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 29 • BACKWARD 1 • Q2/5

Under the 5% rule, a result as extreme as the data occurred in 12% of trials. The decision is

A. the model is proven

B. consistent with the model

C. not consistent with the model

D. no decision

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B1-Q3

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 29 • BACKWARD 1 • Q3/5

968

QUESTION ID

S29-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. consistent with the model

GRADING NOTE 12% ≥ 5%.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B1-Q3

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 29 • BACKWARD 1 • Q3/5

True or false?

Under the 5% rule, 4% of trials means the model is consistent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B1-Q4

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 29 • BACKWARD 1 • Q4/5

969

QUESTION ID

S29-B1-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 4% < 5%.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B1-Q4

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 29 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Which proportion of simulated results gives the strongest evidence against a model?

A. 0.001

B. 0.2

C. 0.04

D. 0.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B1-Q5

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 29 • BACKWARD 1 • Q5/5

970

QUESTION ID

S29-B1-Q5

TYPE

Multiple choice

CORRECT ANSWER

A. 0.001

GRADING NOTE Smallest.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B1-Q5

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 29 • BACKWARD 1 • Q5/5

SET 29 • HSS-IC.B.5

BACKWARD 2

Explain random assignment • High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3

True or false?

Random assignment tends to make the treatment groups similar before treatment, although chance differences remain.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B2-Q1

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 29 • BACKWARD 2 • Q1/5

QUESTION ID

S29-B2-Q1

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE It tends to balance other variables between the groups; it does not make them identical.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S29-B2-Q1**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 29 • BACKWARD 2 • Q1/5

In an experiment, plants were assigned to two plant foods at random. Why?

A. so that more of the plants grow well

B. so that no plant needs to be measured

C. so growth differences can be credited to food

D. so the two group means come out equal

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B2-Q2

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 29 • BACKWARD 2 • Q2/5

973

QUESTION ID

S29-B2-Q2

TYPE

Multiple choice

CORRECT ANSWER

C. so growth differences can be credited to food

GRADING NOTE Purpose.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S29-B2-Q2**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 29 • BACKWARD 2 • Q2/5

True or false?

In an experiment, subjects choose which treatment they receive.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B2-Q3

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 29 • BACKWARD 2 • Q3/5

974

QUESTION ID

S29-B2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The researcher assigns.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B2-Q3

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 29 • BACKWARD 2 • Q3/5

Ten plants are numbered 1-10. Which assignment is random?

A. put the 5 tallest plants on food A

B. let the gardener choose

C. draw 5 numbers from a hat for food A

D. put plants 1-5 on food A

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B2-Q4

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 29 • BACKWARD 2 • Q4/5

975

QUESTION ID

S29-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. draw 5 numbers from a hat for food A

GRADING NOTE Chance.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S29-B2-Q4**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 29 • BACKWARD 2 • Q4/5

True or false?

Random assignment is what allows an experiment to support a conclusion about cause.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B2-Q5

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 29 • BACKWARD 2 • Q5/5

976

QUESTION ID
S29-B2-Q5

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Key idea.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B2-Q5

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 29 • BACKWARD 2 • Q5/5

SET 29 • HSS-IC.B.5

BACKWARD 3

Compute means of small groups • Grade 6 • Statistics & Probability • 6.SP.B.5.c

ANSWER KEY Reduced copy of the question page, with its answer

Find the mean of 14, 11, 13, 12, 15.

Mean: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B3-Q1

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 29 • BACKWARD 3 • Q1/5

978

QUESTION ID

S29-B3-Q1

TYPE

Short answer

CORRECT ANSWER

13

GRADING NOTE 65/5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B3-Q1

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 29 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the mean of 10, 12, 9, 11, 8.

Mean: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B3-Q2

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 29 • BACKWARD 3 • Q2/5

979

QUESTION ID

S29-B3-Q2

TYPE

Short answer

CORRECT ANSWER

10

GRADING NOTE 50/5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B3-Q2

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 29 • BACKWARD 3 • Q2/5

True or false?

The difference of the means 13 and 10 is 3.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B3-Q3

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 29 • BACKWARD 3 • Q3/5

980

QUESTION ID

S29-B3-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE 13 − 10.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B3-Q3

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 29 • BACKWARD 3 • Q3/5

Find the mean of 2.5, 3, 3.5, 5.

A. 3

B. 3.5

C. 3.25

D. 14

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B3-Q4

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 29 • BACKWARD 3 • Q4/5

981

QUESTION ID

S29-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 3.5

GRADING NOTE 14/4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B3-Q4

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 29 • BACKWARD 3 • Q4/5

True or false?

The mean of 8, 9, 10, 11, 12 is 11.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B3-Q5

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 29 • BACKWARD 3 • Q5/5

982

QUESTION ID

S29-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 10.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B3-Q5

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 29 • BACKWARD 3 • Q5/5

SET 29 • HSS-IC.B.5

FORWARD 1

Use differences in either direction • High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5

(Hypothetical simulation results.) In 200 re-randomizations, differences of 3 or more occurred 4 times and differences of −3 or less occurred 5 times. What proportion is at least 3 away from 0 in either direction? Give a decimal.

Proportion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F1-Q1

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 29 • FORWARD 1 • Q1/5

984

QUESTION ID

S29-F1-Q1

TYPE

Short answer

CORRECT ANSWER

0.045

GRADING NOTE 9/200.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S29-F1-Q1**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 29 • FORWARD 1 • Q1/5

True or false?

Under a 5% rule, a proportion of 0.045 means the difference is significant.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F1-Q2

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 29 • FORWARD 1 • Q2/5

985

QUESTION ID

S29-F1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $0.045 < 0.05$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F1-Q2

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 29 • FORWARD 1 • Q2/5

With more plants per group, re-randomized differences tend to be

A. always positive

B. closer to 0

C. farther from 0

D. exactly 0

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F1-Q3

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 29 • FORWARD 1 • Q3/5

986

QUESTION ID

S29-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. closer to 0

GRADING NOTE Less variability.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S29-F1-Q3**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 29 • FORWARD 1 • Q3/5

Treatment mean 31, control mean 28. What is the difference (treatment minus control)?

Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F1-Q4

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 29 • FORWARD 1 • Q4/5

987

QUESTION ID

S29-F1-Q4

TYPE

Short answer

CORRECT ANSWER

3

GRADING NOTE 31 − 28.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S29-F1-Q4**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 29 • FORWARD 1 • Q4/5

True or false?

A significant difference proves that the treatment helps every individual.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F1-Q5

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 29 • FORWARD 1 • Q5/5

988

QUESTION ID

S29-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It is about the groups on average.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F1-Q5

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 29 • FORWARD 1 • Q5/5

SET 29 • HSS-IC.B.5

FORWARD 2

Evaluate a report of an experiment • High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6

A company reports: "Our drink improved scores in a study." The study gave the drink to 10 volunteers and compared them with 10 who chose not to drink it. What is the main flaw?

A. Drinks cannot be studied in an experiment.

B. Subjects chose their own group: no random assignment.

C. Twenty subjects is too many for a study.

D. Scores of two groups cannot be compared.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F2-Q1

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 29 • FORWARD 2 • Q1/5

990

QUESTION ID

S29-F2-Q1

TYPE

Multiple choice

CORRECT ANSWER

B. Subjects chose their own group: no random assignment.

GRADING NOTE Self-selection.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F2-Q1

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 29 • FORWARD 2 • Q1/5

True or false?

A report should say whether a difference could plausibly be due to chance.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F2-Q2

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 29 • FORWARD 2 • Q2/5

991

QUESTION ID
S29-F2-Q2

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Significance.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F2-Q2

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 29 • FORWARD 2 • Q2/5

(Hypothetical simulation results.) A report says a new method raised mean scores from 70 to 71, "a significant improvement." Re-randomization shows such differences 40% of the time. What is the best response?

A. The new method actually lowered scores.

B. Proven true: the study was an experiment.

C. Supported: 71 is higher than 70.

D. Not supported: chance often gives such differences.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F2-Q3

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 29 • FORWARD 2 • Q3/5

992

QUESTION ID
S29-F2-Q3

TYPE
Multiple choice

CORRECT ANSWER
D. Not supported: chance often gives such differences.

GRADING NOTE 40% ≥ 5%.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S29-F2-Q3**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 29 • FORWARD 2 • Q3/5

True or false?

A randomized experiment on adults automatically applies to children.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F2-Q4

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 29 • FORWARD 2 • Q4/5

993

QUESTION ID

S29-F2-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Scope is the subjects.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F2-Q4

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 29 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Which detail most strengthens a report's causal claim?

A. subjects were randomly assigned to treatments

B. the study had a long and technical name

C. the researchers felt confident in the result

D. the report used colorful, eye-catching graphs

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F2-Q5

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 29 • FORWARD 2 • Q5/5

994

QUESTION ID
S29-F2-Q5

TYPE
Multiple choice

CORRECT ANSWER
A. subjects were randomly assigned to treatments

GRADING NOTE Design.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F2-Q5

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 29 • FORWARD 2 • Q5/5

Evaluate the conclusions of reports based on data

MAIN	Probability & Statistics main question	**HSS-IC.B.6**
BACKWARD 1	Know what each design can support	**HSS-IC.B.3**
BACKWARD 2	Recognize lurking variables	**HSS-ID.C.9**
FORWARD 1	Identify the type of bias in a report	**HSS-IC.B.6**
FORWARD 2	Weigh several flaws and decide what a report shows	**HSS-IC.B.6**

SET 30 • HSS-IC.B.6

MAIN

Probability & Statistics main question • High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6

A report says: "Teens who eat breakfast get better grades, so eating breakfast raises grades." The data came from a survey of 2,000 randomly chosen teens. What is the main problem?

A. A survey shows association, not that breakfast causes better grades.

B. A random sample of 2,000 teens is far too small for any survey.

C. Grades cannot be compared between different groups of teens.

D. Random samples cannot be used to describe a population of teens.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-M-Q1

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 30 • MAIN • Q1/5

997

QUESTION ID

S30-M-Q1

TYPE

Multiple choice

CORRECT ANSWER

A. A survey shows association, not that breakfast causes better grades.

GRADING NOTE Cause needs an experiment.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-M-Q1

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 30 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A magazine asked readers to mail in whether they liked their jobs; 80% of the 3,000 replies said no. The magazine reports that "80% of workers dislike their jobs." What is the main problem?

A. Readers chose to reply, so the sample is likely biased.

B. Three thousand replies is too few to report.

C. The percent 80% was computed incorrectly.

D. Job satisfaction cannot be surveyed.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-M-Q2

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 30 • MAIN • Q2/5

998

QUESTION ID

S30-M-Q2

TYPE

Multiple choice

CORRECT ANSWER

A. Readers chose to reply, so the sample is likely biased.

GRADING NOTE Voluntary response.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-M-Q2

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 30 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

A survey asked, "Don't you agree that the unfair new tax should be repealed?" What is the problem?

A. The question is too short to be answered.

B. The wording pushes people toward one answer.

C. The sample must have been too large.

D. Tax questions cannot appear in surveys.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-M-Q3

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 30 • MAIN • Q3/5

999

QUESTION ID
S30-M-Q3

TYPE
Multiple choice

CORRECT ANSWER
B. The wording pushes people toward one answer.

GRADING NOTE Leading question: response bias.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-M-Q3

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 30 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A randomized experiment found that a reading program raised scores of 3rd graders in one district. Which conclusion is best supported?

A. The program will raise scores for students of every age everywhere.

B. The program caused higher scores for students like those in the study.

C. The program is associated with scores but did not cause any change.

D. Nothing can be concluded, because only one district was studied.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-M-Q4

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 30 • MAIN • Q4/5

1000

QUESTION ID
S30-M-Q4

TYPE
Multiple choice

CORRECT ANSWER
B. The program caused higher scores for students like those in the study.

GRADING NOTE Cause for the subjects; limited generalization.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-M-Q4

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 30 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A report on 12 patients says a new treatment "works for everyone," because 9 of the 12 improved. Which criticism is most important?

A. Twelve patients is too many to study at once.

B. The fraction 9/12 should have been written 12/9.

C. No comparison group, and 3 of the 12 did not improve.

D. Improvement cannot be measured in patients.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-M-Q5

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 30 • MAIN • Q5/5

1001

QUESTION ID

S30-M-Q5

TYPE

Multiple choice

CORRECT ANSWER

C. No comparison group, and 3 of the 12 did not improve.

GRADING NOTE No control; and "everyone" contradicts the data.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S30-M-Q5**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 30 • MAIN • Q5/5

SET 30 • HSS-IC.B.6

BACKWARD 1

Know what each design can support • High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3

True or false?

Of a sample survey, an observational study and a randomized experiment, the randomized experiment is the design built to support a cause-and-effect conclusion.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B1-Q1

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 30 • BACKWARD 1 • Q1/5

1003

QUESTION ID

S30-B1-Q1

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Random assignment makes the groups comparable; the other two designs do not assign treatments.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B1-Q1

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 30 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Which design lets results be generalized to a population?

A. a group of volunteers

B. a convenience sample

C. an experiment on friends

D. a random sample from that population

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B1-Q2

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 30 • BACKWARD 1 • Q2/5

1004

QUESTION ID

S30-B1-Q2

TYPE

Multiple choice

CORRECT ANSWER

D. a random sample from that population

GRADING NOTE Random sampling.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B1-Q2

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 30 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

An observational study can prove cause if it is large enough.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B1-Q3

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 30 • BACKWARD 1 • Q3/5

1005

QUESTION ID

S30-B1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Size does not help.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B1-Q3

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 30 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A study used volunteers and random assignment. Its conclusions can include

A. nothing at all

B. cause, for the whole population

C. cause, for subjects like the volunteers

D. association only

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B1-Q4

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 30 • BACKWARD 1 • Q4/5

1006

QUESTION ID

S30-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. cause, for subjects like the volunteers

GRADING NOTE Assignment yes, sampling no.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S30-B1-Q4**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 30 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

Random sampling and random assignment serve different purposes.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B1-Q5

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 30 • BACKWARD 1 • Q5/5

1007

QUESTION ID

S30-B1-Q5

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Generalize versus cause.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B1-Q5

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 30 • BACKWARD 1 • Q5/5

SET 30 • HSS-IC.B.6

BACKWARD 2

Recognize lurking variables • High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9

Cities with more police officers have more crimes. What is a likely lurking variable?

A. the number of officers

B. crime reports

C. city population

D. police uniforms

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B2-Q1

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9 • Set 30 • BACKWARD 2 • Q1/5

1009

QUESTION ID

S30-B2-Q1

TYPE

Multiple choice

CORRECT ANSWER

C. city population

GRADING NOTE Size drives both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S30-B2-Q1**

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9 • Set 30 • BACKWARD 2 • Q1/5

True or false?

A lurking variable affects both variables in an association.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B2-Q2

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9 • Set 30 • BACKWARD 2 • Q2/5

1010

QUESTION ID
S30-B2-Q2

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B2-Q2

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9 • Set 30 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Students who take music lessons score higher in math. Which is a possible lurking variable?

A. family income

B. the music teacher's name

C. the number of notes played

D. math test length

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B2-Q3

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9 • Set 30 • BACKWARD 2 • Q3/5

1011

QUESTION ID

S30-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. family income

GRADING NOTE May affect both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B2-Q3

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9 • Set 30 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

If an association is strong, there can be no lurking variable.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B2-Q4

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9 • Set 30 • BACKWARD 2 • Q4/5

1012

QUESTION ID

S30-B2-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Strength does not exclude one.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B2-Q4

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9 • Set 30 • BACKWARD 2 • Q4/5

True or false?

Random assignment protects an experiment from lurking variables.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B2-Q5

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9 • Set 30 • BACKWARD 2 • Q5/5

1013

QUESTION ID

S30-B2-Q5

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE It balances them.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B2-Q5

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9 • Set 30 • BACKWARD 2 • Q5/5

SET 30 • HSS-IC.B.6

FORWARD 1

Identify the type of bias in a report • High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6

ANSWER KEY Reduced copy of the question page, with its answer

A phone survey calls only landlines during work hours. Which bias is most likely?

A. no bias, because calls are made at random

B. response bias from the wording of questions

C. voluntary response bias from self-selection

D. undercoverage of people away or with only cell phones

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F1-Q1

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 30 • FORWARD 1 • Q1/5

1015

QUESTION ID

S30-F1-Q1

TYPE

Multiple choice

CORRECT ANSWER

D. undercoverage of people away or with only cell phones

GRADING NOTE Some groups cannot be reached.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F1-Q1

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 30 • FORWARD 1 • Q1/5

Many people chosen for a survey refuse to answer. Which bias is this?

A. nonresponse bias

B. measurement error

C. wording bias

D. voluntary response bias

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F1-Q2

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 30 • FORWARD 1 • Q2/5

1016

QUESTION ID

S30-F1-Q2

TYPE

Multiple choice

CORRECT ANSWER

A. nonresponse bias

GRADING NOTE Chosen but missing.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S30-F1-Q2**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 30 • FORWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

A larger sample removes bias caused by a poor sampling method.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F1-Q3

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 30 • FORWARD 1 • Q3/5

1017

QUESTION ID

S30-F1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It only reduces variability.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F1-Q3

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 30 • FORWARD 1 • Q3/5

A survey of mall shoppers is used to describe all residents' shopping habits. Which bias is this?

A. random sampling error only, with no bias

B. a convenience sample that is not representative

C. wording bias from a leading question

D. nonresponse bias from refusals to answer

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F1-Q4

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 30 • FORWARD 1 • Q4/5

1018

QUESTION ID

S30-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. a convenience sample that is not representative

GRADING NOTE Easy to reach, not representative.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S30-F1-Q4**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 30 • FORWARD 1 • Q4/5

True or false?

People may give socially acceptable answers instead of true ones when asked sensitive questions.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F1-Q5

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 30 • FORWARD 1 • Q5/5

1019

QUESTION ID

S30-F1-Q5

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Response bias.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S30-F1-Q5**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 30 • FORWARD 1 • Q5/5

SET 30 • HSS-IC.B.6

FORWARD 2

Weigh several flaws and decide what a report shows • High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6

ANSWER KEY Reduced copy of the question page, with its answer

An app maker reports: "Users who used our app for 30 days lost 8 pounds on average." The users were volunteers and there was no comparison group. Which conclusion is justified?

A. Only that these volunteers lost weight; the app's role is unclear.

B. The app causes weight loss in everyone who uses it.

C. Every single user of the app lost exactly 8 pounds.

D. The study has shown that the app has no effect on weight.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F2-Q1

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 30 • FORWARD 2 • Q1/5

1021

QUESTION ID

S30-F2-Q1

TYPE

Multiple choice

CORRECT ANSWER

A. Only that these volunteers lost weight; the app's role is unclear.

GRADING NOTE No control, no random sample.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S30-F2-Q1**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 30 • FORWARD 2 • Q1/5

True or false?

A report that gives no information about how its sample was chosen should be treated with caution.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F2-Q2

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 30 • FORWARD 2 • Q2/5

1022

QUESTION ID

S30-F2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Cannot judge bias.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F2-Q2

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 30 • FORWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Which change would most improve the app study?

A. ask the users whether they liked the app

B. randomly assign volunteers to use the app or not

C. extend the study from 30 days to 31 days

D. report the weight loss as a percent instead

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F2-Q3

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 30 • FORWARD 2 • Q3/5

1023

QUESTION ID

S30-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. randomly assign volunteers to use the app or not

GRADING NOTE Comparison and random assignment.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S30-F2-Q3**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 30 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

A study funded by a company is always wrong about that company's product.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F2-Q4

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 30 • FORWARD 2 • Q4/5

1024

QUESTION ID

S30-F2-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Funding is a reason to check the design, not proof of error.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F2-Q4

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 30 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Two reports disagree. Report 1 is a randomized experiment with 400 subjects; report 2 is a voluntary online poll of 5,000. Which deserves more weight for a causal claim?

A. equal weight

B. report 1

C. report 2

D. neither

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F2-Q5

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 30 • FORWARD 2 • Q5/5

1025

QUESTION ID

S30-F2-Q5

TYPE

Multiple choice

CORRECT ANSWER

B. report 1

GRADING NOTE Design beats size.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F2-Q5

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 30 • FORWARD 2 • Q5/5

SET 31 • HSS-IC.B.6 • MAKING INFERENCES AND JUSTIFYING CONCLUSIONS

Evaluate the statistics and displays in reports

MAIN	Probability & Statistics main question	**HSS-IC.B.6**
BACKWARD 1	Use margins of error to compare results	**HSS-IC.B.4**
BACKWARD 2	Compute percent change	**7.RP.A.3**
FORWARD 1	Spot misleading features of displays	**HSS-IC.B.6**
FORWARD 2	Reason about conditional rates in reports	**HSS-CP.A.5**

SET 31 • HSS-IC.B.6

MAIN

Probability & Statistics main question • High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6

ANSWER KEY Reduced copy of the question page, with its answer

A bar graph's vertical axis starts at 50. One bar shows 52 and another shows 55. How tall is the second bar compared with the first (ratio of drawn heights), and what is the ratio of the actual values? Give both ratios to two decimal places.

Ratio of drawn heights: _____

Ratio of values: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-M-Q1

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 31 • MAIN • Q1/5

1028

QUESTION ID

S31-M-Q1

TYPE

Short answer

CORRECT ANSWER

2.50; 1.06

GRADING NOTE Form required: both, to two decimal places. (55 − 50)/(52 − 50) = 2.5; 55/52 = 1.0577.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-M-Q1

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 31 • MAIN • Q1/5

An ad says a risk "increased 80%," from 1 in 10,000 to 1.8 in 10,000. Which statement is correct?

A. The absolute risk rose by only 0.8 in 10,000.

B. Most people will now be affected by it.

C. The risk is now 80 out of every 100 people.

D. The risk went from 1% of people to 80%.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-M-Q2

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 31 • MAIN • Q2/5

1029

QUESTION ID

S31-M-Q2

TYPE

Multiple choice

CORRECT ANSWER

A. The absolute risk rose by only 0.8 in 10,000.

GRADING NOTE Relative change sounds large; absolute change is tiny.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-M-Q2

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 31 • MAIN • Q2/5

A poll shows candidate A at 51% and B at 49%, with a margin of error of 3 percentage points. A headline says "Poll shows A has a reliable lead among all voters!" What is the best evaluation?

A. Misleading: the gap is well within the margin of error.

B. Correct: A has a majority of all the voters.

C. Misleading: B is certainly ahead of A.

D. Correct: 51% is more than 49%, so A leads.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-M-Q3

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 31 • MAIN • Q3/5

1030

QUESTION ID

S31-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. Misleading: the gap is well within the margin of error.

GRADING NOTE 48%−54% and 46%−52% overlap a lot, so the poll does not show a reliable lead. The others are false: 51% vs 49% is a sample result, not the population; a sample does not show a majority of all voters; and B is not certainly ahead.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-M-Q3

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 31 • MAIN • Q3/5

A report gives the mean income in a town as $95,000, but most households earn under $50,000 and a few earn millions. What is the best evaluation?

A. The report must have added incomes incorrectly.

B. The mean is the best summary for skewed data.

C. The range is the best measure of center.

D. The median would describe a typical household better.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-M-Q4

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 31 • MAIN • Q4/5

1031

QUESTION ID

S31-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. The median would describe a typical household better.

GRADING NOTE Right-skewed: the mean is pulled up.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-M-Q4

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 31 • MAIN • Q4/5

A pictograph shows sales doubling by drawing a square icon twice as tall and twice as wide. How many times as large is the new icon's area?

Times as large: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-M-Q5

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 31 • MAIN • Q5/5

1032

QUESTION ID

S31-M-Q5

TYPE

Short answer

CORRECT ANSWER

4

GRADING NOTE 2 · 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-M-Q5

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 31 • MAIN • Q5/5

SET 31 • HSS-IC.B.6

BACKWARD 1

Use margins of error to compare results • High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4

ANSWER KEY Reduced copy of the question page, with its answer

An estimate is 51% with a margin of error of 3 points. Give the interval.

From: _____

To: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B1-Q1

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 31 • BACKWARD 1 • Q1/5

1034

QUESTION ID

S31-B1-Q1

TYPE

Short answer

CORRECT ANSWER

48%; 54%

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S31-B1-Q1**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 31 • BACKWARD 1 • Q1/5

True or false?

If two intervals overlap a lot, a cautious reading is not to claim from them which value is larger.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B1-Q2

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 31 • BACKWARD 1 • Q2/5

1035

QUESTION ID

S31-B1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Overlap is only a cautious reading; a formal comparison would use the margin of error of the difference.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B1-Q2

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 31 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Which pair of estimates (each ± 2 points) clearly differs?

A. 48% and 50%

B. 40% and 50%

C. 49% and 51%

D. 45% and 47%

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B1-Q3

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 31 • BACKWARD 1 • Q3/5

1036

QUESTION ID

S31-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 40% and 50%

GRADING NOTE 38-42 and 48-52 do not overlap.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S31-B1-Q3**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 31 • BACKWARD 1 • Q3/5

True or false?

A margin of error of 3 points means the estimate is off by exactly 3 points.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B1-Q4

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 31 • BACKWARD 1 • Q4/5

1037

QUESTION ID

S31-B1-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It bounds plausible values.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B1-Q4

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 31 • BACKWARD 1 • Q4/5

An interval is 46% to 52%. What is the estimate at its center?

Estimate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B1-Q5

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 31 • BACKWARD 1 • Q5/5

1038

QUESTION ID

S31-B1-Q5

TYPE

Short answer

CORRECT ANSWER

49%

GRADING NOTE Midpoint.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S31-B1-Q5**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 31 • BACKWARD 1 • Q5/5

SET 31 • HSS-IC.B.6

BACKWARD 2

Compute percent change • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3

ANSWER KEY Reduced copy of the question page, with its answer

A value rises from 1 to 1.8. What is the percent increase?

Percent: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B2-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 31 • BACKWARD 2 • Q1/5

1040

QUESTION ID

S31-B2-Q1

TYPE

Short answer

CORRECT ANSWER

80%

GRADING NOTE 0.8/1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B2-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 31 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A price rises from $40 to $50. What is the percent increase?

Percent: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B2-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 31 • BACKWARD 2 • Q2/5

1041

QUESTION ID

S31-B2-Q2

TYPE

Short answer

CORRECT ANSWER

25%

GRADING NOTE 10/40.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B2-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 31 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

A rise from 2 to 3 is a 50% increase.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B2-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 31 • BACKWARD 2 • Q3/5

1042

QUESTION ID

S31-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE 1/2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B2-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 31 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A count falls from 200 to 150. What is the percent decrease?

A. 33%

B. 75%

C. 50%

D. 25%

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B2-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 31 • BACKWARD 2 • Q4/5

1043

QUESTION ID

S31-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 25%

GRADING NOTE 50/200.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B2-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 31 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

A rise from 50 to 55 is a 5% increase.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B2-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 31 • BACKWARD 2 • Q5/5

1044

QUESTION ID

S31-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 5/50 = 10%.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B2-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 31 • BACKWARD 2 • Q5/5

SET 31 • HSS-IC.B.6

FORWARD 1

Spot misleading features of displays • High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6

A line graph of monthly sales has a vertical axis from 980 to 1,000. What effect does this have?

A. Large changes look very small.

B. It hides the months.

C. Small changes look very large.

D. It has no effect.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F1-Q1

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 31 • FORWARD 1 • Q1/5

1046

QUESTION ID

S31-F1-Q1

TYPE

Multiple choice

CORRECT ANSWER

C. Small changes look very large.

GRADING NOTE Truncated axis.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S31-F1-Q1**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 31 • FORWARD 1 • Q1/5

True or false?

A pie chart whose percents add to 130% is misleading or in error.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F1-Q2

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 31 • FORWARD 1 • Q2/5

1047

QUESTION ID

S31-F1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Parts of one whole add to 100%.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F1-Q2

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 31 • FORWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

A graph's vertical axis starts at 90. Bars show 92 and 98. What is the ratio of the drawn heights?

Ratio: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F1-Q3

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 31 • FORWARD 1 • Q3/5

1048

QUESTION ID

S31-F1-Q3

TYPE

Short answer

CORRECT ANSWER

4

GRADING NOTE 8/2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S31-F1-Q3**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 31 • FORWARD 1 • Q3/5

True or false?

Using a 3-D effect on a pie chart never changes how large slices appear.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F1-Q4

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 31 • FORWARD 1 • Q4/5

1049

QUESTION ID

S31-F1-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Perspective distorts.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F1-Q4

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 31 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Which display choice is most honest for comparing two counts of 52 and 55?

A. a bar graph with the axis starting at 51

B. a 3-D pie chart

C. a bar graph with the axis starting at 0

D. icons of different widths

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F1-Q5

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 31 • FORWARD 1 • Q5/5

1050

QUESTION ID

S31-F1-Q5

TYPE

Multiple choice

CORRECT ANSWER

C. a bar graph with the axis starting at 0

GRADING NOTE Zero baseline.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F1-Q5

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 31 • FORWARD 1 • Q5/5

SET 31 • HSS-IC.B.6

FORWARD 2

Reason about conditional rates in reports • High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.5

ANSWER KEY Reduced copy of the question page, with its answer

A report says "most car accidents happen within 5 miles of home, so driving near home is dangerous." What is the main flaw?

A. Car accidents cannot be counted accurately by anyone.

B. A distance of five miles is too short to be studied.

C. Cars are always safer when driven far from home.

D. It uses counts without the amount of driving near and far from home.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F2-Q1

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.5 • Set 31 • FORWARD 2 • Q1/5

1052

QUESTION ID

S31-F2-Q1

TYPE

Multiple choice

CORRECT ANSWER

D. It uses counts without the amount of driving near and far from home.

GRADING NOTE A fair comparison needs exposure, such as accidents per mile driven near and far from home; the report gives counts only. The others are unsupported claims.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F2-Q1

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.5 • Set 31 • FORWARD 2 • Q1/5

True or false?

Knowing that 90% of people in a hospital are sick does not tell you the chance that a sick person is in the hospital.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F2-Q2

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.5 • Set 31 • FORWARD 2 • Q2/5

1053

QUESTION ID

S31-F2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE P(sick given in the hospital) differs from P(in the hospital given sick).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F2-Q2

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.5 • Set 31 • FORWARD 2 • Q2/5

A test catches 95% of people who have a rare disease. A report concludes that 95% of people who test positive have it. What is the error?

A. Rare diseases cannot be tested for at all.

B. It confuses P(positive given disease) with P(disease given positive).

C. A test must catch 100% of cases before it is reported.

D. The 95% should have been written as a fraction.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F2-Q3

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.5 • Set 31 • FORWARD 2 • Q3/5

1054

QUESTION ID
S31-F2-Q3

TYPE
Multiple choice

CORRECT ANSWER
B. It confuses P(positive given disease) with P(disease given positive).

GRADING NOTE Conditional probabilities are not symmetric.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F2-Q3

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.5 • Set 31 • FORWARD 2 • Q3/5

True or false?

A higher number of accidents involving red cars proves that red cars are more dangerous.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F2-Q4

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.5 • Set 31 • FORWARD 2 • Q4/5

1055

QUESTION ID

S31-F2-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It depends on how many red cars there are.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F2-Q4

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.5 • Set 31 • FORWARD 2 • Q4/5

To compare the safety of two car colors fairly, which statistic is needed?

A. the number of different car colors sold

B. the average price of the cars of each color

C. accidents per car (or per mile) for each color

D. the total number of accidents for each color

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F2-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.5 • Set 31 • FORWARD 2 • Q5/5

1056

QUESTION ID

S31-F2-Q5

TYPE

Multiple choice

CORRECT ANSWER

C. accidents per car (or per mile) for each color

GRADING NOTE A rate.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F2-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.5 • Set 31 • FORWARD 2 • Q5/5

Describe sample spaces and events

MAIN	Probability & Statistics main question	**HSS-CP.A.1**
BACKWARD 1	List a sample space with a list, table or tree	**7.SP.C.8.b**
BACKWARD 2	Find probabilities in a uniform model	**7.SP.C.7.a**
FORWARD 1	Describe events in multi-stage sample spaces	**HSS-CP.A.1**
FORWARD 2	Combine events with "and", "or" and "not"	**HSS-CP.A.1**

SET 32 • HSS-CP.A.1

MAIN

Probability & Statistics main question • High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1

ANSWER KEY Reduced copy of the question page, with its answer

A fair coin is flipped and a fair six-sided die is rolled, independently. How many outcomes are in the sample space?

Outcomes: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-M-Q1

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 32 • MAIN • Q1/5

1059

QUESTION ID

S32-M-Q1

TYPE

Short answer

CORRECT ANSWER

12

GRADING NOTE 2 · 6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S32-M-Q1**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 32 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Two fair coins are flipped independently; the sample space is HH, HT, TH, TT. How many outcomes are in the event "at least one head"?

Outcomes: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-M-Q2

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 32 • MAIN • Q2/5

1060

QUESTION ID

S32-M-Q2

TYPE

Short answer

CORRECT ANSWER

3

GRADING NOTE HH, HT, TH.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S32-M-Q2**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 32 • MAIN • Q2/5

A fair coin is flipped twice, independently, and the sequence of results is recorded. Which is the sample space?

A. {H, T}

B. {HH, HT, TT}

C. {HH, TT}

D. {HH, HT, TH, TT}

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-M-Q3

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 32 • MAIN • Q3/5

1061

QUESTION ID

S32-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. {HH, HT, TH, TT}

GRADING NOTE Order matters: HT and TH differ.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S32-M-Q3**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 32 • MAIN • Q3/5

Two fair dice are rolled independently. How many of the 36 equally likely outcomes are in the event "the sum is 7"?

Outcomes: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-M-Q4

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 32 • MAIN • Q4/5

1062

QUESTION ID

S32-M-Q4

TYPE

Short answer

CORRECT ANSWER

6

GRADING NOTE (1,6), (2,5), (3,4), (4,3), (5,2), (6,1).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S32-M-Q4**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 32 • MAIN • Q4/5

A spinner has 8 equal sections numbered 1 to 8. Event A is "an even number greater than 3". How many outcomes are in A, and what is P(A)? Give a fraction in lowest terms.

Outcomes in A: _____

P(A): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-M-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 32 • MAIN • Q5/5

1063

QUESTION ID

S32-M-Q5

TYPE

Short answer

CORRECT ANSWER

3; 3/8

GRADING NOTE Form required: both. {4, 6, 8}.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S32-M-Q5**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 32 • MAIN • Q5/5

SET 32 • HSS-CP.A.1

BACKWARD 1

List a sample space with a list, table or tree • Grade 7 • Statistics & Probability • 7.SP.C.8.b

ANSWER KEY Reduced copy of the question page, with its answer

A bag has a red and a blue marble; a fair coin is flipped. How many outcomes (color, coin) are there?

Outcomes: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B1-Q1

Grade 7 • Statistics & Probability • 7.SP.C.8.b • Set 32 • BACKWARD 1 • Q1/5

1065

QUESTION ID

S32-B1-Q1

TYPE

Short answer

CORRECT ANSWER

4

GRADING NOTE 2 · 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B1-Q1

Grade 7 • Statistics & Probability • 7.SP.C.8.b • Set 32 • BACKWARD 1 • Q1/5

True or false?

A tree diagram for flipping a coin 3 times (heads or tails each time) has 8 final branches.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B1-Q2

Grade 7 • Statistics & Probability • 7.SP.C.8.b • Set 32 • BACKWARD 1 • Q2/5

1066

QUESTION ID

S32-B1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE 2 · 2 · 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B1-Q2

Grade 7 • Statistics & Probability • 7.SP.C.8.b • Set 32 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

A sandwich shop offers 3 breads and 4 fillings. How many bread-filling choices are there?

A. 12

B. 81

C. 7

D. 34

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B1-Q3

Grade 7 • Statistics & Probability • 7.SP.C.8.b • Set 32 • BACKWARD 1 • Q3/5

1067

QUESTION ID

S32-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 12

GRADING NOTE $3 \cdot 4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B1-Q3

Grade 7 • Statistics & Probability • 7.SP.C.8.b • Set 32 • BACKWARD 1 • Q3/5

True or false?

The sample space for rolling two six-sided dice (as ordered pairs) has 12 outcomes.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B1-Q4

Grade 7 • Statistics & Probability • 7.SP.C.8.b • Set 32 • BACKWARD 1 • Q4/5

1068

QUESTION ID

S32-B1-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 36.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B1-Q4

Grade 7 • Statistics & Probability • 7.SP.C.8.b • Set 32 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A spinner with 3 equal colors is spun twice. How many outcomes are in the sample space?

Outcomes: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B1-Q5

Grade 7 • Statistics & Probability • 7.SP.C.8.b • Set 32 • BACKWARD 1 • Q5/5

1069

QUESTION ID

S32-B1-Q5

TYPE

Short answer

CORRECT ANSWER

9

GRADING NOTE 3 · 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B1-Q5

Grade 7 • Statistics & Probability • 7.SP.C.8.b • Set 32 • BACKWARD 1 • Q5/5

SET 32 • HSS-CP.A.1

BACKWARD 2

Find probabilities in a uniform model • Grade 7 • Statistics & Probability • 7.SP.C.7.a

ANSWER KEY Reduced copy of the question page, with its answer

A fair die is rolled. What is P(a number greater than 4)? Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B2-Q1

Grade 7 • Statistics & Probability • 7.SP.C.7.a • Set 32 • BACKWARD 2 • Q1/5

1071

QUESTION ID

S32-B2-Q1

TYPE

Short answer

CORRECT ANSWER

1/3

GRADING NOTE {5, 6}.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B2-Q1

Grade 7 • Statistics & Probability • 7.SP.C.7.a • Set 32 • BACKWARD 2 • Q1/5

True or false?

In a uniform model, every outcome has the same probability.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B2-Q2

Grade 7 • Statistics & Probability • 7.SP.C.7.a • Set 32 • BACKWARD 2 • Q2/5

1072

QUESTION ID

S32-B2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B2-Q2

Grade 7 • Statistics & Probability • 7.SP.C.7.a • Set 32 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

A letter is chosen at random from the word MATH. What is P(vowel)?

A. 1/2

B. 1/3

C. 1/4

D. 3/4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B2-Q3

Grade 7 • Statistics & Probability • 7.SP.C.7.a • Set 32 • BACKWARD 2 • Q3/5

1073

QUESTION ID

S32-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 1/4

GRADING NOTE A only.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B2-Q3

Grade 7 • Statistics & Probability • 7.SP.C.7.a • Set 32 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A number is chosen at random from 1 to 20. What is P(multiple of 5)? Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B2-Q4

Grade 7 • Statistics & Probability • 7.SP.C.7.a • Set 32 • BACKWARD 2 • Q4/5

1074

QUESTION ID

S32-B2-Q4

TYPE

Short answer

CORRECT ANSWER

1/5

GRADING NOTE 5, 10, 15, 20.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B2-Q4

Grade 7 • Statistics & Probability • 7.SP.C.7.a • Set 32 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

A spinner with sections of different sizes is a uniform model.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B2-Q5

Grade 7 • Statistics & Probability • 7.SP.C.7.a • Set 32 • BACKWARD 2 • Q5/5

1075

QUESTION ID

S32-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Outcomes are not equally likely.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B2-Q5

Grade 7 • Statistics & Probability • 7.SP.C.7.a • Set 32 • BACKWARD 2 • Q5/5

SET 32 • HSS-CP.A.1

FORWARD 1

Describe events in multi-stage sample spaces • High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Three fair coins are flipped independently. How many outcomes are in the sample space, and how many have exactly 2 heads?

Outcomes: _____

Exactly 2 heads: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F1-Q1

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 32 • FORWARD 1 • Q1/5

1077

QUESTION ID

S32-F1-Q1

TYPE

Short answer

CORRECT ANSWER

8; 3

GRADING NOTE Form required: both. HHT, HTH, THH.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S32-F1-Q1**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 32 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Three fair coins are flipped independently. What is P(exactly 2 heads)? Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F1-Q2

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 32 • FORWARD 1 • Q2/5

1078

QUESTION ID
S32-F1-Q2

TYPE
Short answer

CORRECT ANSWER
3/8

GRADING NOTE 3 of 8.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S32-F1-Q2**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 32 • FORWARD 1 • Q2/5

True or false?

For two fair dice rolled independently, the event "doubles" contains 6 outcomes.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F1-Q3

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 32 • FORWARD 1 • Q3/5

1079

QUESTION ID

S32-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE (1,1) ... (6,6).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F1-Q3

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 32 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

For two fair dice rolled independently, which event contains exactly 4 outcomes?

A. the sum is 7

B. the sum is 12

C. doubles

D. the sum is 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F1-Q4

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 32 • FORWARD 1 • Q4/5

1080

QUESTION ID

S32-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. the sum is 5

GRADING NOTE (1,4), (2,3), (3,2), (4,1). Sum 7 has 6; doubles 6; sum 12 has 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S32-F1-Q4**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 32 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Two fair dice are rolled independently. How many outcomes have a sum of 10 or more?

Outcomes: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F1-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 32 • FORWARD 1 • Q5/5

1081

QUESTION ID

S32-F1-Q5

TYPE

Short answer

CORRECT ANSWER

6

GRADING NOTE Sum 10: 3; sum 11: 2; sum 12: 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S32-F1-Q5**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 32 • FORWARD 1 • Q5/5

SET 32 • HSS-CP.A.1

FORWARD 2

Combine events with "and", "or" and "not" • High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1

ANSWER KEY Reduced copy of the question page, with its answer

A fair six-sided die is rolled. A = "even", B = "greater than 3". How many outcomes are in "A and B"?

Outcomes: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F2-Q1

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 32 • FORWARD 2 • Q1/5

1083

QUESTION ID

S32-F2-Q1

TYPE

Short answer

CORRECT ANSWER

2

GRADING NOTE {4, 6}.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S32-F2-Q1**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 32 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A fair six-sided die is rolled. A = "even", B = "greater than 3". How many outcomes are in "A or B"?

Outcomes: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F2-Q2

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 32 • FORWARD 2 • Q2/5

1084

QUESTION ID

S32-F2-Q2

TYPE

Short answer

CORRECT ANSWER

4

GRADING NOTE {2, 4, 5, 6}.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S32-F2-Q2**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 32 • FORWARD 2 • Q2/5

True or false?

The event "not A" contains every outcome of the sample space that is not in A.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F2-Q3

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 32 • FORWARD 2 • Q3/5

1085

QUESTION ID

S32-F2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Complement.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F2-Q3

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 32 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A fair six-sided die is rolled. A = "even". Which set is "not A"?

A. {2, 4, 6}

B. {1, 2, 3}

C. { }

D. {1, 3, 5}

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F2-Q4

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 32 • FORWARD 2 • Q4/5

1086

QUESTION ID

S32-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. {1, 3, 5}

GRADING NOTE Odd outcomes.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S32-F2-Q4**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 32 • FORWARD 2 • Q4/5

True or false?

The event "A and B" always contains at least as many outcomes as "A or B".

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F2-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 32 • FORWARD 2 • Q5/5

1087

QUESTION ID

S32-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Never more.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F2-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 32 • FORWARD 2 • Q5/5

Unions, intersections and complements of events

MAIN	Probability & Statistics main question	**HSS-CP.A.1**
BACKWARD 1	Count the outcomes in an event	**HSS-CP.A.1**
BACKWARD 2	Find a probability by counting	**7.SP.C.8.a**
FORWARD 1	Use a Venn diagram of counts	**HSS-CP.A.1**
FORWARD 2	Apply the Addition Rule	**HSS-CP.B.7**

SET 33 • HSS-CP.A.1

MAIN

Probability & Statistics main question • High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1

ANSWER KEY Reduced copy of the question page, with its answer

A fair die is rolled. A = "even" and B = "greater than 3". How many outcomes are in A and B, and what is P(A and B)? Give a fraction in lowest terms.

Outcomes: _____

P(A and B): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-M-Q1

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 33 • MAIN • Q1/5

1090

QUESTION ID

S33-M-Q1

TYPE

Short answer

CORRECT ANSWER

2; 1/3

GRADING NOTE Form required: both. {4, 6}.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S33-M-Q1**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 33 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A fair die is rolled. A = "even" and B = "greater than 3". How many outcomes are in A or B, and what is P(A or B)? Give a fraction in lowest terms.

Outcomes: _____

P(A or B): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-M-Q2

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 33 • MAIN • Q2/5

1091

QUESTION ID

S33-M-Q2

TYPE

Short answer

CORRECT ANSWER

4; 2/3

GRADING NOTE Form required: both. {2, 4, 5, 6}.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S33-M-Q2**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 33 • MAIN • Q2/5

A fair die is rolled. A = "even". What is P(not A)? Give a fraction in lowest terms.

P(not A): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-M-Q3

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 33 • MAIN • Q3/5

1092

QUESTION ID

S33-M-Q3

TYPE

Short answer

CORRECT ANSWER

1/2

GRADING NOTE {1, 3, 5}.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S33-M-Q3**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 33 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A fair die is rolled. A = "even" and B = "greater than 3". Which describes the event {1, 3}?

A. A or B

B. A and B

C. not (A or B)

D. not A

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-M-Q4

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 33 • MAIN • Q4/5

1093

QUESTION ID

S33-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. not (A or B)

GRADING NOTE Neither even nor greater than 3. "not A" is {1, 3, 5}.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S33-M-Q4**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 33 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A card is drawn at random from a standard 52-card deck. H = "heart", F = "face card (jack, queen or king)". Find P(H and F) and P(H or F). Give a fraction in lowest terms.

P(H and F): _____

P(H or F): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-M-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 33 • MAIN • Q5/5

1094

QUESTION ID

S33-M-Q5

TYPE

Short answer

CORRECT ANSWER

3/52; 11/26

GRADING NOTE Form required: both. 3 heart face cards; 13 + 12 − 3 = 22, and 22/52 = 11/26.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-M-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 33 • MAIN • Q5/5

SET 33 • HSS-CP.A.1

BACKWARD 1

Count the outcomes in an event • High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1

ANSWER KEY Reduced copy of the question page, with its answer

A number from 1 to 10 is chosen. How many outcomes are in "multiple of 3"?

Outcomes: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B1-Q1

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 33 • BACKWARD 1 • Q1/5

1096

QUESTION ID

S33-B1-Q1

TYPE

Short answer

CORRECT ANSWER

3

GRADING NOTE 3, 6, 9.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S33-B1-Q1**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 33 • BACKWARD 1 • Q1/5

True or false?

For one roll of a fair six-sided die, the event "less than 3" is {1, 2}.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B1-Q2

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 33 • BACKWARD 1 • Q2/5

1097

QUESTION ID

S33-B1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE List.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B1-Q2

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 33 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

A letter is chosen from the word PROBABILITY. How many outcomes (letter positions) are in "the letter B"?

A. 11

B. 1

C. 3

D. 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B1-Q3

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 33 • BACKWARD 1 • Q3/5

1098

QUESTION ID

S33-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 2

GRADING NOTE B appears twice.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B1-Q3

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 33 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

How many cards in a standard 52-card deck are face cards?

Cards: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B1-Q4

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 33 • BACKWARD 1 • Q4/5

1099

QUESTION ID
S33-B1-Q4

TYPE
Short answer

CORRECT ANSWER
12

GRADING NOTE 3 per suit · 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S33-B1-Q4**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 33 • BACKWARD 1 • Q4/5

True or false?

A standard deck has 4 aces and 13 hearts, so it has 17 cards that are aces or hearts.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B1-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 33 • BACKWARD 1 • Q5/5

1100

QUESTION ID

S33-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The ace of hearts is counted twice: 16.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B1-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 33 • BACKWARD 1 • Q5/5

SET 33 • HSS-CP.A.1

BACKWARD 2

Find a probability by counting • Grade 7 • Statistics & Probability • 7.SP.C.8.a

ANSWER KEY Reduced copy of the question page, with its answer

Two fair coins are flipped independently. What is P(two heads)? Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B2-Q1

Grade 7 • Statistics & Probability • 7.SP.C.8.a • Set 33 • BACKWARD 2 • Q1/5

1102

QUESTION ID

S33-B2-Q1

TYPE

Short answer

CORRECT ANSWER

1/4

GRADING NOTE 1 of 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B2-Q1

Grade 7 • Statistics & Probability • 7.SP.C.8.a • Set 33 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A fair six-sided die is rolled. What is P(1 or 2)? Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B2-Q2

Grade 7 • Statistics & Probability • 7.SP.C.8.a • Set 33 • BACKWARD 2 • Q2/5

1103

QUESTION ID

S33-B2-Q2

TYPE

Short answer

CORRECT ANSWER

1/3

GRADING NOTE 2 of 6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B2-Q2

Grade 7 • Statistics & Probability • 7.SP.C.8.a • Set 33 • BACKWARD 2 • Q2/5

True or false?

The probability of an event plus the probability of its complement is 1.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B2-Q3

Grade 7 • Statistics & Probability • 7.SP.C.8.a • Set 33 • BACKWARD 2 • Q3/5

1104

QUESTION ID

S33-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Complements.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B2-Q3

Grade 7 • Statistics & Probability • 7.SP.C.8.a • Set 33 • BACKWARD 2 • Q3/5

A card is drawn at random from a standard 52-card deck. What is P(king)?

A. 1/4

B. 1/52

C. 4/13

D. 1/13

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B2-Q4

Grade 7 • Statistics & Probability • 7.SP.C.8.a • Set 33 • BACKWARD 2 • Q4/5

1105

QUESTION ID

S33-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 1/13

GRADING NOTE 4/52.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B2-Q4

Grade 7 • Statistics & Probability • 7.SP.C.8.a • Set 33 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

If P(A) = 0.3, then P(not A) = 0.3.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B2-Q5

Grade 7 • Statistics & Probability • 7.SP.C.8.a • Set 33 • BACKWARD 2 • Q5/5

1106

QUESTION ID

S33-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 0.7.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B2-Q5

Grade 7 • Statistics & Probability • 7.SP.C.8.a • Set 33 • BACKWARD 2 • Q5/5

SET 33 • HSS-CP.A.1

FORWARD 1

Use a Venn diagram of counts • High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1

Of 30 students, 18 play soccer, 12 play basketball and 5 play both. How many play soccer only, and how many play neither?

Soccer only: _____

Neither: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F1-Q1

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 33 • FORWARD 1 • Q1/5

1108

QUESTION ID

S33-F1-Q1

TYPE

Short answer

CORRECT ANSWER

13; 5

GRADING NOTE Form required: both. 18 − 5; 30 − (18 + 12 − 5).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S33-F1-Q1**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 33 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Of 30 students, 18 play soccer, 12 play basketball and 5 play both. A student is chosen at random. What is P(soccer or basketball)? Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F1-Q2

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 33 • FORWARD 1 • Q2/5

1109

QUESTION ID

S33-F1-Q2

TYPE

Short answer

CORRECT ANSWER

5/6

GRADING NOTE 25/30.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S33-F1-Q2**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 33 • FORWARD 1 • Q2/5

True or false?

In a Venn diagram, the overlap of two circles shows "A and B".

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F1-Q3

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 33 • FORWARD 1 • Q3/5

1110

QUESTION ID

S33-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Intersection.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F1-Q3

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 33 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Of 30 students, 18 play soccer, 12 play basketball and 5 play both. What is P(not soccer)?

A. 3/5

B. 2/5

C. 7/30

D. 1/6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F1-Q4

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 33 • FORWARD 1 • Q4/5

1111

QUESTION ID

S33-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 2/5

GRADING NOTE 12/30.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S33-F1-Q4**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 33 • FORWARD 1 • Q4/5

True or false?

Of 30 students, 18 play soccer, 12 play basketball and 5 play both. Adding 18 and 12 gives the number who play soccer or basketball.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F1-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 33 • FORWARD 1 • Q5/5

1112

QUESTION ID

S33-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The 5 are counted twice.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F1-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 33 • FORWARD 1 • Q5/5

SET 33 • HSS-CP.A.1

FORWARD 2

Apply the Addition Rule • High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.7

ANSWER KEY Reduced copy of the question page, with its answer

P(A) = 0.5, P(B) = 0.4 and P(A and B) = 0.2. Find P(A or B).

P(A or B): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F2-Q1

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.7 • Set 33 • FORWARD 2 • Q1/5

1114

QUESTION ID

S33-F2-Q1

TYPE

Short answer

CORRECT ANSWER

0.7

GRADING NOTE 0.5 + 0.4 − 0.2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S33-F2-Q1**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.7 • Set 33 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A card is drawn at random from a standard 52-card deck. Find P(king or heart). Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F2-Q2

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.7 • Set 33 • FORWARD 2 • Q2/5

1115

QUESTION ID

S33-F2-Q2

TYPE

Short answer

CORRECT ANSWER

4/13

GRADING NOTE 4/52 + 13/52 − 1/52 = 16/52.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S33-F2-Q2**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.7 • Set 33 • FORWARD 2 • Q2/5

True or false?

If A and B cannot happen together, P(A or B) = P(A) + P(B).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F2-Q3

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.7 • Set 33 • FORWARD 2 • Q3/5

1116

QUESTION ID

S33-F2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE P(A and B) = 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F2-Q3

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.7 • Set 33 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

P(A) = 0.6, P(B) = 0.3, P(A or B) = 0.7. What is P(A and B)?

A. 0.9

B. 0.2

C. 0.1

D. 0.18

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F2-Q4

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.7 • Set 33 • FORWARD 2 • Q4/5

1117

QUESTION ID

S33-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 0.2

GRADING NOTE 0.6 + 0.3 − 0.7.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S33-F2-Q4**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.7 • Set 33 • FORWARD 2 • Q4/5

True or false?

P(A or B) can be greater than 1 if A and B are both likely.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F2-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.7 • Set 33 • FORWARD 2 • Q5/5

1118

QUESTION ID

S33-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Never above 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F2-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.7 • Set 33 • FORWARD 2 • Q5/5

Decide independence with the product rule

MAIN	Probability & Statistics main question	**HSS-CP.A.2**
BACKWARD 1	Multiply probabilities	**5.NF.B.4**
BACKWARD 2	Find P(A and B) by counting	**HSS-CP.A.1**
FORWARD 1	Use independence for several events	**HSS-CP.A.2**
FORWARD 2	Interpret independence with conditional probability	**HSS-CP.A.3**

SET 34 • HSS-CP.A.2

MAIN

Probability & Statistics main question • High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.2

ANSWER KEY Reduced copy of the question page, with its answer

P(A) = 0.4, P(B) = 0.5 and P(A and B) = 0.2. Find P(A)·P(B), and decide whether A and B are independent (enter yes or no).

P(A)·P(B): _____

Independent (yes or no): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-M-Q1

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.2 • Set 34 • MAIN • Q1/5

1121

QUESTION ID

S34-M-Q1

TYPE

Short answer

CORRECT ANSWER

0.2; yes

GRADING NOTE Form required: both. 0.2 = 0.2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S34-M-Q1**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.2 • Set 34 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

P(A) = 0.3, P(B) = 0.6 and P(A and B) = 0.25. Find P(A)·P(B), and decide whether A and B are independent (enter yes or no).

P(A)·P(B): _____

Independent (yes or no): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-M-Q2

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.2 • Set 34 • MAIN • Q2/5

1122

QUESTION ID

S34-M-Q2

TYPE

Short answer

CORRECT ANSWER

0.18; no

GRADING NOTE Form required: both. 0.18 ≠ 0.25.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S34-M-Q2**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.2 • Set 34 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

A and B are independent with P(A) = 0.7 and P(B) = 0.2. Find P(A and B).

P(A and B): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-M-Q3

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.2 • Set 34 • MAIN • Q3/5

1123

QUESTION ID
S34-M-Q3

TYPE
Short answer

CORRECT ANSWER
0.14

GRADING NOTE 0.7 · 0.2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S34-M-Q3**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.2 • Set 34 • MAIN • Q3/5

A fair six-sided die is rolled. A = "even" and B = "1 or 2". Are A and B independent?

A. No: A and B share the outcome 2.

B. No: P(A and B) = 1/6, not 1/2.

C. Yes: P(A and B) = 1/6 = (1/2)(1/3).

D. Yes: A and B have no outcomes in common.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-M-Q4

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.2 • Set 34 • MAIN • Q4/5

1124

QUESTION ID

S34-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. Yes: P(A and B) = 1/6 = (1/2)(1/3).

GRADING NOTE A and B = {2}. Sharing an outcome does not decide independence.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-M-Q4

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.2 • Set 34 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

In the survey table, a person is chosen at random. M = "man" and Y = "said yes". Find P(M and Y), and decide whether M and Y are independent (enter yes or no).

	Yes	No	Total
Men	24	16	40
Women	36	24	60
Total	60	40	100

P(M and Y): _____

Independent (yes or no): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-M-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.2 • Set 34 • MAIN • Q5/5

1125

QUESTION ID

S34-M-Q5

TYPE

Short answer

CORRECT ANSWER

0.24; yes

GRADING NOTE Form required: both. 24/100; P(M)·P(Y) = 0.4 · 0.6 = 0.24.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S34-M-Q5**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.2 • Set 34 • MAIN • Q5/5

SET 34 • HSS-CP.A.2

BACKWARD 1

Multiply probabilities • Grade 5 • Number & Operations–Fractions • 5.NF.B.4

ANSWER KEY Reduced copy of the question page, with its answer

Compute (1/2)(1/3). Give a fraction in lowest terms.

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B1-Q1

Grade 5 • Number & Operations–Fractions • 5.NF.B.4 • Set 34 • BACKWARD 1 • Q1/5

1127

QUESTION ID

S34-B1-Q1

TYPE

Short answer

CORRECT ANSWER

1/6

GRADING NOTE 1/6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B1-Q1

Grade 5 • Number & Operations–Fractions • 5.NF.B.4 • Set 34 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Compute 0.4 · 0.5.

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B1-Q2

Grade 5 • Number & Operations–Fractions • 5.NF.B.4 • Set 34 • BACKWARD 1 • Q2/5

1128

QUESTION ID

S34-B1-Q2

TYPE

Short answer

CORRECT ANSWER

0.2

GRADING NOTE Decimal product.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B1-Q2

Grade 5 • Number & Operations–Fractions • 5.NF.B.4 • Set 34 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

(2/3)(3/4) = 1/2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B1-Q3

Grade 5 • Number & Operations–Fractions • 5.NF.B.4 • Set 34 • BACKWARD 1 • Q3/5

1129

QUESTION ID

S34-B1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE 6/12.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B1-Q3

Grade 5 • Number & Operations–Fractions • 5.NF.B.4 • Set 34 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Compute 0.3 · 0.6.

A. 1.8

B. 0.36

C. 0.18

D. 0.9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B1-Q4

Grade 5 • Number & Operations–Fractions • 5.NF.B.4 • Set 34 • BACKWARD 1 • Q4/5

1130

QUESTION ID

S34-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 0.18

GRADING NOTE 3 · 6 = 18, two decimal places.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B1-Q4

Grade 5 • Number & Operations–Fractions • 5.NF.B.4 • Set 34 • BACKWARD 1 • Q4/5

True or false?

$0.7 \cdot 0.2 = 1.4$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B1-Q5

Grade 5 • Number & Operations–Fractions • 5.NF.B.4 • Set 34 • BACKWARD 1 • Q5/5

1131

QUESTION ID

S34-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 0.14.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B1-Q5

Grade 5 • Number & Operations–Fractions • 5.NF.B.4 • Set 34 • BACKWARD 1 • Q5/5

SET 34 • HSS-CP.A.2

BACKWARD 2

Find P(A and B) by counting • High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1

ANSWER KEY Reduced copy of the question page, with its answer

A fair six-sided die is rolled. What is P(even and greater than 3)? Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B2-Q1

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 34 • BACKWARD 2 • Q1/5

1133

QUESTION ID

S34-B2-Q1

TYPE

Short answer

CORRECT ANSWER

1/3

GRADING NOTE {4, 6}.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S34-B2-Q1**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 34 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A fair coin is flipped and a fair die is rolled, independently. What is P(heads and 6)? Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B2-Q2

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 34 • BACKWARD 2 • Q2/5

1134

QUESTION ID

S34-B2-Q2

TYPE

Short answer

CORRECT ANSWER

1/12

GRADING NOTE 1 of 12.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S34-B2-Q2**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 34 • BACKWARD 2 • Q2/5

True or false?

For one roll of a fair six-sided die, P(odd and even) = 0.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B2-Q3

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 34 • BACKWARD 2 • Q3/5

1135

QUESTION ID

S34-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE No outcome is both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B2-Q3

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 34 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Two fair coins are flipped independently. What is P(first is heads and second is tails)?

A. 3/4

B. 1/4

C. 1

D. 1/2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B2-Q4

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 34 • BACKWARD 2 • Q4/5

1136

QUESTION ID

S34-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 1/4

GRADING NOTE HT only.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S34-B2-Q4**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 34 • BACKWARD 2 • Q4/5

True or false?

For one roll of a fair six-sided die, P(less than 3 and odd) = 1/3.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B2-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 34 • BACKWARD 2 • Q5/5

1137

QUESTION ID

S34-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE {1}: 1/6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B2-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 34 • BACKWARD 2 • Q5/5

SET 34 • HSS-CP.A.2

FORWARD 1

Use independence for several events • High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.2

ANSWER KEY Reduced copy of the question page, with its answer

A fair coin is flipped 3 times, independently. What is P(all three heads)? Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F1-Q1

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.2 • Set 34 • FORWARD 1 • Q1/5

1139

QUESTION ID

S34-F1-Q1

TYPE

Short answer

CORRECT ANSWER

1/8

GRADING NOTE $(1/2)^3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S34-F1-Q1**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.2 • Set 34 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Each of 3 independent lights works with probability 0.9. What is P(all 3 work), and P(at least one fails)?

All work: _____

At least one fails: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F1-Q2

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.2 • Set 34 • FORWARD 1 • Q2/5

1140

QUESTION ID

S34-F1-Q2

TYPE

Short answer

CORRECT ANSWER

0.729; 0.271

GRADING NOTE Form required: both. 0.9^3; $1 - 0.729$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F1-Q2

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.2 • Set 34 • FORWARD 1 • Q2/5

True or false?

For independent events, P(A and B and C) = P(A)·P(B)·P(C).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F1-Q3

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.2 • Set 34 • FORWARD 1 • Q3/5

1141

QUESTION ID

S34-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Extends the product rule.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F1-Q3

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.2 • Set 34 • FORWARD 1 • Q3/5

A fair die is rolled twice, independently. What is P(two sixes)?

A. 1/6

B. 1/36

C. 2/6

D. 1/12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F1-Q4

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.2 • Set 34 • FORWARD 1 • Q4/5

1142

QUESTION ID

S34-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 1/36

GRADING NOTE (1/6)(1/6).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S34-F1-Q4**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.2 • Set 34 • FORWARD 1 • Q4/5

True or false?

If P(A and B) = P(A)·P(B), then A and B cannot happen together.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F1-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.2 • Set 34 • FORWARD 1 • Q5/5

1143

QUESTION ID

S34-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Independent events can happen together; that describes mutually exclusive events.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F1-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.2 • Set 34 • FORWARD 1 • Q5/5

SET 34 • HSS-CP.A.2

FORWARD 2

Interpret independence with conditional probability • High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3

ANSWER KEY Reduced copy of the question page, with its answer

P(A and B) = 0.2 and P(B) = 0.5. Find P(A given B).

P(A given B): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F2-Q1

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 34 • FORWARD 2 • Q1/5

1145

QUESTION ID

S34-F2-Q1

TYPE

Short answer

CORRECT ANSWER

0.4

GRADING NOTE 0.2/0.5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S34-F2-Q1**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 34 • FORWARD 2 • Q1/5

True or false?

If A and B are independent, P(A given B) = P(A).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F2-Q2

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 34 • FORWARD 2 • Q2/5

1146

QUESTION ID

S34-F2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Knowing B does not change A.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F2-Q2

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 34 • FORWARD 2 • Q2/5

P(A) = 0.4 and P(A given B) = 0.4. What can be concluded?

A. A and B are independent.

B. P(B) = 0.4.

C. B always happens.

D. A and B are mutually exclusive.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F2-Q3

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 34 • FORWARD 2 • Q3/5

1147

QUESTION ID

S34-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. A and B are independent.

GRADING NOTE Equal conditional and unconditional probability.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S34-F2-Q3**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 34 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

P(A) = 0.3, P(B) = 0.6 and P(A and B) = 0.18. Find P(A given B).

P(A given B): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F2-Q4

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 34 • FORWARD 2 • Q4/5

1148

QUESTION ID

S34-F2-Q4

TYPE

Short answer

CORRECT ANSWER

0.3

GRADING NOTE 0.18/0.6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S34-F2-Q4**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 34 • FORWARD 2 • Q4/5

True or false?

If P(A given B) is greater than P(A), then A and B are independent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F2-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 34 • FORWARD 2 • Q5/5

1149

QUESTION ID

S34-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE B changes A's chance.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F2-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 34 • FORWARD 2 • Q5/5

Compute conditional probability as P(A and B)/P(B)

MAIN	Probability & Statistics main question	**HSS-CP.A.3**
BACKWARD 1	Divide decimals and fractions	**6.NS.B.3**
BACKWARD 2	Count outcomes in a restricted sample space	**HSS-CP.A.1**
FORWARD 1	Compute conditional probability in both directions	**HSS-CP.A.3**
FORWARD 2	Test independence with conditional probability	**HSS-CP.A.3**

SET 35 • HSS-CP.A.3

MAIN

Probability & Statistics main question • High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3

P(A and B) = 0.12 and P(B) = 0.4. Find P(A given B).

P(A given B): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-M-Q1

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 35 • MAIN • Q1/5

1152

QUESTION ID

S35-M-Q1

TYPE

Short answer

CORRECT ANSWER

0.3

GRADING NOTE 0.12/0.4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S35-M-Q1**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 35 • MAIN • Q1/5

A fair die is rolled. What is P(6 given the roll is even)? Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-M-Q2

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 35 • MAIN • Q2/5

1153

QUESTION ID

S35-M-Q2

TYPE

Short answer

CORRECT ANSWER

1/3

GRADING NOTE P(6)/P(even) = (1/6)/(1/2).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-M-Q2

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 35 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Two fair dice are rolled independently. What is P(doubles given the sum is 8)? Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-M-Q3

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 35 • MAIN • Q3/5

1154

QUESTION ID
S35-M-Q3

TYPE
Short answer

CORRECT ANSWER
1/5

GRADING NOTE Sum 8: (2,6), (3,5), (4,4), (5,3), (6,2); one is doubles.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S35-M-Q3**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 35 • MAIN • Q3/5

On a given day, P(rain) = 0.3 and P(rain and wind) = 0.12. What is P(wind given rain)?

A. 0.4

B. 2.5

C. 0.12

D. 0.036

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-M-Q4

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 35 • MAIN • Q4/5

1155

QUESTION ID

S35-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 0.4

GRADING NOTE 0.12/0.3. 0.036 multiplies; 2.5 inverts the division.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-M-Q4

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 35 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

P(B) = 0.25 and P(A given B) = 0.6. Find P(A and B).

P(A and B): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-M-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 35 • MAIN • Q5/5

1156

QUESTION ID

S35-M-Q5

TYPE

Short answer

CORRECT ANSWER

0.15

GRADING NOTE 0.6 · 0.25.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S35-M-Q5**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 35 • MAIN • Q5/5

SET 35 • HSS-CP.A.3

BACKWARD 1

Divide decimals and fractions • Grade 6 • The Number System • 6.NS.B.3

ANSWER KEY Reduced copy of the question page, with its answer

Compute 0.12 ÷ 0.4.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B1-Q1

Grade 6 • The Number System • 6.NS.B.3 • Set 35 • BACKWARD 1 • Q1/5

1158

QUESTION ID

S35-B1-Q1

TYPE

Short answer

CORRECT ANSWER

0.3

GRADING NOTE 1.2 ÷ 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B1-Q1

Grade 6 • The Number System • 6.NS.B.3 • Set 35 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Compute 0.18 ÷ 0.6.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B1-Q2

Grade 6 • The Number System • 6.NS.B.3 • Set 35 • BACKWARD 1 • Q2/5

1159

QUESTION ID

S35-B1-Q2

TYPE

Short answer

CORRECT ANSWER

0.3

GRADING NOTE 1.8 ÷ 6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B1-Q2

Grade 6 • The Number System • 6.NS.B.3 • Set 35 • BACKWARD 1 • Q2/5

True or false?

$0.2 \div 0.5 = 0.4$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B1-Q3

Grade 6 • The Number System • 6.NS.B.3 • Set 35 • BACKWARD 1 • Q3/5

1160

QUESTION ID

S35-B1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE 2 ÷ 5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B1-Q3

Grade 6 • The Number System • 6.NS.B.3 • Set 35 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Compute 0.05 ÷ 0.2.

A. 0.25

B. 0.01

C. 2.5

D. 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B1-Q4

Grade 6 • The Number System • 6.NS.B.3 • Set 35 • BACKWARD 1 • Q4/5

1161

QUESTION ID

S35-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 0.25

GRADING NOTE 0.5 ÷ 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B1-Q4

Grade 6 • The Number System • 6.NS.B.3 • Set 35 • BACKWARD 1 • Q4/5

True or false?

$0.3 \div 0.6 = 2.$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B1-Q5

Grade 6 • The Number System • 6.NS.B.3 • Set 35 • BACKWARD 1 • Q5/5

1162

QUESTION ID

S35-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 0.5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B1-Q5

Grade 6 • The Number System • 6.NS.B.3 • Set 35 • BACKWARD 1 • Q5/5

SET 35 • HSS-CP.A.3

BACKWARD 2

Count outcomes in a restricted sample space • High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1

A fair six-sided die is rolled and shows an even number. How many outcomes remain possible?

Outcomes: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B2-Q1

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 35 • BACKWARD 2 • Q1/5

1164

QUESTION ID

S35-B2-Q1

TYPE

Short answer

CORRECT ANSWER

3

GRADING NOTE {2, 4, 6}.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S35-B2-Q1**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 35 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Two fair dice are rolled independently and the sum is 8. How many outcomes give a sum of 8?

Outcomes: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B2-Q2

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 35 • BACKWARD 2 • Q2/5

1165

QUESTION ID

S35-B2-Q2

TYPE

Short answer

CORRECT ANSWER

5

GRADING NOTE (2,6) to (6,2).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S35-B2-Q2**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 35 • BACKWARD 2 • Q2/5

True or false?

A card is drawn from a standard 52-card deck. Knowing that it is a heart leaves 13 possible cards.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B2-Q3

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 35 • BACKWARD 2 • Q3/5

1166

QUESTION ID
S35-B2-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE The hearts.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B2-Q3

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 35 • BACKWARD 2 • Q3/5

A fair six-sided die is rolled and shows a number greater than 2. How many of the remaining outcomes are odd?

A. 2

B. 3

C. 4

D. 1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B2-Q4

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 35 • BACKWARD 2 • Q4/5

1167

QUESTION ID

S35-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 2

GRADING NOTE {3, 4, 5, 6}: 3 and 5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S35-B2-Q4**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 35 • BACKWARD 2 • Q4/5

True or false?

A card is drawn from a standard 52-card deck. Knowing that it is a face card leaves 13 possible cards.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B2-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 35 • BACKWARD 2 • Q5/5

1168

QUESTION ID
S35-B2-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE 12.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B2-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 35 • BACKWARD 2 • Q5/5

SET 35 • HSS-CP.A.3

FORWARD 1

Compute conditional probability in both directions • High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3

ANSWER KEY Reduced copy of the question page, with its answer

A card is drawn at random from a standard 52-card deck. Find P(king given face card) and P(face card given king). Give a fraction in lowest terms.

P(king given face card): _____

P(face card given king): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F1-Q1

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 35 • FORWARD 1 • Q1/5

1170

QUESTION ID

S35-F1-Q1

TYPE

Short answer

CORRECT ANSWER

1/3; 1

GRADING NOTE Form required: both. 4/12 and 4/4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S35-F1-Q1**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 35 • FORWARD 1 • Q1/5

True or false?

P(A given B) and P(B given A) are always equal.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F1-Q2

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 35 • FORWARD 1 • Q2/5

1171

QUESTION ID

S35-F1-Q2

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The king example shows otherwise.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F1-Q2

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 35 • FORWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

P(A) = 0.5, P(B) = 0.2 and P(A and B) = 0.1. Find P(A given B) and P(B given A).

P(A given B): _____

P(B given A): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F1-Q3

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 35 • FORWARD 1 • Q3/5

1172

QUESTION ID

S35-F1-Q3

TYPE

Short answer

CORRECT ANSWER

0.5; 0.2

GRADING NOTE Form required: both. 0.1/0.2 and 0.1/0.5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S35-F1-Q3**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 35 • FORWARD 1 • Q3/5

Which formula gives P(B given A)?

A. P(A)/P(B)

B. P(A)·P(B)

C. P(A and B)/P(A)

D. P(A and B)/P(B)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F1-Q4

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 35 • FORWARD 1 • Q4/5

QUESTION ID

S35-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. P(A and B)/P(A)

GRADING NOTE Divide by the condition's probability.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S35-F1-Q4**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 35 • FORWARD 1 • Q4/5

True or false?

P(A given B) = P(B given A) exactly when P(A) = P(B), provided P(A and B) > 0.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F1-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 35 • FORWARD 1 • Q5/5

1174

QUESTION ID

S35-F1-Q5

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Both equal P(A and B) divided by equal numbers.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F1-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 35 • FORWARD 1 • Q5/5

SET 35 • HSS-CP.A.3

FORWARD 2

Test independence with conditional probability • High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3

ANSWER KEY Reduced copy of the question page, with its answer

P(A) = 0.6, P(B) = 0.5, P(A and B) = 0.3. Find P(A given B), and decide whether A and B are independent (enter yes or no).

P(A given B): _____

Independent (yes or no): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F2-Q1

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 35 • FORWARD 2 • Q1/5

1176

QUESTION ID

S35-F2-Q1

TYPE

Short answer

CORRECT ANSWER

0.6; yes

GRADING NOTE Form required: both. 0.3/0.5 = P(A).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S35-F2-Q1**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 35 • FORWARD 2 • Q1/5

True or false?

If P(A given B) = P(A), then also P(B given A) = P(B), provided P(A) > 0.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F2-Q2

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 35 • FORWARD 2 • Q2/5

1177

QUESTION ID

S35-F2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Symmetric.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F2-Q2

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 35 • FORWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Two cards are drawn at random, without replacement, from a standard 52-card deck. Are "first card is an ace" and "second card is an ace" independent?

A. No: two aces can never be drawn one after another.

B. Yes: the deck always contains exactly 4 aces.

C. Yes: each card in the deck is equally likely to be drawn.

D. No: the first draw changes the chance for the second.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F2-Q3

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 35 • FORWARD 2 • Q3/5

1178

QUESTION ID
S35-F2-Q3

TYPE
Multiple choice

CORRECT ANSWER
D. No: the first draw changes the chance for the second.

GRADING NOTE P(second ace given first ace) = 3/51 ≠ 4/52.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F2-Q3

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 35 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

P(A) = 0.4, P(B) = 0.5, P(A and B) = 0.1. Find P(A given B).

P(A given B): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F2-Q4

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 35 • FORWARD 2 • Q4/5

1179

QUESTION ID
S35-F2-Q4

TYPE
Short answer

CORRECT ANSWER
0.2

GRADING NOTE 0.1/0.5 ≠ 0.4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S35-F2-Q4**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 35 • FORWARD 2 • Q4/5

True or false?

If P(A given B) = 0.2 and P(A) = 0.4, then A and B are independent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F2-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 35 • FORWARD 2 • Q5/5

1180

QUESTION ID

S35-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Not equal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F2-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 35 • FORWARD 2 • Q5/5

Interpret independence through conditional probability

MAIN	Probability & Statistics main question	**HSS-CP.A.3**
BACKWARD 1	Compute a conditional probability	**HSS-CP.A.3**
BACKWARD 2	Test independence with the product rule	**HSS-CP.A.2**
FORWARD 1	Check both directions of independence	**HSS-CP.A.3**
FORWARD 2	Approximate conditional probabilities from data	**HSS-CP.A.4**

SET 36 • HSS-CP.A.3

MAIN

Probability & Statistics main question • High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3

ANSWER KEY Reduced copy of the question page, with its answer

P(A) = 0.5, P(B) = 0.4 and P(A and B) = 0.2. Find P(A given B), and decide whether A and B are independent (enter yes or no).

P(A given B): _____

Independent (yes or no): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-M-Q1

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 36 • MAIN • Q1/5

1183

QUESTION ID

S36-M-Q1

TYPE

Short answer

CORRECT ANSWER

0.5; yes

GRADING NOTE Form required: both. 0.2/0.4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S36-M-Q1**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 36 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

P(A) = 0.3, P(B) = 0.5 and P(A and B) = 0.1. Find P(A given B), and decide whether A and B are independent (enter yes or no).

P(A given B): _____

Independent (yes or no): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-M-Q2

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 36 • MAIN • Q2/5

1184

QUESTION ID

S36-M-Q2

TYPE

Short answer

CORRECT ANSWER

0.2; no

GRADING NOTE Form required: both. 0.1/0.5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S36-M-Q2**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 36 • MAIN • Q2/5

A and B are independent. Which statement is correct?

A. Whenever B happens, then A must also happen.

B. P(A) and P(B) must be equal to each other.

C. A and B can never happen at the same time as each other.

D. Knowing that B happened does not change the probability of A.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-M-Q3

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 36 • MAIN • Q3/5

1185

QUESTION ID

S36-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. Knowing that B happened does not change the probability of A.

GRADING NOTE Meaning of independence.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S36-M-Q3**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 36 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Two cards are drawn at random from a standard 52-card deck. Find P(second is an ace given the first is an ace) when the first card is replaced, and when it is not replaced. Give a fraction in lowest terms.

With replacement: _____

Without replacement: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-M-Q4

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 36 • MAIN • Q4/5

1186

QUESTION ID

S36-M-Q4

TYPE

Short answer

CORRECT ANSWER

1/13; 1/17

GRADING NOTE Form required: both. 4/52 and 3/51.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S36-M-Q4**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 36 • MAIN • Q4/5

Which pair of events is independent?

A. a student studies; the student passes the test

B. first card drawn is an ace; second card (not replaced) is an ace

C. it rains today; the ground is wet today

D. a die shows 6; a separate coin shows heads

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-M-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 36 • MAIN • Q5/5

1187

QUESTION ID

S36-M-Q5

TYPE

Multiple choice

CORRECT ANSWER

D. a die shows 6; a separate coin shows heads

GRADING NOTE Separate devices do not affect each other.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S36-M-Q5**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 36 • MAIN • Q5/5

SET 36 • HSS-CP.A.3

BACKWARD 1

Compute a conditional probability • High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3

ANSWER KEY Reduced copy of the question page, with its answer

P(A and B) = 0.06 and P(B) = 0.3. Find P(A given B).

P(A given B): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B1-Q1

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 36 • BACKWARD 1 • Q1/5

1189

QUESTION ID

S36-B1-Q1

TYPE

Short answer

CORRECT ANSWER

0.2

GRADING NOTE 0.06/0.3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S36-B1-Q1**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 36 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

P(A and B) = 0.25 and P(A) = 0.5. Find P(B given A).

P(B given A): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B1-Q2

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 36 • BACKWARD 1 • Q2/5

1190

QUESTION ID

S36-B1-Q2

TYPE

Short answer

CORRECT ANSWER

0.5

GRADING NOTE 0.25/0.5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S36-B1-Q2**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 36 • BACKWARD 1 • Q2/5

True or false?

P(A given B) = P(A and B)/P(B).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B1-Q3

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 36 • BACKWARD 1 • Q3/5

1191

QUESTION ID

S36-B1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B1-Q3

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 36 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

P(A and B) = 0.1, P(B) = 0.4. Find P(A given B).

A. 0.4

B. 0.25

C. 4

D. 0.04

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B1-Q4

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 36 • BACKWARD 1 • Q4/5

1192

QUESTION ID

S36-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 0.25

GRADING NOTE 0.1/0.4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S36-B1-Q4**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 36 • BACKWARD 1 • Q4/5

True or false?

P(A given B) = P(A and B)·P(B).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B1-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 36 • BACKWARD 1 • Q5/5

1193

QUESTION ID

S36-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Divide, not multiply.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B1-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 36 • BACKWARD 1 • Q5/5

SET 36 • HSS-CP.A.3

BACKWARD 2

Test independence with the product rule • High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.2

True or false?

If P(A) = 0.5, P(B) = 0.4 and P(A and B) = 0.2, A and B are independent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B2-Q1

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.2 • Set 36 • BACKWARD 2 • Q1/5

1195

QUESTION ID

S36-B2-Q1

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $0.5 \cdot 0.4 = 0.2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B2-Q1

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.2 • Set 36 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

P(A) = 0.2 and P(B) = 0.5. If A and B are independent, what is P(A and B)?

P(A and B): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B2-Q2

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.2 • Set 36 • BACKWARD 2 • Q2/5

1196

QUESTION ID

S36-B2-Q2

TYPE

Short answer

CORRECT ANSWER

0.1

GRADING NOTE Product.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S36-B2-Q2**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.2 • Set 36 • BACKWARD 2 • Q2/5

P(A) = 0.3, P(B) = 0.4, P(A and B) = 0.2. Are A and B independent?

A. no, because 0.2 > 0

B. yes, because 0.3 < 0.4

C. yes, because 0.2 < 1

D. no, because 0.3 · 0.4 = 0.12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B2-Q3

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.2 • Set 36 • BACKWARD 2 • Q3/5

1197

QUESTION ID

S36-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. no, because 0.3 · 0.4 = 0.12

GRADING NOTE 0.12 ≠ 0.2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S36-B2-Q3**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.2 • Set 36 • BACKWARD 2 • Q3/5

True or false?

Mutually exclusive events with positive probabilities are independent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B2-Q4

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.2 • Set 36 • BACKWARD 2 • Q4/5

1198

QUESTION ID

S36-B2-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $P(A \text{ and } B) = 0 \neq P(A)\cdot P(B)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B2-Q4

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.2 • Set 36 • BACKWARD 2 • Q4/5

P(A) = 0.6 and P(B) = 0.5 for independent events. Find P(A and B).

P(A and B): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B2-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.2 • Set 36 • BACKWARD 2 • Q5/5

1199

QUESTION ID

S36-B2-Q5

TYPE

Short answer

CORRECT ANSWER

0.3

GRADING NOTE 0.6 · 0.5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B2-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.2 • Set 36 • BACKWARD 2 • Q5/5

SET 36 • HSS-CP.A.3

FORWARD 1

Check both directions of independence • High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3

ANSWER KEY Reduced copy of the question page, with its answer

P(A) = 0.6, P(B) = 0.25 and P(A and B) = 0.15. Find P(A given B) and P(B given A).

P(A given B): _____

P(B given A): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F1-Q1

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 36 • FORWARD 1 • Q1/5

1201

QUESTION ID

S36-F1-Q1

TYPE

Short answer

CORRECT ANSWER

0.6; 0.25

GRADING NOTE Form required: both. Both equal the unconditional values.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S36-F1-Q1**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 36 • FORWARD 1 • Q1/5

True or false?

If P(A given B) = P(A), then A and B are independent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F1-Q2

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 36 • FORWARD 1 • Q2/5

1202

QUESTION ID

S36-F1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Definition (when P(B) > 0).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F1-Q2

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 36 • FORWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

P(A) = 0.5, P(A given B) = 0.5, and P(B) = 0.3. What is P(B given A)?

A. 0.15

B. 0.6

C. 0.5

D. 0.3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F1-Q3

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 36 • FORWARD 1 • Q3/5

1203

QUESTION ID

S36-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 0.3

GRADING NOTE Independence works both ways.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S36-F1-Q3**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 36 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A and B are independent, P(A) = 0.7 and P(B) = 0.4. Find P(A given B).

P(A given B): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F1-Q4

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 36 • FORWARD 1 • Q4/5

1204

QUESTION ID

S36-F1-Q4

TYPE

Short answer

CORRECT ANSWER

0.7

GRADING NOTE Independence.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S36-F1-Q4**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 36 • FORWARD 1 • Q4/5

True or false?

If P(B given A) = 0.25 and P(B) = 0.5, then A and B are independent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F1-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 36 • FORWARD 1 • Q5/5

1205

QUESTION ID

S36-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Not equal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F1-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 36 • FORWARD 1 • Q5/5

SET 36 • HSS-CP.A.3

FORWARD 2

Approximate conditional probabilities from data • High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4

ANSWER KEY Reduced copy of the question page, with its answer

In the survey table, a person is chosen at random. Find P(yes given man) and P(yes).

	Yes	No	Total
Men	24	16	40
Women	36	24	60
Total	60	40	100

P(yes given man): _____

P(yes): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F2-Q1

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 36 • FORWARD 2 • Q1/5

1207

QUESTION ID

S36-F2-Q1

TYPE

Short answer

CORRECT ANSWER

0.6; 0.6

GRADING NOTE Form required: both. 24/40 and 60/100.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S36-F2-Q1**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 36 • FORWARD 2 • Q1/5

True or false?

In the survey table, gender and answer are independent.

	Yes	No	Total
Men	24	16	40
Women	36	24	60
Total	60	40	100

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F2-Q2

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 36 • FORWARD 2 • Q2/5

1208

QUESTION ID

S36-F2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE P(yes given man) = P(yes).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F2-Q2

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 36 • FORWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

In the driving table, find P(drives given junior) and P(drives).

	Drives	Does not drive	Total
Juniors	60	60	120
Seniors	30	50	80
Total	90	110	200

P(drives given junior): _____

P(drives): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F2-Q3

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 36 • FORWARD 2 • Q3/5

1209

QUESTION ID

S36-F2-Q3

TYPE

Short answer

CORRECT ANSWER

0.5; 0.45

GRADING NOTE Form required: both. 60/120 and 90/200.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S36-F2-Q3**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 36 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

In the driving table, grade and driving are independent.

	Drives	Does not drive	Total
Juniors	60	60	120
Seniors	30	50	80
Total	90	110	200

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F2-Q4

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 36 • FORWARD 2 • Q4/5

1210

QUESTION ID

S36-F2-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $0.5 \neq 0.45$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F2-Q4

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 36 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

In the driving table, which is P(junior given drives)?

	Drives	Does not drive	Total
Juniors	60	60	120
Seniors	30	50	80
Total	90	110	200

A. 1/2 **B.** 3/10 **C.** 2/3 **D.** 9/20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F2-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 36 • FORWARD 2 • Q5/5

1211

QUESTION ID
S36-F2-Q5

TYPE
Multiple choice

CORRECT ANSWER
C. 2/3

GRADING NOTE 60/90. 1/2 reverses the condition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F2-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 36 • FORWARD 2 • Q5/5

Construct two-way frequency tables

MAIN	Probability & Statistics main question	**HSS-CP.A.4**
BACKWARD 1	Fill in a two-way frequency table	**8.SP.A.4**
BACKWARD 2	Read joint and marginal frequencies	**HSS-ID.B.5**
FORWARD 1	Check statements against a built table	**HSS-CP.A.4**
FORWARD 2	Use the table as a sample space	**HSS-CP.A.4**

SET 37 • HSS-CP.A.4

MAIN

Probability & Statistics main question • High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4

ANSWER KEY Reduced copy of the question page, with its answer

Of 200 students, 120 are juniors and 80 are seniors. 90 students drive to school, including 60 juniors. Complete the two-way table: juniors who do not drive, seniors who drive, seniors who do not drive.

Juniors, do not drive: _____

Seniors, drive: _____

Seniors, do not drive: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-M-Q1

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 37 • MAIN • Q1/5

1214

QUESTION ID

S37-M-Q1

TYPE

Short answer

CORRECT ANSWER

60; 30; 50

GRADING NOTE Form required: all three. 120 − 60; 90 − 60; 80 − 30.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S37-M-Q1**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 37 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Of 150 adults, 90 own a car, 40 own a bike, and 25 own both. In a two-way table (car yes/no by bike yes/no), how many own a car but no bike, and how many own neither?

Car, no bike: _____

Neither: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-M-Q2

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 37 • MAIN • Q2/5

1215

QUESTION ID
S37-M-Q2

TYPE
Short answer

CORRECT ANSWER
65; 45

GRADING NOTE Form required: both. 90 − 25; 150 − (90 + 40 − 25).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S37-M-Q2**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 37 • MAIN • Q2/5

Of 200 students, 120 are juniors, and 60 of the juniors drive. What percent of juniors drive?

Percent: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-M-Q3

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 37 • MAIN • Q3/5

1216

QUESTION ID

S37-M-Q3

TYPE

Short answer

CORRECT ANSWER

50%

GRADING NOTE 60/120.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S37-M-Q3**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 37 • MAIN • Q3/5

80 patients took part in a study. Which description matches the table?

	Improved	Did not	Total
Drug	35	15	50
Placebo	12	18	30
Total	47	33	80

A. 50 patients got the placebo and the other 30 got the drug.

B. 50 patients improved, and 35 of those improved patients got the drug.

C. 50 got the drug (35 improved); 12 of the placebo patients improved.

D. 35 patients got the drug and 12 patients got a placebo instead.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-M-Q4

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 37 • MAIN • Q4/5

1217

QUESTION ID

S37-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 50 got the drug (35 improved); 12 of the placebo patients improved.

GRADING NOTE Drug row 35 + 15 = 50; placebo row 12 + 18 = 30. 47 improved, not 50.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-M-Q4

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 37 • MAIN • Q4/5

80 patients: 50 got a drug and 30 a placebo; 35 drug patients and 12 placebo patients improved. How many drug patients and how many placebo patients did not improve?

Drug, did not: _____

Placebo, did not: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-M-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 37 • MAIN • Q5/5

1218

QUESTION ID

S37-M-Q5

TYPE

Short answer

CORRECT ANSWER

15; 18

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-M-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 37 • MAIN • Q5/5

SET 37 • HSS-CP.A.4

BACKWARD 1

Fill in a two-way frequency table • Grade 8 • Statistics & Probability • 8.SP.A.4

ANSWER KEY Reduced copy of the question page, with its answer

A table has row totals 40 and 60. One cell in the first row is 25. What is the other cell in that row?

Cell: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B1-Q1

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 37 • BACKWARD 1 • Q1/5

1220

QUESTION ID

S37-B1-Q1

TYPE

Short answer

CORRECT ANSWER

15

GRADING NOTE 40 − 25.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B1-Q1

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 37 • BACKWARD 1 • Q1/5

True or false?

In a two-way table, the column totals add to the grand total.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B1-Q2

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 37 • BACKWARD 1 • Q2/5

1221

QUESTION ID

S37-B1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Every object counted once.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B1-Q2

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 37 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Of 50 people, 30 are adults. 12 of the adults wear glasses. How many adults do not?

A. 18

B. 12

C. 38

D. 20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B1-Q3

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 37 • BACKWARD 1 • Q3/5

1222

GRADING NOTE 30 − 12.

QUESTION ID

S37-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 18

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B1-Q3

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 37 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A two-way table with two columns classifies 100 objects. One column total is 45. What is the other column total?

Total: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B1-Q4

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 37 • BACKWARD 1 • Q4/5

1223

QUESTION ID

S37-B1-Q4

TYPE

Short answer

CORRECT ANSWER

55

GRADING NOTE 100 − 45.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B1-Q4

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 37 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

A cell can be found from its row total and the other cells in that row.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B1-Q5

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 37 • BACKWARD 1 • Q5/5

1224

QUESTION ID

S37-B1-Q5

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Subtraction.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B1-Q5

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 37 • BACKWARD 1 • Q5/5

SET 37 • HSS-CP.A.4

BACKWARD 2

Read joint and marginal frequencies • High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.5

ANSWER KEY Reduced copy of the question page, with its answer

In the drug table, how many patients improved in total?

	Improved	Did not	Total
Drug	35	15	50
Placebo	12	18	30
Total	47	33	80

Improved: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B2-Q1

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.5 • Set 37 • BACKWARD 2 • Q1/5

1226

QUESTION ID

S37-B2-Q1

TYPE

Short answer

CORRECT ANSWER

47

GRADING NOTE 35 + 12.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S37-B2-Q1**

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.5 • Set 37 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

In the drug table, what fraction of all patients got the drug and improved? Give lowest terms.

	Improved	Did not	Total
Drug	35	15	50
Placebo	12	18	30
Total	47	33	80

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B2-Q2

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.5 • Set 37 • BACKWARD 2 • Q2/5

1227

QUESTION ID

S37-B2-Q2

TYPE

Short answer

CORRECT ANSWER

7/16

GRADING NOTE 35/80.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S37-B2-Q2**

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.5 • Set 37 • BACKWARD 2 • Q2/5

True or false?

In the drug table, 30 patients got the placebo.

	Improved	Did not	Total
Drug	35	15	50
Placebo	12	18	30
Total	47	33	80

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B2-Q3

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.5 • Set 37 • BACKWARD 2 • Q3/5

1228

QUESTION ID

S37-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE 12 + 18.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B2-Q3

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.5 • Set 37 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

In the drug table, which number is a marginal total?

	Improved	Did not	Total
Drug	35	15	50
Placebo	12	18	30
Total	47	33	80

A. 12 **B.** 50 **C.** 35 **D.** 15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B2-Q4

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.5 • Set 37 • BACKWARD 2 • Q4/5

1229

QUESTION ID

S37-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 50

GRADING NOTE The drug row total.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S37-B2-Q4**

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.5 • Set 37 • BACKWARD 2 • Q4/5

True or false?

In the drug table, 35/47 is the fraction of drug patients who improved.

	Improved	Did not	Total
Drug	35	15	50
Placebo	12	18	30
Total	47	33	80

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B2-Q5

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.5 • Set 37 • BACKWARD 2 • Q5/5

1230

QUESTION ID

S37-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 35/50; 35/47 is the fraction of improvers who got the drug.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B2-Q5

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.5 • Set 37 • BACKWARD 2 • Q5/5

SET 37 • HSS-CP.A.4

FORWARD 1

Check statements against a built table • High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4

True or false?

In the drug table, a greater percent of drug patients than placebo patients improved.

	Improved	Did not	Total
Drug	35	15	50
Placebo	12	18	30
Total	47	33	80

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F1-Q1

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 37 • FORWARD 1 • Q1/5

1232

QUESTION ID

S37-F1-Q1

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE 70% versus 40%.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S37-F1-Q1**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 37 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

In the drug table, what percent of drug patients improved, and what percent of placebo patients improved?

	Improved	Did not	Total
Drug	35	15	50
Placebo	12	18	30
Total	47	33	80

Drug: _____

Placebo: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F1-Q2

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 37 • FORWARD 1 • Q2/5

1233

QUESTION ID

S37-F1-Q2

TYPE

Short answer

CORRECT ANSWER

70%; 40%

GRADING NOTE Form required: both. 35/50 and 12/30.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S37-F1-Q2**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 37 • FORWARD 1 • Q2/5

A report says "most patients who improved got the drug." Using the drug table, is this correct?

	Improved	Did not	Total
Drug	35	15	50
Placebo	12	18	30
Total	47	33	80

A. No: only 35 patients got the drug.

B. Yes: 70% of all patients improved.

C. Yes: 35 of the 47 who improved got the drug.

D. No: 50 patients got the drug.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F1-Q3

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 37 • FORWARD 1 • Q3/5

1234

QUESTION ID

S37-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. Yes: 35 of the 47 who improved got the drug.

GRADING NOTE 35/47 ≈ 74%. Only 47 of 80 improved (58.75%).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F1-Q3

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 37 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Of 120 voters, 70 are under 40. 45 of all voters support a plan, including 30 of those under 40. How many voters 40 or older do not support it?

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F1-Q4

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 37 • FORWARD 1 • Q4/5

1235

QUESTION ID

S37-F1-Q4

TYPE

Short answer

CORRECT ANSWER

35

GRADING NOTE 50 older; 15 support; 35 do not.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S37-F1-Q4**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 37 • FORWARD 1 • Q4/5

True or false?

A two-way table can be built when each object is classified by only one category.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F1-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 37 • FORWARD 1 • Q5/5

1236

QUESTION ID

S37-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It needs two categorical variables.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S37-F1-Q5**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 37 • FORWARD 1 • Q5/5

SET 37 • HSS-CP.A.4

FORWARD 2

Use the table as a sample space • High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4

ANSWER KEY Reduced copy of the question page, with its answer

A patient is chosen at random from the drug table. What is P(improved)? Give a fraction in lowest terms.

	Improved	Did not	Total
Drug	35	15	50
Placebo	12	18	30
Total	47	33	80

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F2-Q1

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 37 • FORWARD 2 • Q1/5

1238

QUESTION ID

S37-F2-Q1

TYPE

Short answer

CORRECT ANSWER

47/80

GRADING NOTE 47/80.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S37-F2-Q1**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 37 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A patient is chosen at random from the drug table. What is P(improved given drug)? Give a fraction in lowest terms.

	Improved	Did not	Total
Drug	35	15	50
Placebo	12	18	30
Total	47	33	80

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F2-Q2

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 37 • FORWARD 2 • Q2/5

1239

QUESTION ID

S37-F2-Q2

TYPE

Short answer

CORRECT ANSWER

7/10

GRADING NOTE 35/50.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S37-F2-Q2**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 37 • FORWARD 2 • Q2/5

True or false?

In the drug table, improving and treatment are independent.

	Improved	Did not	Total
Drug	35	15	50
Placebo	12	18	30
Total	47	33	80

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F2-Q3

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 37 • FORWARD 2 • Q3/5

1240

QUESTION ID

S37-F2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $7/10 \neq 47/80$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F2-Q3

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 37 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A patient is chosen at random from the drug table. What is P(drug given improved)?

	Improved	Did not	Total
Drug	35	15	50
Placebo	12	18	30
Total	47	33	80

A. 47/80 **B.** 35/47 **C.** 7/16 **D.** 35/50

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F2-Q4

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 37 • FORWARD 2 • Q4/5

1241

QUESTION ID
S37-F2-Q4

TYPE
Multiple choice

CORRECT ANSWER
B. 35/47

GRADING NOTE Condition on the improved column.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S37-F2-Q4**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 37 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A patient is chosen at random from the drug table. What is P(placebo and did not improve)? Give a fraction in lowest terms.

	Improved	Did not	Total
Drug	35	15	50
Placebo	12	18	30
Total	47	33	80

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F2-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 37 • FORWARD 2 • Q5/5

1242

QUESTION ID

S37-F2-Q5

TYPE

Short answer

CORRECT ANSWER

9/40

GRADING NOTE 18/80.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S37-F2-Q5**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 37 • FORWARD 2 • Q5/5

Use two-way tables to find conditional probabilities and judge independence

MAIN	Probability & Statistics main question	**HSS-CP.A.4**
BACKWARD 1	Compute conditional relative frequencies	**HSS-ID.B.5**
BACKWARD 2	Compare P(A given B) with P(A)	**HSS-CP.A.3**
FORWARD 1	Use larger two-way tables	**HSS-CP.A.4**
FORWARD 2	Explain table results in everyday language	**HSS-CP.A.5**

SET 38 • HSS-CP.A.4

MAIN

Probability & Statistics main question • High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4

A student is chosen at random from the driving table. Find P(drives given junior) and P(drives given senior). Give decimals.

	Drives	Does not drive	Total
Juniors	60	60	120
Seniors	30	50	80
Total	90	110	200

Given junior: _____

Given senior: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-M-Q1

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 38 • MAIN • Q1/5

1245

QUESTION ID

S38-M-Q1

TYPE

Short answer

CORRECT ANSWER

0.5; 0.375

GRADING NOTE Form required: both. 60/120 and 30/80.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-M-Q1

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 38 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

In the driving table, are "drives" and "junior" independent?

	Drives	Does not drive	Total
Juniors	60	60	120
Seniors	30	50	80
Total	90	110	200

A. No: P(drives given junior) ≠ P(drives given senior).

B. Yes: every student is a junior or a senior.

C. Yes: 60 juniors drive and 60 do not.

D. No: more juniors than seniors were surveyed.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-M-Q2

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 38 • MAIN • Q2/5

1246

QUESTION ID

S38-M-Q2

TYPE

Multiple choice

CORRECT ANSWER

A. No: P(drives given junior) ≠ P(drives given senior).

GRADING NOTE 0.5 versus 0.375. Group sizes alone do not decide.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-M-Q2

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 38 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

A student is chosen at random from the sports table. Find P(plays a sport) and P(plays a sport given 9th grade). Give decimals.

	Plays a sport	Does not	Total
9th grade	30	20	50
10th grade	42	28	70
Total	72	48	120

P(sport): _____

P(sport given 9th): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-M-Q3

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 38 • MAIN • Q3/5

1247

QUESTION ID

S38-M-Q3

TYPE

Short answer

CORRECT ANSWER

0.6; 0.6

GRADING NOTE Form required: both. 72/120 and 30/50.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-M-Q3

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 38 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

In the sports table, are "plays a sport" and "9th grade" independent?

	Plays a sport	Does not	Total
9th grade	30	20	50
10th grade	42	28	70
Total	72	48	120

A. Yes: P(sport given 9th) = P(sport) = 0.6.

B. No: 30 is not equal to 42.

C. No: the grades have different sizes.

D. Yes: more students play than do not.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-M-Q4

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 38 • MAIN • Q4/5

1248

QUESTION ID

S38-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. Yes: P(sport given 9th) = P(sport) = 0.6.

GRADING NOTE Compare rates, not counts.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-M-Q4

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 38 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A student who drives is chosen at random from the driving table. What is P(junior given drives)? Give a fraction in lowest terms.

	Drives	Does not drive	Total
Juniors	60	60	120
Seniors	30	50	80
Total	90	110	200

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-M-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 38 • MAIN • Q5/5

1249

QUESTION ID

S38-M-Q5

TYPE

Short answer

CORRECT ANSWER

2/3

GRADING NOTE 60/90.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S38-M-Q5**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 38 • MAIN • Q5/5

SET 38 • HSS-CP.A.4

BACKWARD 1

Compute conditional relative frequencies • High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.5

ANSWER KEY Reduced copy of the question page, with its answer

In the driving table, what fraction of seniors drive? Give lowest terms.

	Drives	Does not drive	Total
Juniors	60	60	120
Seniors	30	50	80
Total	90	110	200

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B1-Q1

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.5 • Set 38 • BACKWARD 1 • Q1/5

1251

QUESTION ID

S38-B1-Q1

TYPE

Short answer

CORRECT ANSWER

3/8

GRADING NOTE 30/80.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S38-B1-Q1**

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.5 • Set 38 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

In the sports table, 42/70 of 10th graders play a sport.

	Plays a sport	Does not	Total
9th grade	30	20	50
10th grade	42	28	70
Total	72	48	120

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B1-Q2

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.5 • Set 38 • BACKWARD 1 • Q2/5

1252

QUESTION ID

S38-B1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Row 10th.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B1-Q2

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.5 • Set 38 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

In the sports table, what fraction of students who play a sport are in 10th grade?

	Plays a sport	Does not	Total
9th grade	30	20	50
10th grade	42	28	70
Total	72	48	120

A. 5/12 **B.** 7/12 **C.** 7/20 **D.** 3/5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B1-Q3

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.5 • Set 38 • BACKWARD 1 • Q3/5

1253

QUESTION ID

S38-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 7/12

GRADING NOTE 42/72.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B1-Q3

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.5 • Set 38 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

In the driving table, what percent of all students drive?

	Drives	Does not drive	Total
Juniors	60	60	120
Seniors	30	50	80
Total	90	110	200

Percent: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B1-Q4

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.5 • Set 38 • BACKWARD 1 • Q4/5

1254

QUESTION ID

S38-B1-Q4

TYPE

Short answer

CORRECT ANSWER

45%

GRADING NOTE 90/200.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S38-B1-Q4**

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.5 • Set 38 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

A row conditional frequency divides by the grand total.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B1-Q5

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.5 • Set 38 • BACKWARD 1 • Q5/5

1255

QUESTION ID
S38-B1-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE By the row total.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B1-Q5

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.5 • Set 38 • BACKWARD 1 • Q5/5

SET 38 • HSS-CP.A.4

BACKWARD 2

Compare P(A given B) with P(A) • High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3

True or false?

If P(A given B) = P(A), A and B are independent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B2-Q1

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 38 • BACKWARD 2 • Q1/5

1257

QUESTION ID

S38-B2-Q1

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B2-Q1

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 38 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

P(A) = 0.45 and P(A given B) = 0.5. Are A and B independent?

A. only if P(B) = 0.5

B. yes

C. no

D. cannot tell

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B2-Q2

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 38 • BACKWARD 2 • Q2/5

1258

QUESTION ID

S38-B2-Q2

TYPE

Multiple choice

CORRECT ANSWER

C. no

GRADING NOTE Not equal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S38-B2-Q2**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 38 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

P(A and B) = 0.3 and P(B) = 0.5. Find P(A given B).

P(A given B): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B2-Q3

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 38 • BACKWARD 2 • Q3/5

1259

QUESTION ID

S38-B2-Q3

TYPE

Short answer

CORRECT ANSWER

0.6

GRADING NOTE 0.3/0.5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S38-B2-Q3**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 38 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

P(A given B) is always less than P(A).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B2-Q4

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 38 • BACKWARD 2 • Q4/5

1260

QUESTION ID

S38-B2-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It can be greater, equal or less.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B2-Q4

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 38 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

P(A and B) = 0.18 and P(B) = 0.3. If P(A) = 0.6, are A and B independent?

A. yes

B. only if P(A) = 0.3

C. cannot tell

D. no

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B2-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 38 • BACKWARD 2 • Q5/5

1261

QUESTION ID

S38-B2-Q5

TYPE

Multiple choice

CORRECT ANSWER

A. yes

GRADING NOTE 0.18/0.3 = 0.6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S38-B2-Q5**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 38 • BACKWARD 2 • Q5/5

SET 38 • HSS-CP.A.4

FORWARD 1

Use larger two-way tables • High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4

ANSWER KEY Reduced copy of the question page, with its answer

A student is chosen at random from the music table. Find P(rock) and P(rock given high school). Give decimals.

	Pop	Rock	Other	Total
Middle school	16	24	10	50
High school	24	36	15	75
Total	40	60	25	125

P(rock): _____

P(rock given high school): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F1-Q1

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 38 • FORWARD 1 • Q1/5

1263

QUESTION ID

S38-F1-Q1

TYPE

Short answer

CORRECT ANSWER

0.48; 0.48

GRADING NOTE Form required: both. 60/125 and 36/75.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S38-F1-Q1**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 38 • FORWARD 1 • Q1/5

True or false?

In the music table, school level and favorite type of music are independent.

	Pop	Rock	Other	Total
Middle school	16	24	10	50
High school	24	36	15	75
Total	40	60	25	125

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F1-Q2

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 38 • FORWARD 1 • Q2/5

1264

QUESTION ID

S38-F1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Each row has the same proportions 0.32, 0.48, 0.2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F1-Q2

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 38 • FORWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

In the music table, what is P(middle school given pop)? Give a fraction in lowest terms.

	Pop	Rock	Other	Total
Middle school	16	24	10	50
High school	24	36	15	75
Total	40	60	25	125

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F1-Q3

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 38 • FORWARD 1 • Q3/5

1265

QUESTION ID

S38-F1-Q3

TYPE

Short answer

CORRECT ANSWER

2/5

GRADING NOTE 16/40.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S38-F1-Q3**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 38 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

In the music table, what is P(other given middle school)?

	Pop	Rock	Other	Total
Middle school	16	24	10	50
High school	24	36	15	75
Total	40	60	25	125

A. 1/5 **B.** 2/25 **C.** 2/5 **D.** 1/3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F1-Q4

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 38 • FORWARD 1 • Q4/5

1266

QUESTION ID

S38-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 1/5

GRADING NOTE 10/50.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F1-Q4

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 38 • FORWARD 1 • Q4/5

True or false?

In the music table, P(pop given middle school) is greater than P(pop given high school).

	Pop	Rock	Other	Total
Middle school	16	24	10	50
High school	24	36	15	75
Total	40	60	25	125

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F1-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 38 • FORWARD 1 • Q5/5

1267

QUESTION ID

S38-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Both 0.32.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F1-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 38 • FORWARD 1 • Q5/5

SET 38 • HSS-CP.A.4

FORWARD 2

Explain table results in everyday language • High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.5

In the driving table, P(drives given junior) = 0.5 and P(drives given senior) = 0.375. Which sentence says this in everyday language?

	Drives	Does not drive	Total
Juniors	60	60	120
Seniors	30	50	80
Total	90	110	200

A. Most of the students who drive to school are seniors.

B. Half of the juniors drive, but fewer than 4 in 10 seniors drive.

C. Juniors and seniors are equally likely to drive.

D. Half of all the students who drive are juniors.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F2-Q1

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.5 • Set 38 • FORWARD 2 • Q1/5

1269

QUESTION ID

S38-F2-Q1

TYPE

Multiple choice

CORRECT ANSWER

B. Half of the juniors drive, but fewer than 4 in 10 seniors drive.

GRADING NOTE Condition on grade.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F2-Q1

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.5 • Set 38 • FORWARD 2 • Q1/5

True or false?

"Among students who drive, two-thirds are juniors" describes P(junior given drives).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F2-Q2

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.5 • Set 38 • FORWARD 2 • Q2/5

1270

QUESTION ID

S38-F2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE The condition is "drives".

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F2-Q2

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.5 • Set 38 • FORWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Which statement describes independence in the sports table?

	Plays a sport	Does not	Total
9th grade	30	20	50
10th grade	42	28	70
Total	72	48	120

A. A student's grade does not change the chance of playing a sport.

B. Every student in the table plays exactly one sport.

C. The two grades have the same number of students.

D. Students in 10th grade play more sports than 9th graders.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F2-Q3

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.5 • Set 38 • FORWARD 2 • Q3/5

1271

QUESTION ID

S38-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. A student's grade does not change the chance of playing a sport.

GRADING NOTE Same rate in each grade.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F2-Q3

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.5 • Set 38 • FORWARD 2 • Q3/5

True or false?

"Most juniors drive" and "most drivers are juniors" mean the same thing.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F2-Q4

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.5 • Set 38 • FORWARD 2 • Q4/5

1272

QUESTION ID

S38-F2-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Different conditions.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F2-Q4

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.5 • Set 38 • FORWARD 2 • Q4/5

"Among seniors, 3 in 8 drive." Which probability is this?

A. P(senior given drives)

B. P(drives given senior)

C. P(senior and drives)

D. P(senior)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F2-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.5 • Set 38 • FORWARD 2 • Q5/5

1273

QUESTION ID

S38-F2-Q5

TYPE

Multiple choice

CORRECT ANSWER

B. P(drives given senior)

GRADING NOTE Condition: senior.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F2-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.5 • Set 38 • FORWARD 2 • Q5/5

Conditional probability and independence in everyday language

MAIN	Probability & Statistics main question	**HSS-CP.A.5**
BACKWARD 1	Compute and read conditional probabilities	**HSS-CP.A.3**
BACKWARD 2	Read conditional rates in a table	**HSS-CP.A.4**
FORWARD 1	Translate statements and compute	**HSS-CP.A.5**
FORWARD 2	Find a conditional probability as a fraction of outcomes	**HSS-CP.B.6**

SET 39 • HSS-CP.A.5

MAIN

Probability & Statistics main question • High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.5

Which statement describes a conditional probability?

A. One in four cards in a deck is a heart.

B. Among smokers, the chance of lung disease is higher.

C. A fair coin lands heads half of the time.

D. The chance of rain tomorrow is 30 percent.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-M-Q1

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.5 • Set 39 • MAIN • Q1/5

1276

QUESTION ID

S39-M-Q1

TYPE

Multiple choice

CORRECT ANSWER

B. Among smokers, the chance of lung disease is higher.

GRADING NOTE "Among smokers" names the condition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S39-M-Q1**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.5 • Set 39 • MAIN • Q1/5

"Most people with lung disease were smokers." Which probability does this sentence say is high?

A. P(lung disease)

B. P(smoker given lung disease)

C. P(smoker and lung disease)

D. P(lung disease given smoker)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-M-Q2

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.5 • Set 39 • MAIN • Q2/5

1277

QUESTION ID

S39-M-Q2

TYPE

Multiple choice

CORRECT ANSWER

B. P(smoker given lung disease)

GRADING NOTE The condition is "people with lung disease".

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S39-M-Q2**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.5 • Set 39 • MAIN • Q2/5

Which pair of events is most likely to be independent?

A. heavy rain today; a baseball game being canceled

B. a student's shirt color; the school bus being late

C. studying hard for a test; passing that test

D. a person's height; the person's shoe size

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-M-Q3

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.5 • Set 39 • MAIN • Q3/5

1278

QUESTION ID

S39-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. a student's shirt color; the school bus being late

GRADING NOTE No plausible influence.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S39-M-Q3**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.5 • Set 39 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A fair coin has landed heads 5 times in a row. What is the probability of heads on the next flip?

A. 1/32

B. 1/2

C. less than 1/2

D. more than 1/2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-M-Q4

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.5 • Set 39 • MAIN • Q4/5

1279

QUESTION ID

S39-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 1/2

GRADING NOTE Flips are independent; 1/32 is P(5 heads in a row) before flipping.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S39-M-Q4**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.5 • Set 39 • MAIN • Q4/5

In a town, 40% of adults own a dog, and among dog owners 70% walk daily. What percent of all adults are dog owners who walk daily?

Percent: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-M-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.5 • Set 39 • MAIN • Q5/5

1280

QUESTION ID

S39-M-Q5

TYPE

Short answer

CORRECT ANSWER

28%

GRADING NOTE 0.4 · 0.7.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S39-M-Q5**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.5 • Set 39 • MAIN • Q5/5

SET 39 • HSS-CP.A.5

BACKWARD 1

Compute and read conditional probabilities • High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3

ANSWER KEY Reduced copy of the question page, with its answer

P(A and B) = 0.28 and P(A) = 0.4. Find P(B given A).

P(B given A): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B1-Q1

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 39 • BACKWARD 1 • Q1/5

1282

QUESTION ID

S39-B1-Q1

TYPE

Short answer

CORRECT ANSWER

0.7

GRADING NOTE 0.28/0.4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S39-B1-Q1**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 39 • BACKWARD 1 • Q1/5

True or false?

"The probability of B given A" means the chance of B when A is known to happen.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B1-Q2

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 39 • BACKWARD 1 • Q2/5

1283

QUESTION ID

S39-B1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Meaning.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B1-Q2

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 39 • BACKWARD 1 • Q2/5

P(passing given studying) is the probability that

A. a student passes

B. a student passes, among students who study

C. a student studies, among students who pass

D. a student both studies and passes

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B1-Q3

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 39 • BACKWARD 1 • Q3/5

1284

QUESTION ID

S39-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. a student passes, among students who study

GRADING NOTE Condition: studying.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B1-Q3

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 39 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

P(B) = 0.5 and P(A given B) = 0.3. Find P(A and B).

P(A and B): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B1-Q4

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 39 • BACKWARD 1 • Q4/5

1285

QUESTION ID

S39-B1-Q4

TYPE

Short answer

CORRECT ANSWER

0.15

GRADING NOTE 0.5 · 0.3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S39-B1-Q4**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 39 • BACKWARD 1 • Q4/5

True or false?

P(rain given clouds) is the same as P(clouds given rain).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B1-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 39 • BACKWARD 1 • Q5/5

1286

QUESTION ID

S39-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Different conditions.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B1-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 39 • BACKWARD 1 • Q5/5

SET 39 • HSS-CP.A.5

BACKWARD 2

Read conditional rates in a table • High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4

ANSWER KEY Reduced copy of the question page, with its answer

In the driving table, what fraction of juniors drive? Give lowest terms.

	Drives	Does not drive	Total
Juniors	60	60	120
Seniors	30	50	80
Total	90	110	200

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B2-Q1

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 39 • BACKWARD 2 • Q1/5

1288

QUESTION ID

S39-B2-Q1

TYPE

Short answer

CORRECT ANSWER

1/2

GRADING NOTE 60/120.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S39-B2-Q1**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 39 • BACKWARD 2 • Q1/5

True or false?

In the driving table, "among drivers, 2 in 3 are juniors" is correct.

	Drives	Does not drive	Total
Juniors	60	60	120
Seniors	30	50	80
Total	90	110	200

A. True **B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B2-Q2

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 39 • BACKWARD 2 • Q2/5

1289

QUESTION ID
S39-B2-Q2

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE 60/90.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B2-Q2

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 39 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

In the driving table, which sentence is correct?

	Drives	Does not drive	Total
Juniors	60	60	120
Seniors	30	50	80
Total	90	110	200

A. Among seniors, 3 in 8 drive.

B. Among seniors, 1 in 2 drive.

C. Among juniors, 3 in 8 drive.

D. Among drivers, 3 in 8 are seniors.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B2-Q3

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 39 • BACKWARD 2 • Q3/5

1290

QUESTION ID

S39-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. Among seniors, 3 in 8 drive.

GRADING NOTE 30/80. Drivers who are seniors: 30/90 = 1/3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S39-B2-Q3**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 39 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

In the driving table, what fraction of drivers are seniors? Give lowest terms.

	Drives	Does not drive	Total
Juniors	60	60	120
Seniors	30	50	80
Total	90	110	200

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B2-Q4

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 39 • BACKWARD 2 • Q4/5

1291

QUESTION ID

S39-B2-Q4

TYPE

Short answer

CORRECT ANSWER

1/3

GRADING NOTE 30/90.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S39-B2-Q4**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 39 • BACKWARD 2 • Q4/5

True or false?

In the driving table, "among seniors, most drive" is correct.

	Drives	Does not drive	Total
Juniors	60	60	120
Seniors	30	50	80
Total	90	110	200

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B2-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 39 • BACKWARD 2 • Q5/5

1292

QUESTION ID

S39-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $3/8 < 1/2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S39-B2-Q5**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 39 • BACKWARD 2 • Q5/5

SET 39 • HSS-CP.A.5

FORWARD 1

Translate statements and compute • High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.5

ANSWER KEY Reduced copy of the question page, with its answer

"If it is a school day, there is an 80% chance that Sam takes the bus." Which notation matches?

A. P(bus given school day) = 0.8

B. P(bus) = 0.8

C. P(school day given bus) = 0.8

D. P(bus and school day) = 0.8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F1-Q1

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.5 • Set 39 • FORWARD 1 • Q1/5

1294

QUESTION ID

S39-F1-Q1

TYPE

Multiple choice

CORRECT ANSWER

A. P(bus given school day) = 0.8

GRADING NOTE The condition follows "if".

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F1-Q1

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.5 • Set 39 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

5 of 7 days are school days, and on a school day Sam takes the bus with probability 0.8. What is P(school day and bus)? Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F1-Q2

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.5 • Set 39 • FORWARD 1 • Q2/5

1295

QUESTION ID

S39-F1-Q2

TYPE

Short answer

CORRECT ANSWER

4/7

GRADING NOTE (5/7)(4/5).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S39-F1-Q2**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.5 • Set 39 • FORWARD 1 • Q2/5

True or false?

"Knowing it is a school day changes the chance that Sam takes the bus" means the events are not independent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F1-Q3

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.5 • Set 39 • FORWARD 1 • Q3/5

1296

QUESTION ID
S39-F1-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Dependence.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F1-Q3

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.5 • Set 39 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A test is positive for 95% of people with a disease.
Which sentence states this correctly?

A. 95% of all people have the disease.

B. 95% of all people test positive.

C. Among people with the disease, 95% test positive.

D. Among people who test positive, 95% have the disease.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F1-Q4

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.5 • Set 39 • FORWARD 1 • Q4/5

1297

QUESTION ID
S39-F1-Q4

TYPE
Multiple choice

CORRECT ANSWER
C. Among people with the disease, 95% test positive.

GRADING NOTE Condition: has the disease.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S39-F1-Q4**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.5 • Set 39 • FORWARD 1 • Q4/5

True or false?

"Among people who test positive, 95% have the disease" follows from "95% of people with the disease test positive".

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F1-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.5 • Set 39 • FORWARD 1 • Q5/5

1298

QUESTION ID

S39-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Reversing the condition is not allowed.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F1-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.5 • Set 39 • FORWARD 1 • Q5/5

SET 39 • HSS-CP.A.5

FORWARD 2

Find a conditional probability as a fraction of outcomes • High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.6

ANSWER KEY Reduced copy of the question page, with its answer

A fair six-sided die is rolled. Given the roll is odd, what is P(the roll is 5)? Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F2-Q1

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.6 • Set 39 • FORWARD 2 • Q1/5

1300

GRADING NOTE {1, 3, 5}.

QUESTION ID

S39-F2-Q1

TYPE

Short answer

CORRECT ANSWER

1/3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S39-F2-Q1**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.6 • Set 39 • FORWARD 2 • Q1/5

1300

ANSWER KEY Reduced copy of the question page, with its answer

A card drawn at random from a standard 52-card deck is red. What is P(it is a king)? Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F2-Q2

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.6 • Set 39 • FORWARD 2 • Q2/5

1301

QUESTION ID

S39-F2-Q2

TYPE

Short answer

CORRECT ANSWER

1/13

GRADING NOTE 2 of 26.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S39-F2-Q2**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.6 • Set 39 • FORWARD 2 • Q2/5

True or false?

P(A given B) can be found as (outcomes in both A and B)/(outcomes in B) in a uniform model.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F2-Q3

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.6 • Set 39 • FORWARD 2 • Q3/5

1302

QUESTION ID

S39-F2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Fraction of B.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F2-Q3

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.6 • Set 39 • FORWARD 2 • Q3/5

A number from 1 to 10 is even. What is P(it is greater than 6)?

A. 1/2

B. 2/5

C. 2/3

D. 1/5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F2-Q4

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.6 • Set 39 • FORWARD 2 • Q4/5

1303

QUESTION ID

S39-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 2/5

GRADING NOTE {2, 4, 6, 8, 10}: 8 and 10.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S39-F2-Q4**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.6 • Set 39 • FORWARD 2 • Q4/5

True or false?

Given the roll of a fair six-sided die is even, P(roll is 3) = 1/6.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F2-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.6 • Set 39 • FORWARD 2 • Q5/5

1304

QUESTION ID

S39-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F2-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.6 • Set 39 • FORWARD 2 • Q5/5

Find conditional probability as a fraction of outcomes

MAIN	Probability & Statistics main question	**HSS-CP.B.6**
BACKWARD 1	Count outcomes in events	**HSS-CP.A.1**
BACKWARD 2	Use the definition of conditional probability	**HSS-CP.A.3**
FORWARD 1	Condition on "at least one"	**HSS-CP.B.6**
FORWARD 2	Multiply along a chain of conditions	**HSS-CP.B.8**

SET 40 • HSS-CP.B.6

MAIN

Probability & Statistics main question • High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.6

Two fair dice are rolled independently. The first die shows 5. What is P(the sum is 10 or more)? Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-M-Q1

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.6 • Set 40 • MAIN • Q1/5

1307

QUESTION ID

S40-M-Q1

TYPE

Short answer

CORRECT ANSWER

1/3

GRADING NOTE Second die 5 or 6: 2 of 6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S40-M-Q1**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.6 • Set 40 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A card drawn at random from a standard 52-card deck is red. What is P(it is an ace)? Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-M-Q2

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.6 • Set 40 • MAIN • Q2/5

1308

QUESTION ID
S40-M-Q2

TYPE
Short answer

CORRECT ANSWER
1/13

GRADING NOTE 2 of 26.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S40-M-Q2**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.6 • Set 40 • MAIN • Q2/5

A patient from the drug table is chosen at random and got the placebo. What is P(improved)? Give a fraction in lowest terms.

	Improved	Did not	Total
Drug	35	15	50
Placebo	12	18	30
Total	47	33	80

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-M-Q3

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.6 • Set 40 • MAIN • Q3/5

1309

QUESTION ID

S40-M-Q3

TYPE

Short answer

CORRECT ANSWER

2/5

GRADING NOTE 12 of 30.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S40-M-Q3**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.6 • Set 40 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A number from 1 to 10 is chosen at random and is odd. What is P(it is prime)? Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-M-Q4

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.6 • Set 40 • MAIN • Q4/5

1310

QUESTION ID

S40-M-Q4

TYPE

Short answer

CORRECT ANSWER

3/5

GRADING NOTE {1, 3, 5, 7, 9}: 3, 5, 7.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S40-M-Q4**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.6 • Set 40 • MAIN • Q4/5

From the drug table, P(improved given placebo) = 2/5. What does this mean in the model?

	Improved	Did not	Total
Drug	35	15	50
Placebo	12	18	30
Total	47	33	80

A. Of the patients who improved, 2 in 5 got the placebo.

B. 2 in 5 of all patients got the placebo and improved.

C. Of the patients given the placebo, 2 in 5 improved.

D. 2 in 5 of all patients got the placebo.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-M-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.6 • Set 40 • MAIN • Q5/5

1311

QUESTION ID

S40-M-Q5

TYPE

Multiple choice

CORRECT ANSWER

C. Of the patients given the placebo, 2 in 5 improved.

GRADING NOTE The placebo group is the sample space.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S40-M-Q5**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.6 • Set 40 • MAIN • Q5/5

SET 40 • HSS-CP.B.6

BACKWARD 1

Count outcomes in events • High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1

ANSWER KEY Reduced copy of the question page, with its answer

How many red cards are in a standard 52-card deck?

Cards: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B1-Q1

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 40 • BACKWARD 1 • Q1/5

1313

QUESTION ID

S40-B1-Q1

TYPE

Short answer

CORRECT ANSWER

26

GRADING NOTE Hearts and diamonds.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S40-B1-Q1**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 40 • BACKWARD 1 • Q1/5

Two fair dice rolled independently: how many outcomes have the first die equal to 5?

Outcomes: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B1-Q2

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 40 • BACKWARD 1 • Q2/5

1314

QUESTION ID

S40-B1-Q2

TYPE

Short answer

CORRECT ANSWER

6

GRADING NOTE (5,1) to (5,6).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S40-B1-Q2**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 40 • BACKWARD 1 • Q2/5

True or false?

A standard 52-card deck has 2 red aces.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B1-Q3

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 40 • BACKWARD 1 • Q3/5

1315

QUESTION ID

S40-B1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Hearts and diamonds.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B1-Q3

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 40 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

How many numbers from 1 to 10 are odd and prime?

A. 2

B. 5

C. 3

D. 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B1-Q4

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 40 • BACKWARD 1 • Q4/5

1316

QUESTION ID

S40-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 3

GRADING NOTE 3, 5, 7.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S40-B1-Q4**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 40 • BACKWARD 1 • Q4/5

True or false?

A standard 52-card deck has 4 red kings.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B1-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 40 • BACKWARD 1 • Q5/5

1317

QUESTION ID

S40-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B1-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 40 • BACKWARD 1 • Q5/5

SET 40 • HSS-CP.B.6

BACKWARD 2

Use the definition of conditional probability • High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3

ANSWER KEY Reduced copy of the question page, with its answer

P(A and B) = 1/12 and P(B) = 1/4. Find P(A given B).
Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B2-Q1

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 40 • BACKWARD 2 • Q1/5

1319

QUESTION ID
S40-B2-Q1

TYPE
Short answer

CORRECT ANSWER
1/3

GRADING NOTE (1/12)/(1/4).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S40-B2-Q1**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 40 • BACKWARD 2 • Q1/5

True or false?

In a uniform model, P(A and B)/P(B) equals (count of A and B)/(count of B).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B2-Q2

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 40 • BACKWARD 2 • Q2/5

1320

QUESTION ID

S40-B2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE The total cancels.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B2-Q2

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 40 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

P(A and B) = 0.1 and P(B) = 0.5. Find P(A given B).

A. 5

B. 0.5

C. 0.05

D. 0.2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B2-Q3

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 40 • BACKWARD 2 • Q3/5

1321

QUESTION ID

S40-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 0.2

GRADING NOTE 0.1/0.5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S40-B2-Q3**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 40 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

P(A and B) = 2/36 and P(B) = 6/36. Find P(A given B). Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B2-Q4

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 40 • BACKWARD 2 • Q4/5

1322

QUESTION ID

S40-B2-Q4

TYPE

Short answer

CORRECT ANSWER

1/3

GRADING NOTE 2/6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S40-B2-Q4**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 40 • BACKWARD 2 • Q4/5

True or false?

P(A given B) = P(A and B)/P(A).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B2-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 40 • BACKWARD 2 • Q5/5

1323

QUESTION ID

S40-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Divide by P(B).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B2-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 40 • BACKWARD 2 • Q5/5

SET 40 • HSS-CP.B.6

FORWARD 1

Condition on "at least one" • High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.6

ANSWER KEY Reduced copy of the question page, with its answer

Two fair dice are rolled independently and at least one shows 6. What is P(the sum is 7)? Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F1-Q1

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.6 • Set 40 • FORWARD 1 • Q1/5

1325

QUESTION ID

S40-F1-Q1

TYPE

Short answer

CORRECT ANSWER

2/11

GRADING NOTE 11 outcomes contain a 6; (1,6) and (6,1) sum to 7.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S40-F1-Q1**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.6 • Set 40 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A family has two children; each is equally likely to be a girl or a boy. Given at least one is a girl, what is P(both are girls)? Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F1-Q2

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.6 • Set 40 • FORWARD 1 • Q2/5

1326

QUESTION ID

S40-F1-Q2

TYPE

Short answer

CORRECT ANSWER

1/3

GRADING NOTE GG, GB, BG.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S40-F1-Q2**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.6 • Set 40 • FORWARD 1 • Q2/5

True or false?

Two fair coins are flipped independently. Given at least one shows heads, P(both heads) = 1/2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F1-Q3

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.6 • Set 40 • FORWARD 1 • Q3/5

1327

QUESTION ID

S40-F1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 1/3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F1-Q3

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.6 • Set 40 • FORWARD 1 • Q3/5

Two fair dice are rolled independently and the sum is at least 11. What is P(doubles)?

A. 1/36

B. 1/6

C. 1/2

D. 1/3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F1-Q4

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.6 • Set 40 • FORWARD 1 • Q4/5

1328

QUESTION ID

S40-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 1/3

GRADING NOTE (5,6), (6,5), (6,6).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S40-F1-Q4**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.6 • Set 40 • FORWARD 1 • Q4/5

True or false?

Conditioning on an event removes the outcomes outside it from the sample space.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F1-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.6 • Set 40 • FORWARD 1 • Q5/5

1329

QUESTION ID

S40-F1-Q5

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Restriction.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F1-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.6 • Set 40 • FORWARD 1 • Q5/5

SET 40 • HSS-CP.B.6

FORWARD 2

Multiply along a chain of conditions • High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.8

ANSWER KEY Reduced copy of the question page, with its answer

A bag has 4 red and 2 blue marbles. Two are drawn at random without replacement. What is P(both red)? Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F2-Q1

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.8 • Set 40 • FORWARD 2 • Q1/5

1331

QUESTION ID

S40-F2-Q1

TYPE

Short answer

CORRECT ANSWER

2/5

GRADING NOTE (4/6)(3/5) = 12/30.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F2-Q1

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.8 • Set 40 • FORWARD 2 • Q1/5

True or false?

P(A and B) = P(A)·P(B given A).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F2-Q2

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.8 • Set 40 • FORWARD 2 • Q2/5

1332

QUESTION ID

S40-F2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE General Multiplication Rule.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F2-Q2

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.8 • Set 40 • FORWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

P(A) = 0.8 and P(B given A) = 0.25. Find P(A and B).

P(A and B): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F2-Q3

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.8 • Set 40 • FORWARD 2 • Q3/5

1333

QUESTION ID

S40-F2-Q3

TYPE

Short answer

CORRECT ANSWER

0.2

GRADING NOTE 0.8 · 0.25.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S40-F2-Q3**

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.8 • Set 40 • FORWARD 2 • Q3/5

Two cards are drawn at random, without replacement, from a standard 52-card deck. What is P(second is a heart given the first is a heart)?

A. 1/4

B. 13/52

C. 12/51

D. 13/51

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F2-Q4

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.8 • Set 40 • FORWARD 2 • Q4/5

1334

QUESTION ID
S40-F2-Q4

TYPE
Multiple choice

CORRECT ANSWER
C. 12/51

GRADING NOTE 12 hearts left of 51 cards.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S40-F2-Q4**

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.8 • Set 40 • FORWARD 2 • Q4/5

True or false?

For draws without replacement, P(second red given first red) equals P(first red).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F2-Q5

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.8 • Set 40 • FORWARD 2 • Q5/5

1335

QUESTION ID
S40-F2-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE One red is gone.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F2-Q5

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.8 • Set 40 • FORWARD 2 • Q5/5

Apply the Addition Rule

MAIN	Probability & Statistics main question	**HSS-CP.B.7**
BACKWARD 1	Count a union without double counting	**HSS-CP.A.1**
BACKWARD 2	Add and subtract fractions	**5.NF.A.1**
FORWARD 1	Use the rule in reverse and with exclusive events	**HSS-CP.B.7**
FORWARD 2	Combine the Addition Rule with complements	**HSS-CP.B.7**

SET 41 • HSS-CP.B.7

MAIN

Probability & Statistics main question • High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.7

ANSWER KEY Reduced copy of the question page, with its answer

P(A) = 0.45, P(B) = 0.35 and P(A and B) = 0.15. Find P(A or B).

P(A or B): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-M-Q1

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.7 • Set 41 • MAIN • Q1/5

1338

QUESTION ID

S41-M-Q1

TYPE

Short answer

CORRECT ANSWER

0.65

GRADING NOTE 0.45 + 0.35 − 0.15.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S41-M-Q1**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.7 • Set 41 • MAIN • Q1/5

A card is drawn at random from a standard 52-card deck. Find P(red or king). Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-M-Q2

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.7 • Set 41 • MAIN • Q2/5

1339

QUESTION ID

S41-M-Q2

TYPE

Short answer

CORRECT ANSWER

7/13

GRADING NOTE 26 + 4 − 2 = 28; 28/52.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S41-M-Q2**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.7 • Set 41 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Two fair dice are rolled independently. Find P(doubles or sum of 8). Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-M-Q3

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.7 • Set 41 • MAIN • Q3/5

1340

QUESTION ID
S41-M-Q3

TYPE
Short answer

CORRECT ANSWER
5/18

GRADING NOTE 6 + 5 − 1 = 10 outcomes; 10/36.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S41-M-Q3**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.7 • Set 41 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Of 200 students, 120 take Spanish, 70 take art, and 40 take both. A student is chosen at random. What is P(Spanish or art)?

A. 3/4

B. 7/10

C. 19/20

D. 1/5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-M-Q4

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.7 • Set 41 • MAIN • Q4/5

1341

QUESTION ID

S41-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 3/4

GRADING NOTE 150/200. 19/20 forgets to subtract the 40.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-M-Q4

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.7 • Set 41 • MAIN • Q4/5

For a randomly chosen shopper, P(buys bread or milk) = 0.65. What does this mean?

A. About 65% of shoppers buy bread, milk, or both.

B. About 65% of shoppers buy both bread and milk.

C. About 65% of shoppers buy neither one.

D. About 65% of shoppers buy exactly one.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-M-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.7 • Set 41 • MAIN • Q5/5

1342

QUESTION ID

S41-M-Q5

TYPE

Multiple choice

CORRECT ANSWER

A. About 65% of shoppers buy bread, milk, or both.

GRADING NOTE "Or" includes both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S41-M-Q5**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.7 • Set 41 • MAIN • Q5/5

SET 41 • HSS-CP.B.7

BACKWARD 1

Count a union without double counting • High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1

ANSWER KEY Reduced copy of the question page, with its answer

A fair six-sided die is rolled. How many outcomes are "even or less than 3"?

Outcomes: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B1-Q1

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 41 • BACKWARD 1 • Q1/5

1344

QUESTION ID

S41-B1-Q1

TYPE

Short answer

CORRECT ANSWER

4

GRADING NOTE {1, 2, 4, 6}.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B1-Q1

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 41 • BACKWARD 1 • Q1/5

True or false?

In a standard 52-card deck, the number of cards that are hearts or face cards is 13 + 12 − 3 = 22.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B1-Q2

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 41 • BACKWARD 1 • Q2/5

1345

QUESTION ID

S41-B1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Remove the 3 heart face cards counted twice.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B1-Q2

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 41 • BACKWARD 1 • Q2/5

In a standard 52-card deck, how many cards are aces or spades?

A. 4

B. 13

C. 16

D. 17

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B1-Q3

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 41 • BACKWARD 1 • Q3/5

1346

QUESTION ID

S41-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 16

GRADING NOTE $4 + 13 - 1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B1-Q3

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 41 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

How many numbers from 1 to 20 are multiples of 2 or multiples of 5?

Numbers: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B1-Q4

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 41 • BACKWARD 1 • Q4/5

1347

QUESTION ID

S41-B1-Q4

TYPE

Short answer

CORRECT ANSWER

12

GRADING NOTE 10 + 4 − 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S41-B1-Q4**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 41 • BACKWARD 1 • Q4/5

True or false?

The number of outcomes in "A or B" is always the count of A plus the count of B.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B1-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 41 • BACKWARD 1 • Q5/5

1348

QUESTION ID

S41-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Subtract the overlap.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B1-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 41 • BACKWARD 1 • Q5/5

SET 41 • HSS-CP.B.7

BACKWARD 2

Add and subtract fractions • Grade 5 • Number & Operations–Fractions • 5.NF.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Compute 1/2 + 1/13 − 1/26. Give a fraction in lowest terms.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B2-Q1

Grade 5 • Number & Operations–Fractions • 5.NF.A.1 • Set 41 • BACKWARD 2 • Q1/5

1350

QUESTION ID

S41-B2-Q1

TYPE

Short answer

CORRECT ANSWER

7/13

GRADING NOTE 13/26 + 2/26 − 1/26.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B2-Q1

Grade 5 • Number & Operations–Fractions • 5.NF.A.1 • Set 41 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Compute 1/6 + 5/36 − 1/36. Give a fraction in lowest terms.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B2-Q2

Grade 5 • Number & Operations–Fractions • 5.NF.A.1 • Set 41 • BACKWARD 2 • Q2/5

1351

QUESTION ID
S41-B2-Q2

TYPE
Short answer

CORRECT ANSWER
5/18

GRADING NOTE 6/36 + 5/36 − 1/36.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B2-Q2

Grade 5 • Number & Operations–Fractions • 5.NF.A.1 • Set 41 • BACKWARD 2 • Q2/5

True or false?

1/4 + 1/3 = 7/12.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B2-Q3

Grade 5 • Number & Operations–Fractions • 5.NF.A.1 • Set 41 • BACKWARD 2 • Q3/5

1352

QUESTION ID

S41-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE 3/12 + 4/12.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B2-Q3

Grade 5 • Number & Operations–Fractions • 5.NF.A.1 • Set 41 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Compute 3/5 + 7/20 − 1/5.

A. 17/20

B. 1

C. 3/4

D. 9/20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B2-Q4

Grade 5 • Number & Operations–Fractions • 5.NF.A.1 • Set 41 • BACKWARD 2 • Q4/5

1353

GRADING NOTE 12/20 + 7/20 − 4/20.

QUESTION ID

S41-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 3/4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B2-Q4

Grade 5 • Number & Operations–Fractions • 5.NF.A.1 • Set 41 • BACKWARD 2 • Q4/5

True or false?

1/2 + 1/3 = 2/5.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B2-Q5

Grade 5 • Number & Operations–Fractions • 5.NF.A.1 • Set 41 • BACKWARD 2 • Q5/5

1354

QUESTION ID

S41-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 5/6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B2-Q5

Grade 5 • Number & Operations–Fractions • 5.NF.A.1 • Set 41 • BACKWARD 2 • Q5/5

SET 41 • HSS-CP.B.7

FORWARD 1

Use the rule in reverse and with exclusive events • High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.7

ANSWER KEY Reduced copy of the question page, with its answer

P(A) = 0.5, P(B) = 0.4 and P(A or B) = 0.7. Find P(A and B).

P(A and B): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F1-Q1

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.7 • Set 41 • FORWARD 1 • Q1/5

1356

QUESTION ID

S41-F1-Q1

TYPE

Short answer

CORRECT ANSWER

0.2

GRADING NOTE 0.5 + 0.4 − 0.7.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S41-F1-Q1**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.7 • Set 41 • FORWARD 1 • Q1/5

True or false?

If A and B are mutually exclusive, P(A or B) = P(A) + P(B).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F1-Q2

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.7 • Set 41 • FORWARD 1 • Q2/5

1357

QUESTION ID

S41-F1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE P(A and B) = 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F1-Q2

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.7 • Set 41 • FORWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

A fair six-sided die is rolled. Find P(1 or 6). Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F1-Q3

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.7 • Set 41 • FORWARD 1 • Q3/5

1358

QUESTION ID
S41-F1-Q3

TYPE
Short answer

CORRECT ANSWER
1/3

GRADING NOTE Exclusive: 1/6 + 1/6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S41-F1-Q3**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.7 • Set 41 • FORWARD 1 • Q3/5

P(A) = 0.3, P(B) = 0.5, and A and B are independent. What is P(A or B)?

A. 0.15

B. 0.5

C. 0.65

D. 0.8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F1-Q4

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.7 • Set 41 • FORWARD 1 • Q4/5

1359

QUESTION ID

S41-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 0.65

GRADING NOTE 0.3 + 0.5 − 0.15.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S41-F1-Q4**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.7 • Set 41 • FORWARD 1 • Q4/5

True or false?

If P(A) = 0.6 and P(B) = 0.7, then A and B could be mutually exclusive.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F1-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.7 • Set 41 • FORWARD 1 • Q5/5

1360

QUESTION ID

S41-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE P(A) + P(B) > 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F1-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.7 • Set 41 • FORWARD 1 • Q5/5

SET 41 • HSS-CP.B.7

FORWARD 2

Combine the Addition Rule with complements • High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.7

ANSWER KEY Reduced copy of the question page, with its answer

P(A or B) = 0.65. What is P(neither A nor B)?

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F2-Q1

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.7 • Set 41 • FORWARD 2 • Q1/5

1362

QUESTION ID

S41-F2-Q1

TYPE

Short answer

CORRECT ANSWER

0.35

GRADING NOTE 1 − 0.65.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F2-Q1

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.7 • Set 41 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Of 200 students, 120 take Spanish, 70 take art and 40 take both. What is P(neither)? Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F2-Q2

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.7 • Set 41 • FORWARD 2 • Q2/5

1363

QUESTION ID

S41-F2-Q2

TYPE

Short answer

CORRECT ANSWER

1/4

GRADING NOTE 1 − 3/4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S41-F2-Q2**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.7 • Set 41 • FORWARD 2 • Q2/5

True or false?

P(not (A or B)) = 1 − P(A or B).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F2-Q3

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.7 • Set 41 • FORWARD 2 • Q3/5

1364

QUESTION ID

S41-F2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Complement.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F2-Q3

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.7 • Set 41 • FORWARD 2 • Q3/5

A card is drawn at random from a standard 52-card deck. What is P(neither red nor a king)?

A. 6/13

B. 1/2

C. 7/13

D. 11/26

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F2-Q4

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.7 • Set 41 • FORWARD 2 • Q4/5

1365

QUESTION ID

S41-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 6/13

GRADING NOTE 1 − 7/13.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S41-F2-Q4**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.7 • Set 41 • FORWARD 2 • Q4/5

True or false?

P(neither A nor B) = 1 − P(A) − P(B) in every case.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F2-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.7 • Set 41 • FORWARD 2 • Q5/5

1366

QUESTION ID
S41-F2-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE Only if A and B cannot overlap.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F2-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.7 • Set 41 • FORWARD 2 • Q5/5

Apply the general Multiplication Rule (+)

MAIN	Probability & Statistics main question	**HSS-CP.B.8**
BACKWARD 1	Rearrange the conditional probability formula	**HSS-CP.A.3**
BACKWARD 2	Multiply fractions	**5.NF.B.4**
BACKWARD 3	Multiply for independent events	**HSS-CP.A.2**
FORWARD 1	Combine orders and longer chains	**HSS-CP.B.8**
FORWARD 2	Count arrangements to find probabilities	**HSS-CP.B.9**

SET 42 • HSS-CP.B.8

MAIN

Probability & Statistics main question • High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.8

A bag has 5 red and 3 blue marbles. Two are drawn at random without replacement. What is P(both red)? Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-M-Q1

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.8 • Set 42 • MAIN • Q1/5

1369

QUESTION ID

S42-M-Q1

TYPE

Short answer

CORRECT ANSWER

5/14

GRADING NOTE (5/8)(4/7) = 20/56.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S42-M-Q1**

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.8 • Set 42 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A bag has 5 red and 3 blue marbles. Two are drawn at random without replacement. What is P(first red and second blue)? Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-M-Q2

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.8 • Set 42 • MAIN • Q2/5

1370

QUESTION ID
S42-M-Q2

TYPE
Short answer

CORRECT ANSWER
15/56

GRADING NOTE (5/8)(3/7).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S42-M-Q2**

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.8 • Set 42 • MAIN • Q2/5

Two cards are drawn at random, without replacement, from a standard 52-card deck. What is P(both aces)? Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-M-Q3

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.8 • Set 42 • MAIN • Q3/5

1371

QUESTION ID
S42-M-Q3

TYPE
Short answer

CORRECT ANSWER
1/221

GRADING NOTE (4/52)(3/51).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S42-M-Q3**

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.8 • Set 42 • MAIN • Q3/5

P(A) = 0.6 and P(B given A) = 0.5. Find P(A and B).

P(A and B): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-M-Q4

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.8 • Set 42 • MAIN • Q4/5

1372

QUESTION ID
S42-M-Q4

TYPE
Short answer

CORRECT ANSWER
0.3

GRADING NOTE 0.6 · 0.5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S42-M-Q4**

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.8 • Set 42 • MAIN • Q4/5

A committee has 3 boys and 2 girls. Two members are chosen at random, one after the other. What is P(both girls)?

A. 2/5

B. 1/10

C. 4/25

D. 1/5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-M-Q5

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.8 • Set 42 • MAIN • Q5/5

1373

QUESTION ID
S42-M-Q5

TYPE
Multiple choice

CORRECT ANSWER
B. 1/10

GRADING NOTE (2/5)(1/4). 4/25 assumes replacement.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-M-Q5

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.8 • Set 42 • MAIN • Q5/5

SET 42 • HSS-CP.B.8

BACKWARD 1

Rearrange the conditional probability formula • High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3

True or false?

From P(B given A) = P(A and B)/P(A), it follows that P(A and B) = P(A)·P(B given A).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B1-Q1

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 42 • BACKWARD 1 • Q1/5

1375

QUESTION ID

S42-B1-Q1

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Multiply both sides.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B1-Q1

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 42 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

P(A) = 0.4 and P(B given A) = 0.25. Find P(A and B).

P(A and B): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B1-Q2

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 42 • BACKWARD 1 • Q2/5

1376

QUESTION ID

S42-B1-Q2

TYPE

Short answer

CORRECT ANSWER

0.1

GRADING NOTE Product.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S42-B1-Q2**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 42 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

P(B) = 0.5 and P(A given B) = 0.6. What is P(A and B)?

A. 0.1

B. 1.1

C. 0.83

D. 0.3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B1-Q3

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 42 • BACKWARD 1 • Q3/5

1377

QUESTION ID

S42-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 0.3

GRADING NOTE 0.5 · 0.6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S42-B1-Q3**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 42 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

P(A and B) = 0.12 and P(A) = 0.3. Find P(B given A).

P(B given A): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B1-Q4

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 42 • BACKWARD 1 • Q4/5

1378

QUESTION ID

S42-B1-Q4

TYPE

Short answer

CORRECT ANSWER

0.4

GRADING NOTE 0.12/0.3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B1-Q4

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 42 • BACKWARD 1 • Q4/5

True or false?

P(A and B) = P(A) + P(B given A).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B1-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 42 • BACKWARD 1 • Q5/5

1379

QUESTION ID

S42-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Multiply.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B1-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 42 • BACKWARD 1 • Q5/5

SET 42 • HSS-CP.B.8

BACKWARD 2

Multiply fractions • Grade 5 • Number & Operations–Fractions • 5.NF.B.4

ANSWER KEY Reduced copy of the question page, with its answer

Compute (5/8)(4/7). Give a fraction in lowest terms.

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B2-Q1

Grade 5 • Number & Operations–Fractions • 5.NF.B.4 • Set 42 • BACKWARD 2 • Q1/5

1381

QUESTION ID

S42-B2-Q1

TYPE

Short answer

CORRECT ANSWER

5/14

GRADING NOTE 20/56.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B2-Q1

Grade 5 • Number & Operations–Fractions • 5.NF.B.4 • Set 42 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Compute (4/52)(3/51). Give a fraction in lowest terms.

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B2-Q2

Grade 5 • Number & Operations–Fractions • 5.NF.B.4 • Set 42 • BACKWARD 2 • Q2/5

1382

QUESTION ID

S42-B2-Q2

TYPE

Short answer

CORRECT ANSWER

1/221

GRADING NOTE 12/2652.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B2-Q2

Grade 5 • Number & Operations–Fractions • 5.NF.B.4 • Set 42 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

(2/5)(1/4) = 1/10.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B2-Q3

Grade 5 • Number & Operations–Fractions • 5.NF.B.4 • Set 42 • BACKWARD 2 • Q3/5

1383

QUESTION ID

S42-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE 2/20.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B2-Q3

Grade 5 • Number & Operations–Fractions • 5.NF.B.4 • Set 42 • BACKWARD 2 • Q3/5

Compute (3/8)(2/7).

A. 3/28

B. 6/15

C. 3/56

D. 5/15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B2-Q4

Grade 5 • Number & Operations–Fractions • 5.NF.B.4 • Set 42 • BACKWARD 2 • Q4/5

1384

QUESTION ID

S42-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 3/28

GRADING NOTE 6/56.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B2-Q4

Grade 5 • Number & Operations–Fractions • 5.NF.B.4 • Set 42 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

(1/2)(1/3) = 2/5.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B2-Q5

Grade 5 • Number & Operations–Fractions • 5.NF.B.4 • Set 42 • BACKWARD 2 • Q5/5

1385

QUESTION ID

S42-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 1/6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B2-Q5

Grade 5 • Number & Operations–Fractions • 5.NF.B.4 • Set 42 • BACKWARD 2 • Q5/5

SET 42 • HSS-CP.B.8

BACKWARD 3

Multiply for independent events • High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.2

ANSWER KEY Reduced copy of the question page, with its answer

A fair die is rolled twice, independently. What is P(both 6)? Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B3-Q1

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.2 • Set 42 • BACKWARD 3 • Q1/5

1387

QUESTION ID

S42-B3-Q1

TYPE

Short answer

CORRECT ANSWER

1/36

GRADING NOTE (1/6)(1/6).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S42-B3-Q1**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.2 • Set 42 • BACKWARD 3 • Q1/5

True or false?

With replacement, P(red then red) from 5 red and 3 blue is (5/8)(5/8).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B3-Q2

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.2 • Set 42 • BACKWARD 3 • Q2/5

1388

QUESTION ID

S42-B3-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Independent draws.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B3-Q2

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.2 • Set 42 • BACKWARD 3 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

A fair coin is flipped and a fair die is rolled, independently. P(tails and even)?

A. 2/3

B. 1/4

C. 1/12

D. 1/2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B3-Q3

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.2 • Set 42 • BACKWARD 3 • Q3/5

1389

QUESTION ID

S42-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 1/4

GRADING NOTE (1/2)(1/2).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S42-B3-Q3**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.2 • Set 42 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Independent events have P(A) = 0.5 and P(B) = 0.3.
Find P(A and B).

P(A and B): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B3-Q4

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.2 • Set 42 • BACKWARD 3 • Q4/5

1390

QUESTION ID
S42-B3-Q4

TYPE
Short answer

CORRECT ANSWER
0.15

GRADING NOTE 0.5 · 0.3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S42-B3-Q4**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.2 • Set 42 • BACKWARD 3 • Q4/5

True or false?

Without replacement, draws are independent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B3-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.2 • Set 42 • BACKWARD 3 • Q5/5

1391

QUESTION ID

S42-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Without replacement changes the second draw.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B3-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.2 • Set 42 • BACKWARD 3 • Q5/5

SET 42 • HSS-CP.B.8

FORWARD 1

Combine orders and longer chains • High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.8

ANSWER KEY Reduced copy of the question page, with its answer

A bag has 5 red and 3 blue marbles. Two are drawn at random without replacement. What is P(one of each color)? Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F1-Q1

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.8 • Set 42 • FORWARD 1 • Q1/5

1393

QUESTION ID

S42-F1-Q1

TYPE

Short answer

CORRECT ANSWER

15/28

GRADING NOTE (5/8)(3/7) + (3/8)(5/7) = 30/56.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F1-Q1

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.8 • Set 42 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Three cards are drawn at random, without replacement, from a standard 52-card deck. What is P(all three are hearts)? Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F1-Q2

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.8 • Set 42 • FORWARD 1 • Q2/5

1394

QUESTION ID

S42-F1-Q2

TYPE

Short answer

CORRECT ANSWER

11/850

GRADING NOTE (13/52)(12/51)(11/50) = 1716/132600.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S42-F1-Q2**

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.8 • Set 42 • FORWARD 1 • Q2/5

True or false?

P(A and B) = P(B)·P(A given B) gives the same answer as P(A)·P(B given A).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F1-Q3
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.8 • Set 42 • FORWARD 1 • Q3/5

1395

QUESTION ID
S42-F1-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Both equal P(A and B).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F1-Q3
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.8 • Set 42 • FORWARD 1 • Q3/5

A bag has 4 green and 6 yellow marbles. Two are drawn at random without replacement. What is P(both yellow)?

A. 9/25

B. 2/5

C. 3/5

D. 1/3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F1-Q4

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.8 • Set 42 • FORWARD 1 • Q4/5

1396

QUESTION ID

S42-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 1/3

GRADING NOTE (6/10)(5/9) = 30/90.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S42-F1-Q4**

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.8 • Set 42 • FORWARD 1 • Q4/5

True or false?

P(first red and second blue) equals P(first blue and second red) for 5 red and 3 blue drawn without replacement.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F1-Q5

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.8 • Set 42 • FORWARD 1 • Q5/5

1397

QUESTION ID
S42-F1-Q5

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Both 15/56.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F1-Q5

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.8 • Set 42 • FORWARD 1 • Q5/5

SET 42 • HSS-CP.B.8

FORWARD 2

Count arrangements to find probabilities • High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.9

ANSWER KEY Reduced copy of the question page, with its answer

How many ways can 2 marbles be chosen from 8 (order does not matter)?

Ways: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F2-Q1

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 42 • FORWARD 2 • Q1/5

1399

QUESTION ID

S42-F2-Q1

TYPE

Short answer

CORRECT ANSWER

28

GRADING NOTE $8 \cdot 7/2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S42-F2-Q1**

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 42 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

How many ways can 2 red marbles be chosen from 5 red?

Ways: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F2-Q2

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 42 • FORWARD 2 • Q2/5

1400

QUESTION ID
S42-F2-Q2

TYPE
Short answer

CORRECT ANSWER
10

GRADING NOTE 5·4/2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S42-F2-Q2**

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 42 • FORWARD 2 • Q2/5

True or false?

P(both red) = 10/28 agrees with (5/8)(4/7).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F2-Q3

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 42 • FORWARD 2 • Q3/5

1401

QUESTION ID

S42-F2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Both 5/14.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F2-Q3

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 42 • FORWARD 2 • Q3/5

How many ordered ways can 2 of 8 marbles be drawn one after the other?

A. 16

B. 28

C. 64

D. 56

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F2-Q4

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 42 • FORWARD 2 • Q4/5

1402

QUESTION ID

S42-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 56

GRADING NOTE $8 \cdot 7$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F2-Q4

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 42 • FORWARD 2 • Q4/5

True or false?

Counting with combinations always gives a different probability than the Multiplication Rule.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F2-Q5

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 42 • FORWARD 2 • Q5/5

1403

QUESTION ID

S42-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Both methods agree.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F2-Q5

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 42 • FORWARD 2 • Q5/5

Count ordered arrangements with permutations (+)

MAIN	Probability & Statistics main question	**HSS-CP.B.9**
BACKWARD 1	Count outcomes with a tree or the counting principle	**7.SP.C.8.b**
BACKWARD 2	Multiply several whole numbers	**5.NBT.B.5**
FORWARD 1	Count arrangements with conditions	**HSS-CP.B.9**
FORWARD 2	Count selections where order does not matter	**HSS-CP.B.9**

SET 43 • HSS-CP.B.9

MAIN

Probability & Statistics main question • High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.9

Five runners race. In how many ways can first, second and third place be awarded?

Ways: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-M-Q1

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 43 • MAIN • Q1/5

1406

QUESTION ID

S43-M-Q1

TYPE

Short answer

CORRECT ANSWER

60

GRADING NOTE 5 · 4 · 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S43-M-Q1**

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 43 • MAIN • Q1/5

In how many ways can 6 different books be arranged on a shelf?

Ways: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-M-Q2

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 43 • MAIN • Q2/5

1407

QUESTION ID

S43-M-Q2

TYPE

Short answer

CORRECT ANSWER

720

GRADING NOTE 6!.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S43-M-Q2**

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 43 • MAIN • Q2/5

A 4-digit code uses the digits 0-9 with no digit repeated. How many codes are possible?

Codes: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-M-Q3

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 43 • MAIN • Q3/5

1408

QUESTION ID
S43-M-Q3

TYPE
Short answer

CORRECT ANSWER
5040

GRADING NOTE $10 \cdot 9 \cdot 8 \cdot 7$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-M-Q3

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 43 • MAIN • Q3/5

What is the value of 8P2 (the number of ordered arrangements of 2 objects from 8)?

A. 56

B. 28

C. 16

D. 64

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-M-Q4
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 43 • MAIN • Q4/5

1409

QUESTION ID
S43-M-Q4

TYPE
Multiple choice

CORRECT ANSWER
A. 56

GRADING NOTE 8 · 7. 28 is 8C2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S43-M-Q4**
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 43 • MAIN • Q4/5

A club of 10 members chooses a president, a vice president and a treasurer. How many ways are possible?

Ways: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-M-Q5
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 43 • MAIN • Q5/5

1410

QUESTION ID
S43-M-Q5

TYPE
Short answer

CORRECT ANSWER
720

GRADING NOTE $10 \cdot 9 \cdot 8$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S43-M-Q5**
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 43 • MAIN • Q5/5

SET 43 • HSS-CP.B.9

BACKWARD 1

Count outcomes with a tree or the counting principle • Grade 7 • Statistics & Probability • 7.SP.C.8.b

ANSWER KEY Reduced copy of the question page, with its answer

An outfit is a shirt (4 choices) and pants (3 choices).
How many outfits?

Outfits: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B1-Q1

Grade 7 • Statistics & Probability • 7.SP.C.8.b • Set 43 • BACKWARD 1 • Q1/5

1412

QUESTION ID
S43-B1-Q1

TYPE
Short answer

CORRECT ANSWER
12

GRADING NOTE 4 · 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B1-Q1

Grade 7 • Statistics & Probability • 7.SP.C.8.b • Set 43 • BACKWARD 1 • Q1/5

True or false?

A tree for 3 stages with 2 choices each has 8 paths.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B1-Q2

Grade 7 • Statistics & Probability • 7.SP.C.8.b • Set 43 • BACKWARD 1 • Q2/5

1413

QUESTION ID

S43-B1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE 2^3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B1-Q2

Grade 7 • Statistics & Probability • 7.SP.C.8.b • Set 43 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

A meal is a main (5 choices), a side (3) and a drink (2). How many meals?

A. 30

B. 60

C. 15

D. 10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B1-Q3

Grade 7 • Statistics & Probability • 7.SP.C.8.b • Set 43 • BACKWARD 1 • Q3/5

1414

QUESTION ID

S43-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 30

GRADING NOTE 5 · 3 · 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B1-Q3

Grade 7 • Statistics & Probability • 7.SP.C.8.b • Set 43 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A license code is a letter (26 choices) followed by a digit (10 choices). How many codes?

Codes: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B1-Q4

Grade 7 • Statistics & Probability • 7.SP.C.8.b • Set 43 • BACKWARD 1 • Q4/5

1415

GRADING NOTE 26 · 10.

QUESTION ID

S43-B1-Q4

TYPE

Short answer

CORRECT ANSWER

260

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B1-Q4

Grade 7 • Statistics & Probability • 7.SP.C.8.b • Set 43 • BACKWARD 1 • Q4/5

True or false?

With 3 shirts and 2 pants there are 5 outfits.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B1-Q5

Grade 7 • Statistics & Probability • 7.SP.C.8.b • Set 43 • BACKWARD 1 • Q5/5

1416

QUESTION ID

S43-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B1-Q5

Grade 7 • Statistics & Probability • 7.SP.C.8.b • Set 43 • BACKWARD 1 • Q5/5

SET 43 • HSS-CP.B.9

BACKWARD 2

Multiply several whole numbers • Grade 5 • Number & Operations in Base Ten • 5.NBT.B.5

ANSWER KEY Reduced copy of the question page, with its answer

Compute 10 · 9 · 8.

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B2-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.5 • Set 43 • BACKWARD 2 • Q1/5

1418

QUESTION ID

S43-B2-Q1

TYPE

Short answer

CORRECT ANSWER

720

GRADING NOTE 90 · 8.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B2-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.5 • Set 43 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Compute $6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1$.

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B2-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.5 • Set 43 • BACKWARD 2 • Q2/5

1419

QUESTION ID

S43-B2-Q2

TYPE

Short answer

CORRECT ANSWER

720

GRADING NOTE 6!.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B2-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.5 • Set 43 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$5 \cdot 4 \cdot 3 = 60$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B2-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.5 • Set 43 • BACKWARD 2 • Q3/5

1420

QUESTION ID

S43-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $20 \cdot 3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B2-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.5 • Set 43 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Compute 7 · 6 · 5.

A. 343

B. 180

C. 42

D. 210

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B2-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.5 • Set 43 • BACKWARD 2 • Q4/5

1421

QUESTION ID

S43-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 210

GRADING NOTE 42 · 5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B2-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.5 • Set 43 • BACKWARD 2 • Q4/5

True or false?

$4 \cdot 3 \cdot 2 \cdot 1 = 10$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B2-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.5 • Set 43 • BACKWARD 2 • Q5/5

1422

QUESTION ID

S43-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 24.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B2-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.5 • Set 43 • BACKWARD 2 • Q5/5

SET 43 • HSS-CP.B.9

FORWARD 1

Count arrangements with conditions • High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.9

ANSWER KEY Reduced copy of the question page, with its answer

In how many ways can the letters of MATH be arranged if M must be first?

Ways: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F1-Q1

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 43 • FORWARD 1 • Q1/5

1424

QUESTION ID
S43-F1-Q1

TYPE
Short answer

CORRECT ANSWER
6

GRADING NOTE 3! for the rest.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S43-F1-Q1**

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 43 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

In how many ways can 5 people stand in a line if Ana and Ben must be at the two ends?

Ways: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F1-Q2

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 43 • FORWARD 1 • Q2/5

1425

QUESTION ID
S43-F1-Q2

TYPE
Short answer

CORRECT ANSWER
12

GRADING NOTE 2 ways for the ends · 3! = 12.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S43-F1-Q2**

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 43 • FORWARD 1 • Q2/5

True or false?

nPr = n!/(n − r)!.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F1-Q3

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 43 • FORWARD 1 • Q3/5

1426

QUESTION ID

S43-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Formula.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F1-Q3

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 43 • FORWARD 1 • Q3/5

How many 3-letter arrangements (no repeats) can be made from the letters A, B, C, D, E if the first letter is a vowel?

A. 60

B. 12

C. 48

D. 24

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F1-Q4

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 43 • FORWARD 1 • Q4/5

1427

QUESTION ID

S43-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 24

GRADING NOTE 2 vowels · 4 · 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S43-F1-Q4**

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 43 • FORWARD 1 • Q4/5

True or false?

The number of ways to arrange 4 people in a line is 4 + 3 + 2 + 1.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S43-F1-Q5

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 43 • FORWARD 1 • Q5/5

1428

QUESTION ID
S43-F1-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE $4 \cdot 3 \cdot 2 \cdot 1 = 24$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S43-F1-Q5

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 43 • FORWARD 1 • Q5/5

SET 43 • HSS-CP.B.9

FORWARD 2

Count selections where order does not matter • High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.9

How many ways can 3 of 5 students be chosen for a team (order does not matter)?

Ways: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F2-Q1
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 43 • FORWARD 2 • Q1/5

1430

QUESTION ID
S43-F2-Q1

TYPE
Short answer

CORRECT ANSWER
10

GRADING NOTE 60 ordered / 3! = 10.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S43-F2-Q1**
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 43 • FORWARD 2 • Q1/5

True or false?

Each unordered group of 3 people can be ordered in 6 ways.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F2-Q2

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 43 • FORWARD 2 • Q2/5

1431

QUESTION ID

S43-F2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE 3!.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F2-Q2

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 43 • FORWARD 2 • Q2/5

Which situation is counted with combinations rather than permutations?

A. choosing 3 pizza toppings

B. awarding gold, silver and bronze

C. making a 4-digit code

D. arranging books on a shelf

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F2-Q3

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 43 • FORWARD 2 • Q3/5

1432

QUESTION ID
S43-F2-Q3

TYPE
Multiple choice

CORRECT ANSWER
A. choosing 3 pizza toppings

GRADING NOTE Order of toppings does not matter.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F2-Q3

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 43 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A permutation count is 120 for choosing 3 of 6 in order. How many unordered groups is that?

Groups: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F2-Q4

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 43 • FORWARD 2 • Q4/5

1433

QUESTION ID
S43-F2-Q4

TYPE
Short answer

CORRECT ANSWER
20

GRADING NOTE 120/6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F2-Q4

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 43 • FORWARD 2 • Q4/5

True or false?

There are more combinations than permutations of 3 objects from 6.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F2-Q5

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 43 • FORWARD 2 • Q5/5

1434

QUESTION ID

S43-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Fewer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F2-Q5

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 43 • FORWARD 2 • Q5/5

Count unordered selections with combinations (+)

MAIN	Probability & Statistics main question	**HSS-CP.B.9**
BACKWARD 1	Count ordered arrangements	**HSS-CP.B.9**
BACKWARD 2	Represent and count outcomes of compound events	**7.SP.C.8.b**
FORWARD 1	Count selections from separate groups	**HSS-CP.B.9**
FORWARD 2	Turn counts into probabilities	**HSS-CP.B.9**

SET 44 • HSS-CP.B.9

MAIN

Probability & Statistics main question • High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.9

How many ways can 3 students be chosen from 5 (order does not matter)?

Ways: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S44-M-Q1

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 44 • MAIN • Q1/5

1437

QUESTION ID

S44-M-Q1

TYPE

Short answer

CORRECT ANSWER

10

GRADING NOTE 5C3 = 10.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S44-M-Q1**

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 44 • MAIN • Q1/5

A pizza shop has 8 toppings. How many 3-topping pizzas are possible (no repeated toppings)?

Pizzas: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-M-Q2

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 44 • MAIN • Q2/5

1438

QUESTION ID
S44-M-Q2

TYPE
Short answer

CORRECT ANSWER
56

GRADING NOTE 8C3 = 56.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S44-M-Q2**

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 44 • MAIN • Q2/5

How many 4-person committees can be chosen from 10 people?

Committees: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-M-Q3

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 44 • MAIN • Q3/5

1439

QUESTION ID

S44-M-Q3

TYPE

Short answer

CORRECT ANSWER

210

GRADING NOTE 10C4 = 210.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S44-M-Q3**

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 44 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

What is the value of 6C2?

A. 15

B. 12

C. 36

D. 30

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-M-Q4

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 44 • MAIN • Q4/5

1440

QUESTION ID

S44-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 15

GRADING NOTE 6·5/2. 30 is 6P2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-M-Q4

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 44 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Seven people each shake hands once with every other person. How many handshakes are there?

Handshakes: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-M-Q5

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 44 • MAIN • Q5/5

1441

QUESTION ID
S44-M-Q5

TYPE
Short answer

CORRECT ANSWER
21

GRADING NOTE 7C2 = 21.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-M-Q5

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 44 • MAIN • Q5/5

SET 44 • HSS-CP.B.9

BACKWARD 1

Count ordered arrangements • High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.9

ANSWER KEY Reduced copy of the question page, with its answer

How many ordered ways can 3 of 5 students be lined up?

Ways: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B1-Q1

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 44 • BACKWARD 1 • Q1/5

1443

QUESTION ID
S44-B1-Q1

TYPE
Short answer

CORRECT ANSWER
60

GRADING NOTE $5 \cdot 4 \cdot 3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S44-B1-Q1**

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 44 • BACKWARD 1 • Q1/5

True or false?

5P3 = 60.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B1-Q2

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 44 • BACKWARD 1 • Q2/5

1444

QUESTION ID

S44-B1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE 5 · 4 · 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B1-Q2

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 44 • BACKWARD 1 • Q2/5

How many ordered pairs of different people can be formed from 7 people?

A. 42

B. 21

C. 49

D. 14

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B1-Q3

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 44 • BACKWARD 1 • Q3/5

1445

QUESTION ID

S44-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 42

GRADING NOTE $7 \cdot 6$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B1-Q3

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 44 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

How many ways can 3 people be arranged in order?

Ways: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B1-Q4

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 44 • BACKWARD 1 • Q4/5

1446

QUESTION ID

S44-B1-Q4

TYPE

Short answer

CORRECT ANSWER

6

GRADING NOTE 3!.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B1-Q4

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 44 • BACKWARD 1 • Q4/5

True or false?

10P4 = 210.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B1-Q5

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 44 • BACKWARD 1 • Q5/5

1447

QUESTION ID

S44-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 5040; 210 is 10C4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B1-Q5

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 44 • BACKWARD 1 • Q5/5

SET 44 • HSS-CP.B.9

BACKWARD 2

Represent and count outcomes of compound events • Grade 7 • Statistics & Probability • 7.SP.C.8.b

ANSWER KEY Reduced copy of the question page, with its answer

How many 2-letter groups (order does not matter) can be chosen from the letters A, B, C?

Groups: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B2-Q1

Grade 7 • Statistics & Probability • 7.SP.C.8.b • Set 44 • BACKWARD 2 • Q1/5

1449

GRADING NOTE AB, AC, BC.

QUESTION ID

S44-B2-Q1

TYPE

Short answer

CORRECT ANSWER

3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B2-Q1

Grade 7 • Statistics & Probability • 7.SP.C.8.b • Set 44 • BACKWARD 2 • Q1/5

True or false?

From A, B, C, D, the groups AB and BA are the same group.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B2-Q2

Grade 7 • Statistics & Probability • 7.SP.C.8.b • Set 44 • BACKWARD 2 • Q2/5

1450

QUESTION ID

S44-B2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Unordered.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B2-Q2

Grade 7 • Statistics & Probability • 7.SP.C.8.b • Set 44 • BACKWARD 2 • Q2/5

How many 2-letter groups (order does not matter) can be chosen from A, B, C, D?

A. 8

B. 6

C. 4

D. 12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B2-Q3

Grade 7 • Statistics & Probability • 7.SP.C.8.b • Set 44 • BACKWARD 2 • Q3/5

1451

QUESTION ID

S44-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 6

GRADING NOTE AB, AC, AD, BC, BD, CD.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B2-Q3

Grade 7 • Statistics & Probability • 7.SP.C.8.b • Set 44 • BACKWARD 2 • Q3/5

How many ordered pairs of different letters can be made from A, B, C, D?

Pairs: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B2-Q4

Grade 7 • Statistics & Probability • 7.SP.C.8.b • Set 44 • BACKWARD 2 • Q4/5

1452

QUESTION ID

S44-B2-Q4

TYPE

Short answer

CORRECT ANSWER

12

GRADING NOTE 4 · 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B2-Q4

Grade 7 • Statistics & Probability • 7.SP.C.8.b • Set 44 • BACKWARD 2 • Q4/5

True or false?

An organized list of groups should include both AB and BA when order does not matter.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B2-Q5

Grade 7 • Statistics & Probability • 7.SP.C.8.b • Set 44 • BACKWARD 2 • Q5/5

1453

QUESTION ID

S44-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE List each group once.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B2-Q5

Grade 7 • Statistics & Probability • 7.SP.C.8.b • Set 44 • BACKWARD 2 • Q5/5

SET 44 • HSS-CP.B.9

FORWARD 1

Count selections from separate groups • High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.9

A committee needs 3 of 5 boys and 2 of 4 girls. How many committees are possible?

Committees: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F1-Q1

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 44 • FORWARD 1 • Q1/5

1455

QUESTION ID

S44-F1-Q1

TYPE

Short answer

CORRECT ANSWER

60

GRADING NOTE $10 \cdot 6$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F1-Q1

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 44 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

How many ways can 2 hearts be chosen from the 13 hearts in a standard 52-card deck?

Ways: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F1-Q2

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 44 • FORWARD 1 • Q2/5

1456

QUESTION ID

S44-F1-Q2

TYPE

Short answer

CORRECT ANSWER

78

GRADING NOTE 13C2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F1-Q2

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 44 • FORWARD 1 • Q2/5

True or false?

nCr = n!/(r!(n − r)!).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F1-Q3

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 44 • FORWARD 1 • Q3/5

1457

QUESTION ID

S44-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Formula.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F1-Q3

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 44 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

How many 5-card hands can be chosen from a standard 52-card deck (order does not matter)?

A. 2,598,960

B. 380,204,032

C. 260

D. 311,875,200

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F1-Q4

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 44 • FORWARD 1 • Q4/5

1458

QUESTION ID
S44-F1-Q4

TYPE
Multiple choice

CORRECT ANSWER
A. 2,598,960

GRADING NOTE 52C5. 311,875,200 is 52P5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S44-F1-Q4**

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 44 • FORWARD 1 • Q4/5

True or false?

Choosing 2 from 4 girls and 3 from 5 boys gives 6 + 10 = 16 committees.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F1-Q5

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 44 • FORWARD 1 • Q5/5

1459

QUESTION ID

S44-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Multiply: 60.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F1-Q5

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 44 • FORWARD 1 • Q5/5

SET 44 • HSS-CP.B.9

FORWARD 2

Turn counts into probabilities • High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.9

ANSWER KEY Reduced copy of the question page, with its answer

3 people are chosen at random from 5 boys and 4 girls.
How many groups are possible in all?

Groups: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F2-Q1
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 44 • FORWARD 2 • Q1/5

1461

QUESTION ID
S44-F2-Q1

TYPE
Short answer

CORRECT ANSWER
84

GRADING NOTE 9C3 = 84.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F2-Q1
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 44 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

3 people are chosen at random, without replacement (every group of 3 equally likely), from 5 boys and 4 girls. What is P(all three are girls)? Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F2-Q2

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 44 • FORWARD 2 • Q2/5

1462

QUESTION ID
S44-F2-Q2

TYPE
Short answer

CORRECT ANSWER
1/21

GRADING NOTE 4C3/9C3 = 4/84.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S44-F2-Q2**

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 44 • FORWARD 2 • Q2/5

True or false?

In a uniform model, P(event) = (number of outcomes in the event)/(number of outcomes in all).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F2-Q3

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 44 • FORWARD 2 • Q3/5

1463

QUESTION ID
S44-F2-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Counting probability.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F2-Q3

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 44 • FORWARD 2 • Q3/5

2 cards are drawn at random, without replacement, from a standard 52-card deck. What is P(both hearts)?

A. 1/17

B. 1/16

C. 1/4

D. 13/52

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F2-Q4
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 44 • FORWARD 2 • Q4/5

1464

QUESTION ID
S44-F2-Q4

TYPE
Multiple choice

CORRECT ANSWER
A. 1/17

GRADING NOTE 78/1326.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S44-F2-Q4**
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 44 • FORWARD 2 • Q4/5

True or false?

P(all three are boys) from 5 boys and 4 girls is 3/9.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F2-Q5

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 44 • FORWARD 2 • Q5/5

1465

QUESTION ID

S44-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 10/84 = 5/42.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F2-Q5

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 44 • FORWARD 2 • Q5/5

Compute probabilities with permutations and combinations (+)

MAIN	Probability & Statistics main question	**HSS-CP.B.9**
BACKWARD 1	Count with permutations and combinations	**HSS-CP.B.9**
BACKWARD 2	Multiply conditional probabilities along draws	**HSS-CP.B.8**
FORWARD 1	Use complements with counting	**HSS-CP.B.9**
FORWARD 2	Define a random variable from a counting model	**HSS-MD.A.1**

SET 45 • HSS-CP.B.9

MAIN

Probability & Statistics main question • High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.9

A lottery draws 3 different numbers from 1 to 10 (order does not matter). What is the probability that one ticket with 3 numbers wins? Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-M-Q1

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 45 • MAIN • Q1/5

1468

QUESTION ID
S45-M-Q1

TYPE
Short answer

CORRECT ANSWER
1/120

GRADING NOTE 1/10C3 = 1/120.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S45-M-Q1**

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 45 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A 3-person group is chosen at random from 5 boys and 4 girls. What is P(all girls)? Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-M-Q2

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 45 • MAIN • Q2/5

1469

QUESTION ID
S45-M-Q2

TYPE
Short answer

CORRECT ANSWER
1/21

GRADING NOTE 4/84.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-M-Q2

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 45 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Five people line up in random order. What is P(Ann is first and Bo is second)? Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-M-Q3

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 45 • MAIN • Q3/5

1470

QUESTION ID
S45-M-Q3

TYPE
Short answer

CORRECT ANSWER
1/20

GRADING NOTE 3!/5! = 6/120.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-M-Q3

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 45 • MAIN • Q3/5

Two cards are dealt at random, without replacement, from a standard 52-card deck. What is P(both are hearts)? Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-M-Q4

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 45 • MAIN • Q4/5

1471

QUESTION ID

S45-M-Q4

TYPE

Short answer

CORRECT ANSWER

1/17

GRADING NOTE 13C2/52C2 = 78/1326.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-M-Q4

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 45 • MAIN • Q4/5

A 3-person group is chosen at random from 5 boys and 4 girls. What is P(exactly one girl)?

A. 1/3

B. 5/21

C. 10/21

D. 4/9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-M-Q5

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 45 • MAIN • Q5/5

1472

QUESTION ID

S45-M-Q5

TYPE

Multiple choice

CORRECT ANSWER

C. 10/21

GRADING NOTE 4C1 · 5C2 / 84 = 40/84.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S45-M-Q5**

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 45 • MAIN • Q5/5

SET 45 • HSS-CP.B.9

BACKWARD 1

Count with permutations and combinations • High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.9

ANSWER KEY Reduced copy of the question page, with its answer

Compute 10C3.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B1-Q1

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 45 • BACKWARD 1 • Q1/5

1474

QUESTION ID

S45-B1-Q1

TYPE

Short answer

CORRECT ANSWER

120

GRADING NOTE 720/6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B1-Q1

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 45 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Compute 5!.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B1-Q2

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 45 • BACKWARD 1 • Q2/5

1475

QUESTION ID
S45-B1-Q2

TYPE
Short answer

CORRECT ANSWER
120

GRADING NOTE 5 · 4 · 3 · 2 · 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S45-B1-Q2**

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 45 • BACKWARD 1 • Q2/5

True or false?

9C3 = 84.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B1-Q3

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 45 • BACKWARD 1 • Q3/5

1476

QUESTION ID

S45-B1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE 504/6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B1-Q3

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 45 • BACKWARD 1 • Q3/5

Compute 52C2.

A. 676

B. 1,326

C. 2,652

D. 104

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B1-Q4

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 45 • BACKWARD 1 • Q4/5

1477

QUESTION ID

S45-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 1,326

GRADING NOTE 52 · 51/2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B1-Q4

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 45 • BACKWARD 1 • Q4/5

True or false?

5C2 = 20.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B1-Q5

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 45 • BACKWARD 1 • Q5/5

1478

QUESTION ID

S45-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 10.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B1-Q5

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 45 • BACKWARD 1 • Q5/5

SET 45 • HSS-CP.B.9

BACKWARD 2

Multiply conditional probabilities along draws • High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.8

Using the Multiplication Rule, find P(both hearts) for two cards dealt at random, without replacement, from a standard 52-card deck. Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B2-Q1

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.8 • Set 45 • BACKWARD 2 • Q1/5

1480

QUESTION ID
S45-B2-Q1

TYPE
Short answer

CORRECT ANSWER
1/17

GRADING NOTE (13/52)(12/51).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S45-B2-Q1**

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.8 • Set 45 • BACKWARD 2 • Q1/5

True or false?

(4/9)(3/8)(2/7) = 1/21.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B2-Q2

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.8 • Set 45 • BACKWARD 2 • Q2/5

1481

QUESTION ID

S45-B2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE 24/504.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B2-Q2

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.8 • Set 45 • BACKWARD 2 • Q2/5

From 5 boys and 4 girls, two are chosen in order without replacement. P(both boys)?

A. 25/81

B. 1/4

C. 5/9

D. 5/18

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B2-Q3

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.8 • Set 45 • BACKWARD 2 • Q3/5

1482

QUESTION ID

S45-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 5/18

GRADING NOTE (5/9)(4/8).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S45-B2-Q3**

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.8 • Set 45 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Five people line up at random. P(Ann first)? Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B2-Q4

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.8 • Set 45 • BACKWARD 2 • Q4/5

1483

QUESTION ID
S45-B2-Q4

TYPE
Short answer

CORRECT ANSWER
1/5

GRADING NOTE 1 of 5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S45-B2-Q4**

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.8 • Set 45 • BACKWARD 2 • Q4/5

True or false?

A group has 5 boys and 4 girls. Two people are chosen one after the other, at random and without replacement. Given that the first is a girl, P(second is a girl) = 4/9.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B2-Q5

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.8 • Set 45 • BACKWARD 2 • Q5/5

1484

QUESTION ID
S45-B2-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE 3/8.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B2-Q5

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.8 • Set 45 • BACKWARD 2 • Q5/5

SET 45 • HSS-CP.B.9

FORWARD 1

Use complements with counting • High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.9

ANSWER KEY Reduced copy of the question page, with its answer

A 3-person group is chosen at random from 5 boys and 4 girls. What is P(no girls)? Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F1-Q1

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 45 • FORWARD 1 • Q1/5

1486

QUESTION ID

S45-F1-Q1

TYPE

Short answer

CORRECT ANSWER

5/42

GRADING NOTE 10/84.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F1-Q1

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 45 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A 3-person group is chosen at random from 5 boys and 4 girls. What is P(at least one girl)? Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F1-Q2

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 45 • FORWARD 1 • Q2/5

1487

QUESTION ID
S45-F1-Q2

TYPE
Short answer

CORRECT ANSWER
37/42

GRADING NOTE 1 − 5/42.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S45-F1-Q2**

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 45 • FORWARD 1 • Q2/5

True or false?

P(at least one) = 1 − P(none).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F1-Q3

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 45 • FORWARD 1 • Q3/5

1488

QUESTION ID

S45-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Complement.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F1-Q3

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 45 • FORWARD 1 • Q3/5

Four cards are dealt at random, without replacement, from a standard 52-card deck. What is P(no aces)? (48C4 = 194,580 and 52C4 = 270,725.)

A. 194,580/52

B. 4/52

C. 1/13

D. 38,916/54,145

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F1-Q4

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 45 • FORWARD 1 • Q4/5

1489

QUESTION ID

S45-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 38,916/54,145

GRADING NOTE 194,580/270,725 in lowest terms.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S45-F1-Q4**

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 45 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

P(at least one girl) = 3 · (4/9).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F1-Q5

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 45 • FORWARD 1 • Q5/5

1490

QUESTION ID

S45-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Not a valid method.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F1-Q5

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 45 • FORWARD 1 • Q5/5

SET 45 • HSS-CP.B.9

FORWARD 2

Define a random variable from a counting model • High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.A.1

ANSWER KEY Reduced copy of the question page, with its answer

X = the number of girls in a group of 3 chosen at random (without replacement, every group equally likely) from 5 boys and 4 girls. What values can X take? Give the smallest and the largest.

Smallest: _____

Largest: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F2-Q1

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.1 • Set 45 • FORWARD 2 • Q1/5

1492

QUESTION ID

S45-F2-Q1

TYPE

Short answer

CORRECT ANSWER

0; 3

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S45-F2-Q1**

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.1 • Set 45 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

X = the number of girls in a group of 3 chosen at random (without replacement, every group equally likely) from 5 boys and 4 girls. Find P(X = 0) and P(X = 3). Give a fraction in lowest terms.

P(X = 0): _____

P(X = 3): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F2-Q2

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.1 • Set 45 • FORWARD 2 • Q2/5

1493

QUESTION ID

S45-F2-Q2

TYPE

Short answer

CORRECT ANSWER

5/42; 1/21

GRADING NOTE Form required: both. 10/84 and 4/84.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S45-F2-Q2**

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.1 • Set 45 • FORWARD 2 • Q2/5

True or false?

X = the number of girls in a group of 3 chosen at random (without replacement, every group equally likely) from 5 boys and 4 girls. The probabilities P(X = 0), P(X = 1), P(X = 2), P(X = 3) add to 1.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S45-F2-Q3

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.1 • Set 45 • FORWARD 2 • Q3/5

1494

QUESTION ID

S45-F2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE 10 + 40 + 30 + 4 = 84.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S45-F2-Q3

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.1 • Set 45 • FORWARD 2 • Q3/5

X = the number of girls in a group of 3 chosen at random (without replacement, every group equally likely) from 5 boys and 4 girls. What is P(X = 2)?

A. 5/14

B. 2/9

C. 1/21

D. 10/21

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F2-Q4
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.1 • Set 45 • FORWARD 2 • Q4/5

1495

QUESTION ID
S45-F2-Q4

TYPE
Multiple choice

CORRECT ANSWER
A. 5/14

GRADING NOTE 4C2 · 5C1/84 = 30/84.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S45-F2-Q4**
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.1 • Set 45 • FORWARD 2 • Q4/5

True or false?

A random variable assigns a probability, not a number, to each outcome.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F2-Q5
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.1 • Set 45 • FORWARD 2 • Q5/5

1496

QUESTION ID
S45-F2-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE It assigns a number; the distribution gives probabilities.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F2-Q5
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.1 • Set 45 • FORWARD 2 • Q5/5

Define a random variable and graph its distribution (+)

MAIN	Probability & Statistics main question	**HSS-MD.A.1**
BACKWARD 1	Describe outcomes in a sample space	**HSS-CP.A.1**
BACKWARD 2	Read and draw a distribution display	**HSS-ID.A.1**
FORWARD 1	Find probabilities of ranges of values	**HSS-MD.A.1**
FORWARD 2	Compute the expected value of a random variable	**HSS-MD.A.2**

SET 46 • HSS-MD.A.1

MAIN

Probability & Statistics main question • High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.A.1

Three fair coins are flipped independently. X = number of heads. Find P(X = 0), P(X = 1), P(X = 2) and P(X = 3). Give a fraction in lowest terms.

P(X = 0): _____

P(X = 1): _____

P(X = 2): _____

P(X = 3): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-M-Q1

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.1 • Set 46 • MAIN • Q1/5

1499

QUESTION ID

S46-M-Q1

TYPE

Short answer

CORRECT ANSWER

1/8; 3/8; 3/8; 1/8

GRADING NOTE Form required: all four. Outcomes with 0, 1, 2, 3 heads: 1, 3, 3, 1 of 8.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-M-Q1

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.1 • Set 46 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Two fair dice are rolled independently and X = the sum. How many different values can X take, and what is P(X = 7)? Give a fraction in lowest terms.

Number of values: _____

P(X = 7): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-M-Q2

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.1 • Set 46 • MAIN • Q2/5

1500

QUESTION ID

S46-M-Q2

TYPE

Short answer

CORRECT ANSWER

11; 1/6

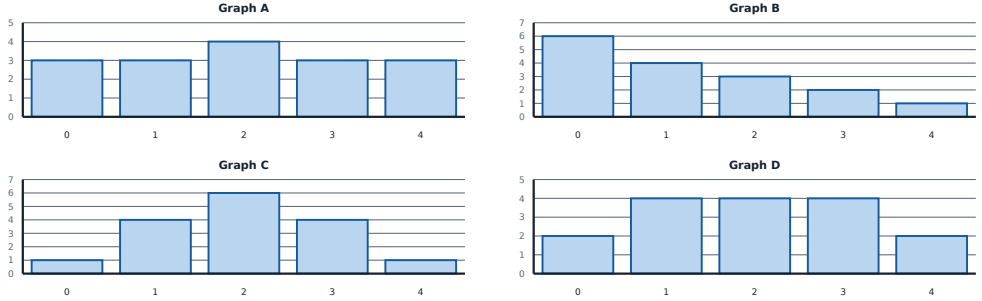
GRADING NOTE Form required: both. 2 to 12; 6 of 36.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S46-M-Q2**

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.1 • Set 46 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Four fair coins are flipped independently and X = number of heads. Each bar graph shows, for X = 0 to 4, how many of the 16 equally likely outcomes give that value. Which graph shows the distribution of X? In each histogram, the horizontal axis shows number of heads and the vertical axis shows frequency.



A. Graph A **B.** Graph B **C.** Graph C **D.** Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-M-Q3

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.1 • Set 46 • MAIN • Q3/5

1501

QUESTION ID
S46-M-Q3

TYPE
Multiple choice

CORRECT ANSWER
C. Graph C

GRADING NOTE Counts 1, 4, 6, 4, 1 (4C0 to 4C4). The others are flat, skewed, or have the wrong middle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-M-Q3

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.1 • Set 46 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Which list could be a probability distribution for X = 1, 2, 3?

A. 0.3, 0.4, 0.4

B. 0.2, 0.2, 0.2

C. 0.5, −0.1, 0.6

D. 0.2, 0.5, 0.3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-M-Q4

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.1 • Set 46 • MAIN • Q4/5

1502

QUESTION ID

S46-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 0.2, 0.5, 0.3

GRADING NOTE Each between 0 and 1 and summing to 1. The others sum to 1.1, include a negative, or sum to 0.6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-M-Q4

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.1 • Set 46 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

X takes the values 0, 1, 2, 3 with P(X = 0) = 0.1, P(X = 1) = 0.35 and P(X = 3) = 0.15. Find P(X = 2).

P(X = 2): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-M-Q5
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.1 • Set 46 • MAIN • Q5/5

1503

QUESTION ID
S46-M-Q5

TYPE
Short answer

CORRECT ANSWER
0.4

GRADING NOTE 1 − 0.6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-M-Q5
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.1 • Set 46 • MAIN • Q5/5

SET 46 • HSS-MD.A.1

BACKWARD 1

Describe outcomes in a sample space • High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Three fair coins are flipped independently. How many outcomes have exactly 1 head?

Outcomes: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B1-Q1

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 46 • BACKWARD 1 • Q1/5

1505

QUESTION ID
S46-B1-Q1

TYPE
Short answer

CORRECT ANSWER
3

GRADING NOTE HTT, THT, TTH.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S46-B1-Q1**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 46 • BACKWARD 1 • Q1/5

True or false?

Three fair coins are flipped independently. The 8 outcomes of the sample space are equally likely.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B1-Q2

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 46 • BACKWARD 1 • Q2/5

1506

QUESTION ID

S46-B1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE 2^3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B1-Q2

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 46 • BACKWARD 1 • Q2/5

Two fair dice are rolled independently. How many outcomes have a sum of 4?

A. 3

B. 2

C. 6

D. 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B1-Q3

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 46 • BACKWARD 1 • Q3/5

1507

QUESTION ID

S46-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 3

GRADING NOTE (1,3), (2,2), (3,1).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S46-B1-Q3**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 46 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Four fair coins are flipped independently. How many outcomes are there?

Outcomes: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B1-Q4

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 46 • BACKWARD 1 • Q4/5

1508

QUESTION ID

S46-B1-Q4

TYPE

Short answer

CORRECT ANSWER

16

GRADING NOTE 2^4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S46-B1-Q4**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 46 • BACKWARD 1 • Q4/5

True or false?

For two fair dice rolled independently, the sums 2 and 12 each come from 2 outcomes.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B1-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 46 • BACKWARD 1 • Q5/5

1509

QUESTION ID

S46-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE One each.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B1-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 46 • BACKWARD 1 • Q5/5

SET 46 • HSS-MD.A.1

BACKWARD 2

Read and draw a distribution display • High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

In a bar graph of a distribution, taller bars show more likely values.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B2-Q1

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1 • Set 46 • BACKWARD 2 • Q1/5

1511

QUESTION ID

S46-B2-Q1

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Height is frequency.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B2-Q1

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1 • Set 46 • BACKWARD 2 • Q1/5

A bar graph shows counts 1, 4, 6, 4, 1 for X = 0 to 4. What is its shape?

A. skewed right

B. uniform

C. skewed left

D. symmetric

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B2-Q2

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1 • Set 46 • BACKWARD 2 • Q2/5

1512

QUESTION ID

S46-B2-Q2

TYPE

Multiple choice

CORRECT ANSWER

D. symmetric

GRADING NOTE Mirror image.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B2-Q2

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1 • Set 46 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

A bar graph has counts 1, 4, 6, 4, 1 for X = 0 to 4. What is the total count?

Total: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B2-Q3

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1 • Set 46 • BACKWARD 2 • Q3/5

1513

QUESTION ID

S46-B2-Q3

TYPE

Short answer

CORRECT ANSWER

16

GRADING NOTE Sum.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S46-B2-Q3**

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1 • Set 46 • BACKWARD 2 • Q3/5

True or false?

A bar graph with counts 6, 4, 3, 2, 1 is symmetric.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B2-Q4

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1 • Set 46 • BACKWARD 2 • Q4/5

1514

QUESTION ID

S46-B2-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Skewed right.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B2-Q4

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1 • Set 46 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Counts are 1, 4, 6, 4, 1 for X = 0 to 4. Which value of X is most likely?

A. 1

B. 2

C. 0

D. 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B2-Q5

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1 • Set 46 • BACKWARD 2 • Q5/5

1515

QUESTION ID

S46-B2-Q5

TYPE

Multiple choice

CORRECT ANSWER

B. 2

GRADING NOTE Tallest bar.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S46-B2-Q5**

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1 • Set 46 • BACKWARD 2 • Q5/5

SET 46 • HSS-MD.A.1

FORWARD 1

Find probabilities of ranges of values • High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.A.1

For X = number of heads in 3 flips, find P(X ≥ 2). Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F1-Q1
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.1 • Set 46 • FORWARD 1 • Q1/5

1517

QUESTION ID
S46-F1-Q1

TYPE
Short answer

CORRECT ANSWER
1/2

GRADING NOTE 3/8 + 1/8.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F1-Q1
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.1 • Set 46 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

For X = sum of two fair dice rolled independently, find P(X ≤ 4). Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F1-Q2
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.1 • Set 46 • FORWARD 1 • Q2/5

1518

QUESTION ID
S46-F1-Q2

TYPE
Short answer

CORRECT ANSWER
1/6

GRADING NOTE 1 + 2 + 3 = 6 outcomes.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F1-Q2
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.1 • Set 46 • FORWARD 1 • Q2/5

True or false?

The probabilities of all values of a random variable add to 1.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F1-Q3
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.1 • Set 46 • FORWARD 1 • Q3/5

1519

QUESTION ID
S46-F1-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Distribution.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F1-Q3
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.1 • Set 46 • FORWARD 1 • Q3/5

X takes the values 0 to 3 with probabilities 0.1, 0.35, 0.4, 0.15. What is P(X > 1)?

A. 0.9

B. 0.45

C. 0.4

D. 0.55

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F1-Q4
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.1 • Set 46 • FORWARD 1 • Q4/5

1520

QUESTION ID
S46-F1-Q4

TYPE
Multiple choice

CORRECT ANSWER
D. 0.55

GRADING NOTE 0.4 + 0.15.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S46-F1-Q4**
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.1 • Set 46 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$P(X \geq 1) = 1 - P(X = 1)$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F1-Q5

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.1 • Set 46 • FORWARD 1 • Q5/5

1521

QUESTION ID

S46-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $1 - P(X = 0)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F1-Q5

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.1 • Set 46 • FORWARD 1 • Q5/5

SET 46 • HSS-MD.A.1

FORWARD 2

Compute the expected value of a random variable • High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.A.2

X = number of heads when 2 fair coins are flipped independently. Find the expected value of X.

E(X): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F2-Q1
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 46 • FORWARD 2 • Q1/5

1523

QUESTION ID
S46-F2-Q1

TYPE
Short answer

CORRECT ANSWER
1

GRADING NOTE 0(1/4) + 1(1/2) + 2(1/4).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S46-F2-Q1**
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 46 • FORWARD 2 • Q1/5

True or false?

The expected value is the probability-weighted average of the values.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F2-Q2
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 46 • FORWARD 2 • Q2/5

1524

QUESTION ID
S46-F2-Q2

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F2-Q2
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 46 • FORWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

X takes 1, 2, 3 with probabilities 0.2, 0.5, 0.3. Find E(X).

E(X): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F2-Q3
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 46 • FORWARD 2 • Q3/5

1525

QUESTION ID
S46-F2-Q3

TYPE
Short answer

CORRECT ANSWER
2.1

GRADING NOTE 0.2 + 1 + 0.9.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F2-Q3
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 46 • FORWARD 2 • Q3/5

X = result of spinning a spinner with 4 equal sections numbered 1 to 4. What is E(X)?

A. 2

B. 4

C. 10

D. 2.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F2-Q4
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 46 • FORWARD 2 • Q4/5

1526

QUESTION ID
S46-F2-Q4

TYPE
Multiple choice

CORRECT ANSWER
D. 2.5

GRADING NOTE 10/4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S46-F2-Q4**
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 46 • FORWARD 2 • Q4/5

True or false?

The expected value must be one of the values X can take.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F2-Q5
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 46 • FORWARD 2 • Q5/5

1527

QUESTION ID
S46-F2-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE E(X) = 2.5 for the spinner.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F2-Q5
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 46 • FORWARD 2 • Q5/5

Calculate and interpret expected value (+)

MAIN	Probability & Statistics main question	**HSS-MD.A.2**
BACKWARD 1	Compute a mean	**6.SP.B.5.c**
BACKWARD 2	Read a probability distribution	**HSS-MD.A.1**
BACKWARD 3	Multiply and add decimals	**6.NS.B.3**
FORWARD 1	Expected value with gains and losses	**HSS-MD.A.2**
FORWARD 2	Find expected values of theoretical distributions	**HSS-MD.A.3**

SET 47 • HSS-MD.A.2

MAIN

Probability & Statistics main question • High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.A.2

X = number of heads when 3 fair coins are flipped independently (table). Find E(X).

x	0	1	2	3
P(X = x)	1/8	3/8	3/8	1/8

E(X): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-M-Q1
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 47 • MAIN • Q1/5

1530

QUESTION ID
S47-M-Q1

TYPE
Short answer

CORRECT ANSWER
1.5

GRADING NOTE 0(1/8) + 1(3/8) + 2(3/8) + 3(1/8) = 12/8.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S47-M-Q1**
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 47 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Find E(X) for the distribution in the table.

x	0	1	2	3
P(X = x)	0.1	0.35	0.4	0.15

E(X): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-M-Q2

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 47 • MAIN • Q2/5

1531

QUESTION ID
S47-M-Q2

TYPE
Short answer

CORRECT ANSWER
1.6

GRADING NOTE 0.35 + 0.8 + 0.45.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-M-Q2

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 47 • MAIN • Q2/5

X = the number rolled on a fair six-sided die. Find E(X).

E(X): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-M-Q3
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 47 • MAIN • Q3/5

1532

QUESTION ID
S47-M-Q3

TYPE
Short answer

CORRECT ANSWER
3.5

GRADING NOTE (1 + 2 + ... + 6)/6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S47-M-Q3**
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 47 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

For the distribution in the table, E(X) = 1.6. What does this mean?

x	0	1	2	3
P(X = x)	0.1	0.35	0.4	0.15

A. The single most likely value of X is 1.6.

B. X will equal exactly 1.6 on most of the trials.

C. Over many trials, the average value of X is about 1.6.

D. X is 1.6 or less with a probability of 1.6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-M-Q4

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 47 • MAIN • Q4/5

1533

QUESTION ID

S47-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. Over many trials, the average value of X is about 1.6.

GRADING NOTE Long-run mean.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-M-Q4

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 47 • MAIN • Q4/5

A game pays $10 with probability 0.2, $2 with probability 0.3, and $0 otherwise. Find the expected payout.

Dollars: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-M-Q5
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 47 • MAIN • Q5/5

1534

QUESTION ID
S47-M-Q5

TYPE
Short answer

CORRECT ANSWER
2.6

GRADING NOTE 2 + 0.6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-M-Q5
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 47 • MAIN • Q5/5

SET 47 • HSS-MD.A.2

BACKWARD 1

Compute a mean • Grade 6 • Statistics & Probability • 6.SP.B.5.c

Find the mean of 1, 2, 3, 4, 5, 6.

Mean: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B1-Q1

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 47 • BACKWARD 1 • Q1/5

1536

QUESTION ID

S47-B1-Q1

TYPE

Short answer

CORRECT ANSWER

3.5

GRADING NOTE 21/6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B1-Q1

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 47 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the mean of 0, 1, 1, 1, 2, 2, 2, 3.

Mean: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B1-Q2

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 47 • BACKWARD 1 • Q2/5

1537

QUESTION ID

S47-B1-Q2

TYPE

Short answer

CORRECT ANSWER

1.5

GRADING NOTE 12/8.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B1-Q2

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 47 • BACKWARD 1 • Q2/5

True or false?

A value listed 3 times counts 3 times in the mean.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B1-Q3

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 47 • BACKWARD 1 • Q3/5

1538

QUESTION ID

S47-B1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Weights.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B1-Q3

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 47 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the mean of 2, 2, 2, 8.

A. 14

B. 5

C. 2

D. 3.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B1-Q4

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 47 • BACKWARD 1 • Q4/5

1539

QUESTION ID

S47-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 3.5

GRADING NOTE 14/4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B1-Q4

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 47 • BACKWARD 1 • Q4/5

True or false?

The mean of 0, 0, 0, 4 is 2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B1-Q5

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 47 • BACKWARD 1 • Q5/5

1540

QUESTION ID

S47-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 4/4 = 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B1-Q5

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 47 • BACKWARD 1 • Q5/5

SET 47 • HSS-MD.A.2

BACKWARD 2

Read a probability distribution • High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.A.1

ANSWER KEY Reduced copy of the question page, with its answer

In the distribution table with values 0 to 3, what is P(X = 2)?

x	0	1	2	3
P(X = x)	0.1	0.35	0.4	0.15

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B2-Q1

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.1 • Set 47 • BACKWARD 2 • Q1/5

1542

QUESTION ID

S47-B2-Q1

TYPE

Short answer

CORRECT ANSWER

0.4

GRADING NOTE Read the table.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B2-Q1

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.1 • Set 47 • BACKWARD 2 • Q1/5

True or false?

A probability distribution lists every value of X with its probability.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B2-Q2
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.1 • Set 47 • BACKWARD 2 • Q2/5

1543

QUESTION ID
S47-B2-Q2

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B2-Q2
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.1 • Set 47 • BACKWARD 2 • Q2/5

In the table (X = number of heads when 3 fair coins are flipped independently), which values of X are equally likely?

x	0	1	2	3
P(X = x)	1/8	3/8	3/8	1/8

A. 2 and 3 **B.** 0 and 1 **C.** all four **D.** 1 and 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B2-Q3
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.1 • Set 47 • BACKWARD 2 • Q3/5

1544

QUESTION ID
S47-B2-Q3

TYPE
Multiple choice

CORRECT ANSWER
D. 1 and 2

GRADING NOTE P(X = 1) = P(X = 2) = 3/8.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B2-Q3
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.1 • Set 47 • BACKWARD 2 • Q3/5

A distribution has P(X = 5) = 0.3 and P(X = 10) = 0.7.
What is P(X = 5) + P(X = 10)?

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B2-Q4
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.1 • Set 47 • BACKWARD 2 • Q4/5

1545

QUESTION ID
S47-B2-Q4

TYPE
Short answer

CORRECT ANSWER
1

GRADING NOTE Total probability.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B2-Q4
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.1 • Set 47 • BACKWARD 2 • Q4/5

True or false?

A distribution may have probabilities adding to 1.2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S47-B2-Q5
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.1 • Set 47 • BACKWARD 2 • Q5/5

1546

QUESTION ID
S47-B2-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE Must add to 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S47-B2-Q5**
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.1 • Set 47 • BACKWARD 2 • Q5/5

SET 47 • HSS-MD.A.2

BACKWARD 3

Multiply and add decimals • Grade 6 • The Number System • 6.NS.B.3

ANSWER KEY Reduced copy of the question page, with its answer

Compute 2(0.4) + 3(0.15).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B3-Q1

Grade 6 • The Number System • 6.NS.B.3 • Set 47 • BACKWARD 3 • Q1/5

1548

QUESTION ID

S47-B3-Q1

TYPE

Short answer

CORRECT ANSWER

1.25

GRADING NOTE 0.8 + 0.45.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B3-Q1

Grade 6 • The Number System • 6.NS.B.3 • Set 47 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Compute 10(0.2) + 2(0.3).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B3-Q2

Grade 6 • The Number System • 6.NS.B.3 • Set 47 • BACKWARD 3 • Q2/5

1549

GRADING NOTE 2 + 0.6.

QUESTION ID

S47-B3-Q2

TYPE

Short answer

CORRECT ANSWER

2.6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B3-Q2

Grade 6 • The Number System • 6.NS.B.3 • Set 47 • BACKWARD 3 • Q2/5

True or false?

1(0.35) + 2(0.4) = 1.15.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B3-Q3

Grade 6 • The Number System • 6.NS.B.3 • Set 47 • BACKWARD 3 • Q3/5

1550

QUESTION ID

S47-B3-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE 0.35 + 0.8.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B3-Q3

Grade 6 • The Number System • 6.NS.B.3 • Set 47 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Compute 5(0.3) + 10(0.7).

A. 8.5

B. 7

C. 1.5

D. 15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B3-Q4

Grade 6 • The Number System • 6.NS.B.3 • Set 47 • BACKWARD 3 • Q4/5

1551

QUESTION ID

S47-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 8.5

GRADING NOTE 1.5 + 7.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B3-Q4

Grade 6 • The Number System • 6.NS.B.3 • Set 47 • BACKWARD 3 • Q4/5

True or false?

$4(0.25) + 2(0.5) = 1.5$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B3-Q5

Grade 6 • The Number System • 6.NS.B.3 • Set 47 • BACKWARD 3 • Q5/5

1552

GRADING NOTE $1 + 1 = 2$.

QUESTION ID

S47-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B3-Q5

Grade 6 • The Number System • 6.NS.B.3 • Set 47 • BACKWARD 3 • Q5/5

SET 47 • HSS-MD.A.2

FORWARD 1

Expected value with gains and losses • High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.A.2

ANSWER KEY Reduced copy of the question page, with its answer

A game: win $5 with probability 0.2, lose $2 with probability 0.8. Find the expected gain.

Dollars: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F1-Q1
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 47 • FORWARD 1 • Q1/5

1554

QUESTION ID
S47-F1-Q1

TYPE
Short answer

CORRECT ANSWER
−0.6

GRADING NOTE 1 − 1.6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S47-F1-Q1**
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 47 • FORWARD 1 • Q1/5

True or false?

A negative expected gain means a player loses money on average over many plays.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F1-Q2
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 47 • FORWARD 1 • Q2/5

1555

QUESTION ID
S47-F1-Q2

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Long run.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F1-Q2
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 47 • FORWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

X has values −3, 0, 6 with probabilities 0.5, 0.2, 0.3. Find E(X).

E(X): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F1-Q3
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 47 • FORWARD 1 • Q3/5

1556

QUESTION ID
S47-F1-Q3

TYPE
Short answer

CORRECT ANSWER
0.3

GRADING NOTE −1.5 + 1.8.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S47-F1-Q3**
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 47 • FORWARD 1 • Q3/5

A game has expected gain −$0.60. About how much does a player expect to lose in 100 plays?

A. $600

B. $6

C. $0.60

D. $60

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F1-Q4
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 47 • FORWARD 1 • Q4/5

1557

QUESTION ID
S47-F1-Q4

TYPE
Multiple choice

CORRECT ANSWER
D. $60

GRADING NOTE 100 · 0.6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S47-F1-Q4**
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 47 • FORWARD 1 • Q4/5

True or false?

If E(X) = 0, the player breaks even on every single play.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F1-Q5
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 47 • FORWARD 1 • Q5/5

1558

QUESTION ID
S47-F1-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE Only on average.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F1-Q5
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 47 • FORWARD 1 • Q5/5

SET 47 • HSS-MD.A.2

FORWARD 2

Find expected values of theoretical distributions • High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.A.3

ANSWER KEY Reduced copy of the question page, with its answer

A student guesses at random on 6 true-false questions (each answer is True or False, and each guess is correct with probability 1/2, independently of the others). X = number correct. Find P(X = 3). Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F2-Q1
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.3 • Set 47 • FORWARD 2 • Q1/5

1560

QUESTION ID
S47-F2-Q1

TYPE
Short answer

CORRECT ANSWER
5/16

GRADING NOTE 6C3/64 = 20/64.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F2-Q1
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.3 • Set 47 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A student guesses at random on 6 true-false questions (each answer is True or False, and each guess is correct with probability 1/2, independently of the others). Find the expected number correct.

E(X): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F2-Q2
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.3 • Set 47 • FORWARD 2 • Q2/5

1561

QUESTION ID
S47-F2-Q2

TYPE
Short answer

CORRECT ANSWER
3

GRADING NOTE 6 · 0.5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F2-Q2
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.3 • Set 47 • FORWARD 2 • Q2/5

True or false?

For n independent trials with success probability p, the expected number of successes is np.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F2-Q3
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.3 • Set 47 • FORWARD 2 • Q3/5

1562

QUESTION ID
S47-F2-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Binomial mean.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F2-Q3
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.3 • Set 47 • FORWARD 2 • Q3/5

A student guesses at random on 10 four-choice questions (each question has 4 choices and one correct answer; each guess picks a choice at random, independently of the others). What is the expected number correct?

A. 2.5

B. 0.25

C. 10

D. 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F2-Q4
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.3 • Set 47 • FORWARD 2 • Q4/5

1563

QUESTION ID
S47-F2-Q4

TYPE
Multiple choice

CORRECT ANSWER
A. 2.5

GRADING NOTE 10 · 1/4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F2-Q4
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.3 • Set 47 • FORWARD 2 • Q4/5

True or false?

A student guesses at random on 4 true-false questions (each answer is True or False, and each guess is correct with probability 1/2, independently of the others). The probability of getting all 4 right is 1/4.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F2-Q5
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.3 • Set 47 • FORWARD 2 • Q5/5

1564

QUESTION ID
S47-F2-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE 1/16.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S47-F2-Q5**
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.3 • Set 47 • FORWARD 2 • Q5/5

Develop a theoretical probability distribution (+)

MAIN	Probability & Statistics main question	**HSS-MD.A.3**
BACKWARD 1	Count arrangements of successes	**HSS-CP.B.9**
BACKWARD 2	Multiply probabilities of independent trials	**HSS-CP.A.2**
FORWARD 1	Build the whole theoretical distribution	**HSS-MD.A.3**
FORWARD 2	Compare a theoretical distribution with data	**HSS-MD.A.4**

SET 48 • HSS-MD.A.3

MAIN

Probability & Statistics main question • High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.A.3

ANSWER KEY Reduced copy of the question page, with its answer

A student guesses at random on 4 true-false questions (each answer is True or False, and each guess is correct with probability 1/2, independently of the others). X = number correct. Find P(X = 2). Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-M-Q1
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.3 • Set 48 • MAIN • Q1/5

1567

QUESTION ID
S48-M-Q1

TYPE
Short answer

CORRECT ANSWER
3/8

GRADING NOTE 4C2 · $(1/2)^4$ = 6/16.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-M-Q1
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.3 • Set 48 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A student guesses at random on 4 true-false questions (each answer is True or False, and each guess is correct with probability 1/2, independently of the others).
X = number correct. Find E(X).

E(X): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-M-Q2
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.3 • Set 48 • MAIN • Q2/5

1568

QUESTION ID
S48-M-Q2

TYPE
Short answer

CORRECT ANSWER
2

GRADING NOTE 4 · 1/2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-M-Q2
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.3 • Set 48 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

A student guesses at random on 5 questions (each question has 4 choices and one correct answer; each guess picks a choice at random, independently of the others). Find P(all 5 correct) and the expected number correct. Give each answer as a fraction in lowest terms.

P(all correct): _____

Expected number correct: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-M-Q3
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.3 • Set 48 • MAIN • Q3/5

1569

QUESTION ID
S48-M-Q3

TYPE
Short answer

CORRECT ANSWER
1/1024; 5/4

GRADING NOTE Form required: both, as fractions. $(1/4)^5$; $5 \cdot 1/4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-M-Q3
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.3 • Set 48 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A student guesses at random on 3 four-choice questions (each question has 4 choices and one correct answer; each guess picks a choice at random, independently of the others). What is P(exactly 1 correct)?

A. 27/64

B. 9/64

C. 1/4

D. 3/4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-M-Q4

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.3 • Set 48 • MAIN • Q4/5

1570

QUESTION ID

S48-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 27/64

GRADING NOTE $3 \cdot (1/4)(3/4)^2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-M-Q4

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.3 • Set 48 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A student guesses at random on 3 four-choice questions (each question has 4 choices and one correct answer; each guess picks a choice at random, independently of the others). What is P(at least 1 correct)? Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-M-Q5
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.3 • Set 48 • MAIN • Q5/5

1571

QUESTION ID
S48-M-Q5

TYPE
Short answer

CORRECT ANSWER
37/64

GRADING NOTE $1 - (3/4)^3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-M-Q5
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.3 • Set 48 • MAIN • Q5/5

SET 48 • HSS-MD.A.3

BACKWARD 1

Count arrangements of successes • High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.9

ANSWER KEY Reduced copy of the question page, with its answer

In how many ways can exactly 2 of 4 answers be correct?

Ways: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B1-Q1

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 48 • BACKWARD 1 • Q1/5

1573

QUESTION ID

S48-B1-Q1

TYPE

Short answer

CORRECT ANSWER

6

GRADING NOTE 4C2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S48-B1-Q1**

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 48 • BACKWARD 1 • Q1/5

In how many ways can exactly 1 of 3 answers be correct?

Ways: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B1-Q2
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 48 • BACKWARD 1 • Q2/5

1574

QUESTION ID
S48-B1-Q2

TYPE
Short answer

CORRECT ANSWER
3

GRADING NOTE 3C1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S48-B1-Q2**
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 48 • BACKWARD 1 • Q2/5

True or false?

5C5 = 1.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B1-Q3

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 48 • BACKWARD 1 • Q3/5

1575

QUESTION ID

S48-B1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE One way to get all correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B1-Q3

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 48 • BACKWARD 1 • Q3/5

In how many ways can exactly 3 of 5 answers be correct?

A. 10

B. 60

C. 15

D. 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B1-Q4

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 48 • BACKWARD 1 • Q4/5

1576

QUESTION ID

S48-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 10

GRADING NOTE 5C3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B1-Q4

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 48 • BACKWARD 1 • Q4/5

True or false?

4C2 = 8.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B1-Q5

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 48 • BACKWARD 1 • Q5/5

1577

QUESTION ID

S48-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B1-Q5

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 48 • BACKWARD 1 • Q5/5

SET 48 • HSS-MD.A.3

BACKWARD 2

Multiply probabilities of independent trials • High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Compute $(1/2)^4$. Give a fraction in lowest terms.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B2-Q1

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.2 • Set 48 • BACKWARD 2 • Q1/5

1579

QUESTION ID

S48-B2-Q1

TYPE

Short answer

CORRECT ANSWER

1/16

GRADING NOTE 1/16.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S48-B2-Q1**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.2 • Set 48 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Compute (1/4)(3/4)(3/4). Give a fraction in lowest terms.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B2-Q2

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.2 • Set 48 • BACKWARD 2 • Q2/5

1580

QUESTION ID
S48-B2-Q2

TYPE
Short answer

CORRECT ANSWER
9/64

GRADING NOTE 9/64.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S48-B2-Q2**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.2 • Set 48 • BACKWARD 2 • Q2/5

True or false?

When each guess is made at random and independently, being right on one question does not change the chance of being right on another.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B2-Q3

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.2 • Set 48 • BACKWARD 2 • Q3/5

1581

QUESTION ID
S48-B2-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE That is what independence means.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B2-Q3

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.2 • Set 48 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

P(correct) = 1/4 on each of 2 independent guesses. P(both correct)?

A. 1/8

B. 1/2

C. 1/16

D. 1/4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B2-Q4

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.2 • Set 48 • BACKWARD 2 • Q4/5

1582

QUESTION ID

S48-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 1/16

GRADING NOTE $(1/4)^2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B2-Q4

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.2 • Set 48 • BACKWARD 2 • Q4/5

True or false?

$(3/4)^3 = 9/12$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B2-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.2 • Set 48 • BACKWARD 2 • Q5/5

1583

QUESTION ID

S48-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 27/64.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B2-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.2 • Set 48 • BACKWARD 2 • Q5/5

SET 48 • HSS-MD.A.3

FORWARD 1

Build the whole theoretical distribution • High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.A.3

ANSWER KEY Reduced copy of the question page, with its answer

A student guesses at random on 3 four-choice questions (each question has 4 choices and one correct answer; each guess picks a choice at random, independently of the others); X = number correct. Find P(X = 0), P(X = 1), P(X = 2), P(X = 3). Give a fraction in lowest terms.

P(X = 0): _____

P(X = 1): _____

P(X = 2): _____

P(X = 3): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F1-Q1

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.3 • Set 48 • FORWARD 1 • Q1/5

1585

QUESTION ID

S48-F1-Q1

TYPE

Short answer

CORRECT ANSWER

27/64; 27/64; 9/64; 1/64

GRADING NOTE Form required: all four. $3Ck\ (1/4)^k\ (3/4)^{3-k}$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F1-Q1

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.3 • Set 48 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A student guesses at random on 3 four-choice questions (each question has 4 choices and one correct answer; each guess picks a choice at random, independently of the others); X = number correct. Find E(X).

E(X): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F1-Q2
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.3 • Set 48 • FORWARD 1 • Q2/5

1586

QUESTION ID
S48-F1-Q2

TYPE
Short answer

CORRECT ANSWER
0.75

GRADING NOTE 3 · 1/4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F1-Q2
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.3 • Set 48 • FORWARD 1 • Q2/5

True or false?

The four probabilities 27/64, 27/64, 9/64, 1/64 add to 1.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F1-Q3
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.3 • Set 48 • FORWARD 1 • Q3/5

1587

QUESTION ID
S48-F1-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE 64/64.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F1-Q3
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.3 • Set 48 • FORWARD 1 • Q3/5

A student guesses at random on 3 four-choice questions (each question has 4 choices and one correct answer; each guess picks a choice at random, independently of the others); X = number correct. Which value of X is least likely?

A. 3

B. 2

C. 0

D. 1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F1-Q4
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.3 • Set 48 • FORWARD 1 • Q4/5

1588

QUESTION ID
S48-F1-Q4

TYPE
Multiple choice

CORRECT ANSWER
A. 3

GRADING NOTE 1/64.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F1-Q4
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.3 • Set 48 • FORWARD 1 • Q4/5

True or false?

A student guesses at random on 3 four-choice questions (each question has 4 choices and one correct answer; each guess picks a choice at random, independently of the others); X = number correct. X = 1 is more likely than X = 0.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F1-Q5
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.3 • Set 48 • FORWARD 1 • Q5/5

1589

QUESTION ID
S48-F1-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE Equal: 27/64 each.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F1-Q5
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.3 • Set 48 • FORWARD 1 • Q5/5

SET 48 • HSS-MD.A.3

FORWARD 2

Compare a theoretical distribution with data • High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.A.4

ANSWER KEY Reduced copy of the question page, with its answer

A fair six-sided die is rolled 60 times, independently. How many sixes does the theoretical distribution predict?

Sixes: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F2-Q1

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.4 • Set 48 • FORWARD 2 • Q1/5

1591

QUESTION ID
S48-F2-Q1

TYPE
Short answer

CORRECT ANSWER
10

GRADING NOTE 60/6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F2-Q1

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.4 • Set 48 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

In 60 rolls, a die showed 6 sixes. What is the empirical probability of a six? Give a decimal.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F2-Q2
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.4 • Set 48 • FORWARD 2 • Q2/5

1592

QUESTION ID
S48-F2-Q2

TYPE
Short answer

CORRECT ANSWER
0.1

GRADING NOTE 6/60.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S48-F2-Q2**
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.4 • Set 48 • FORWARD 2 • Q2/5

True or false?

An empirical probability is found from observed data.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F2-Q3
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.4 • Set 48 • FORWARD 2 • Q3/5

1593

QUESTION ID
S48-F2-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F2-Q3
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.4 • Set 48 • FORWARD 2 • Q3/5

In 100 households, the numbers of TVs gave an average of 1.9. This average is

A. a theoretical probability from a model

B. always equal to 2 for any survey

C. the expected value of the empirical distribution

D. a margin of error for the survey

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F2-Q4
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.4 • Set 48 • FORWARD 2 • Q4/5

1594

QUESTION ID
S48-F2-Q4

TYPE
Multiple choice

CORRECT ANSWER
C. the expected value of the empirical distribution

GRADING NOTE Mean from data.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S48-F2-Q4**
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.4 • Set 48 • FORWARD 2 • Q4/5

True or false?

Empirical and theoretical probabilities for the same event are always equal.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F2-Q5
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.4 • Set 48 • FORWARD 2 • Q5/5

1595

QUESTION ID
S48-F2-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE Data vary.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F2-Q5
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.4 • Set 48 • FORWARD 2 • Q5/5

Develop an empirical probability distribution (+)

MAIN	Probability & Statistics main question	**HSS-MD.A.4**
BACKWARD 1	Find relative frequencies	**7.SP.C.6**
BACKWARD 2	Compute an expected value	**HSS-MD.A.2**
FORWARD 1	Compare empirical and theoretical results	**HSS-MD.A.4**
FORWARD 2	Find the expected payoff of a game of chance	**HSS-MD.B.5.a**

SET 49 • HSS-MD.A.4

MAIN

Probability & Statistics main question • High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.A.4

ANSWER KEY Reduced copy of the question page, with its answer

100 households were surveyed (table). X = number of TVs in a randomly chosen household. Find P(X = 2) and P(X ≥ 3). Give decimals.

TVs in household	0	1	2	3	4
Households	5	30	40	20	5

P(X = 2): _____

P(X ≥ 3): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-M-Q1

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.4 • Set 49 • MAIN • Q1/5

1598

QUESTION ID
S49-M-Q1

TYPE
Short answer

CORRECT ANSWER
0.4; 0.25

GRADING NOTE Form required: both. 40/100; (20 + 5)/100.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-M-Q1

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.4 • Set 49 • MAIN • Q1/5

Using the TV survey, find the expected number of TVs in a household.

TVs in household	0	1	2	3	4
Households	5	30	40	20	5

E(X): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-M-Q2

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.4 • Set 49 • MAIN • Q2/5

1599

QUESTION ID
S49-M-Q2

TYPE
Short answer

CORRECT ANSWER
1.9

GRADING NOTE (30 + 80 + 60 + 20)/100.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S49-M-Q2**

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.4 • Set 49 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

The 100 surveyed households were a random sample of a town's households. Using the sample mean, 1.9 TVs per household, as the estimate, about how many TVs are expected in 500 households of the town?

TVs: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-M-Q3
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.4 • Set 49 • MAIN • Q3/5

1600

QUESTION ID
S49-M-Q3

TYPE
Short answer

CORRECT ANSWER
950

GRADING NOTE 500 · 1.9.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-M-Q3
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.4 • Set 49 • MAIN • Q3/5